
openbricks documentation

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the openbricks contributors

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openbricks is a Pybricks-style MicroPython firmware for **open hardware** — commodity MCUs, commodity motors, commodity sensors.

*Prefer offline reading? This documentation is also available as a single **PDF download**, rebuilt on every deploy. The sidebar names the firmware version these pages document; its version menu lists the archived docs (and PDF) of every earlier firmware release, frozen at /X.Y.Z/.*

Like Pybricks, openbricks is a *firmware you flash to an MCU*, not a library you `pip install` on top of stock MicroPython. The firmware owns the runtime: the 1 kHz motor scheduler, trajectory planner, state observer, and drivebase controller run as native C code inside the image, and the three-layer Python API (drivers → interfaces → robotics) is what `import openbricks` gives you out of the box.

```
from openbricks.drivers.st3032 import ST3032Motor
from openbricks.robotics import DriveBase

left = ST3032Motor(servo_id=1, tx=14, rx=41)
right = ST3032Motor(servo_id=2, tx=14, rx=41, invert=True)

db = DriveBase(left, right, wheel_diameter_mm=56, axle_track_mm=114)
db.straight(500)    # mm
db.turn(90)        # degrees
```

- **Firmware images:** ESP32-S3 (primary) and classic ESP32, prebuilt on every [release](#).
- **Host tooling:** `pipx install openbricks` gets you the [openbricks CLI](#) (flash / run / upload / stop / log over BLE) and, with the `[sim]` extra, a MuJoCo-backed [simulator](#).
- **Source:** github.com/1e0ng/openbricks, MIT licensed.

INSTALLATION

Two things get installed: the **host tooling** on your computer (a CLI that flashes firmware and talks to hubs over BLE) and the **firmware** on the hub itself.

1.1 1. Install the host tooling

```
$ pipx install 'openbricks[sim]' # CLI + MuJoCo simulator
```

or, lighter:

```
$ pipx install openbricks # CLI only (flash / run / log)
```

`pip install` works too; `pipx` is recommended on modern macOS / Linux to avoid the “externally managed environment” error. The package is on PyPI: <https://pypi.org/project/openbricks/>.

The `[sim]` extra adds `mujoco` (~50 MB, native OpenGL) and `numpy`. If you only want to flash and run code on real hardware, skip it — `openbricks sim ...` will print an install hint instead of crashing.

1.2 2. Get a firmware image

Grab a prebuilt image from the [Releases](#) page:

- `openbricks-esp32s3-firmware-<version>.bin` — ESP32-S3 boards (primary target)
- `openbricks-esp32-firmware-<version>.bin` — classic ESP32 boards

(You can also *build the firmware from source*.)

1.3 3. Flash the hub

Connect the board over USB, then flash and name the hub in one step:

```
$ openbricks flash --name RobotA
```

That’s the whole command: the serial port is auto-detected (when exactly one ESP device is connected), the chip type is probed, and the newest matching firmware release is downloaded automatically (cached under `~/.cache/openbricks/firmware`).

- `--name` is the BLE advertising identifier you’ll use later with `openbricks run -n ...`; pick a unique one per hub.

- With several serial devices connected, pass `--port` explicitly (`/dev/ttyUSB0` on Linux, `/dev/cu.usbserial-*` on macOS, `COM5` on Windows).
- To flash a specific/downloaded image instead of the newest release, pass `--firmware path/to/firmware.bin`.

Skip this step entirely if you only want to run code in the *simulator*.

1.4 4. Run your first program

```
$ openbricks list                # find your hub over BLE
$ openbricks run -n RobotA main.py # push a script and stream its output
```

See *the CLI reference* for every command, and *Examples* for programs to start from.

1.5 Troubleshooting

- **Hub doesn't show up in `openbricks list`** — BLE may be toggled off. Short-press the Bluetooth button (GPIO 38 on the S3, GPIO 5 on the classic ESP32); on the ESP32-S3 the onboard LED turns blue when BLE is on, yellow when off. While a program is running the LED flashes that colour at 5 Hz instead of holding it solid — a flashing LED means “robot running”, not a different BLE state. While a host tool is sending to the hub (`openbricks upload`, or `run` staging its program) it flashes **purple** twice as fast, then hands back. A brief flash is the press acknowledgment, shown the moment a program-button press is recognized: **red** for the press that starts a run, **green** for the press that stops one. See `openbricks.hub.ESP32S3DevkitHub`.
- **Serial port permission errors on Linux** — add yourself to the `dialout` group (`sudo usermod -aG dialout $USER`) and re-login.
- **Flash succeeds but the program doesn't start** — check the run log with `openbricks log -n RobotA` for the traceback of the last run.

HARDWARE GUIDE

A starter parts list and wiring notes. Everything here is commodity stuff you can buy on AliExpress / Amazon / Adafruit.

2.1 Recommended starter robot

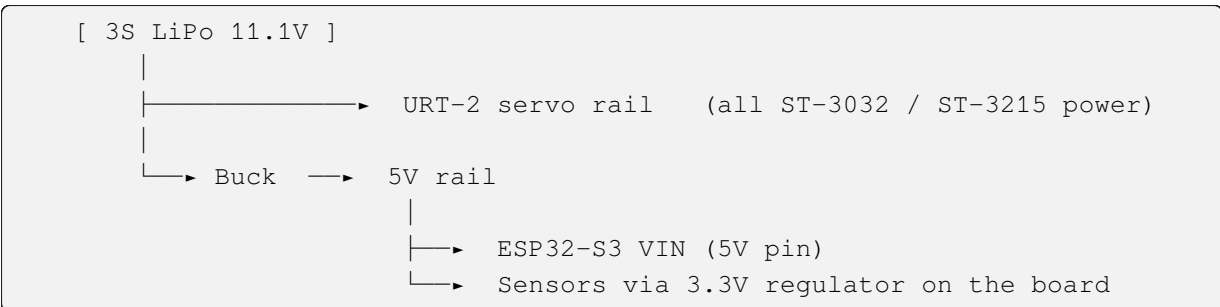
Part	Qty	≈ US\$ each	Notes
ESP32-S3 DevKitC-1 (N16R8)	1	13	Any ESP32-S3 board with ≥10 free GPIOs works; classic ESP32 also supported
Feetech STS3032 serial bus servo (12V)	2	25	Drive wheels, continuous-rotation wheel mode. 10 kg·cm, 148 RPM no-load — datasheet in <code>docs/datasheets/</code> . \$17.55 each from 100 up on Alibaba; boutique resellers ask \$40
Feetech URT-2 serial bus adapter	1	16	One half-duplex bus daisy-chains every servo (\$22 at Kuriosity)
3S LiPo battery (11.1V nominal) + balance charger	1	25–40	Feeds the servo rail directly. ST-3032 brown-out floor is ~9V, so a sagging 2S (7.4V) is not enough. <i>Typical price</i>
Buck converter (12V → 5V, ≥1A)	1	3	Powers the ESP32-S3 and sensors. <i>Typical price</i>
Pololu QTRX-HD-15A reflectance array	1	29	The line-follow bar (ten of its fifteen channels on GPIO 1–10, wiring below); \$29.10 at Pololu
TCS34725 break-out	2–4	3–8	Colour sensor array for zone detection. Adafruit 1334 is discontinued (was \$7.95); generic modules are a few dollars on AliExpress / Amazon
TCA9548A I2C multiplexer break-out	1	7	Required for more than one TCS34725 — its address is fixed at 0x29. Adafruit 2717 \$6.95; clones ≈\$1–2
ICM-45686 break-out	1	32	SPI IMU, read inside the 1 kHz control tick — the heading source for <code>use_gyro</code> . TDK's own EV_ICM-45686 board is \$32.45 at DigiKey; generic modules exist on Amazon
ST-3215 serial bus servo	0–4	—	Optional; good for arms / grippers. Same bus and protocol, but do not share a 6V rail setup — see servo notes
Jumper wires, M3 standoffs, chassis plate	—	—	Your robot, your build

Drive base, IMU, line bar, two colour sensors and the multiplexer come to **roughly US\$190** before battery, charger and chassis. Prices checked 2026-09-05 at single-unit quantity — [Alibaba \(STS3032\)](#), [Kuriosity \(URT-2\)](#), [Pololu 4415](#), [Adafruit 2717](#), [DigiKey EV_ICM-45686](#), [Amazon \(ESP32-S3-DevKitC-1 N16R8\)](#) — and they move; treat the column as a budget, not a quote. The same parts from Taobao / AliExpress typically land at half to two-thirds of these numbers with longer shipping.

A DC-gear-motor build (JGB37-520 / MG370 + H-bridge) is still fully supported — see [Alternative: DC gear motors with encoders](#) at the bottom.

2.2 Power budget

Two rails, one battery:



- Never power the servos from the ESP32's 5V pin or USB — a single ST-3032 stall pulls more than a dev board can source.
- The ST-3032's brown-out floor is ~9V. A 3S pack sags to ~9.9V near empty, which still clears it; a 2S pack does not.
- Tie all grounds together: battery, buck, URT-2, ESP32, sensor breakouts. This sounds obvious but it's the #1 reason new builds misbehave.

2.3 GPIO map (ESP32-S3)

The serial-bus build needs very few pins — that's most of its charm:

Function	GPIO(	Devices on this line
Analog sensors	1–10	The FULL ADC1 bank — the openbricks.drivers.qtr.QTRLineSensor window (wiring below). Buttons and UARTs deliberately live elsewhere so all ten stay analog-capable
I2C0 (SDA, SCL)	15, 16	TCA9548A mux (0x70) + colour sensors behind it, shared bus (an I2C BNO055 IMU at 0x28 fits here too)
UART1 (TX, RX)	14, 41	URT-2 serial bus — every ST-3032 / ST-3215 daisy-chained (RX was GPIO 6 until 1.71.0; it moved so the analog bank stays whole)
Program button	39	Start/stop, polled with an internal pull-up (was GPIO 4 until 1.71.0)
BLE-toggle button	38	Bluetooth on/off (was GPIO 5 until 1.66.3)
WS2812 data	21	Addressable RGB LED strip / ×8 stick DIN (openbricks.drivers.ws2812) — on boards that break out the header corner next to 5 V/GND, this is the free pin beside the reserved 19/20 USB pair
SPI (SCK, MOSI, MISO, CS)	12, 13, 11, 17	ICM-45686 IMU breakout (openbricks.drivers.icm45686.ICM45686) — 3V3 + GND + these four; INT and the other breakout pins stay unwired (the 1 kHz hard tick polls)

Pin gotchas on the ESP32-S3:

- **GPIO 22–25 don't exist** — the pin list is 0–21 then 26–48.
- **GPIO 26–32 (and 33–37 on octal-PSRAM modules) are flash/PSRAM.** Do not use them.

- **GPIO 19/20 are the native-USB D-/D+** and **0/3/45/46 are strapping pins**.
- I2C and UART route through the GPIO matrix, so none of these assignments are fixed — they're the convention the bundled examples use. On a **classic ESP32**, the usual equivalents are I2C on 21/22 and any two free pins for the UART.

The firmware enforces the hard cases at construction time: a driver asked to wire a nonexistent, flash, or USB pin — or a pin the runtime already owns, like the program button — raises `openbricks.pins.ReservedPinError` naming the pin, the role, and the reason, instead of failing somewhere far from the mistake. See `openbricks.pins`.

2.3.1 QTRLineSensor wiring (the standard line-follow window)

`openbricks.drivers.qtr.QTRLineSensor` bakes the rig geometry into the firmware — pins and element positions — so programs just construct it, pick an element, and follow:

```
from openbricks.drivers.qtr import QTRLineSensor
qtr = QTRLineSensor()                    # ten channels, GPIO 1..10
qtr.load_calibration("/qtr.cal")
r = qtr.read()
steer = KP * (50 - r[7].ambient())       # right edge under +16 mm
```

Two layouts, selected with `channels=`:

Ten channels (the default, `QTRLineSensor()`). Ten channels of a QTRX-HD-15A (4 mm pitch) in a skip pattern, left to right as mounted, onto GPIO 1..10 **in order**:

Element index	0	1	2	3	4	5	6	7	8	9
QTR channel	1	3	4	5	7	9	11	12	13	15
GPIO	1	2	3	4	5	6	7	8	9	10
x (mm)	-28	-20	-16	-12	-4	+4	+12	+16	+20	+28

That spans a 56 mm window at spacings 8/4/4/8/8/8/4/4/8 mm (the driver's `positions_mm` carries the true coordinates, so edge interpolation is exact across the unequal gaps). It uses the whole ADC1 bank.

Eight channels (`QTRLineSensor(channels=8)`). Every other channel of the same board (8 mm pitch) onto GPIO 1..8, the same 56 mm window on eight pins:

Element index	0	1	2	3	4	5	6	7
QTR channel	1	3	5	7	9	11	13	15
GPIO	1	2	3	4	5	6	7	8
x (mm)	-28	-20	-12	-4	+4	+12	+20	+28

The point of the eight-channel layout is what it leaves free: GPIO 9 and 10 are the last two ADC1 pins on the ESP32-S3, and they take a **second array** — no new class, a plain `openbricks.drivers.qtr.QTRArray` on those pins (or two `openbricks.drivers.qtr.QTRChannel` probes). `pitch_mm` is the distance between the two QTRX channels you chose, 4 mm per channel step:

Array	Class	GPIO	Notes
Front window	QTRLLineSensor(channels=8)	1–8	QTRX channels 1, 3, 5, 7, 9, 11, 13, 15
Second array	QTRArray(pins=(9, 10), pitch_mm=8.0)	9, 10	any two channels of a second QTRX board, e.g. 7 and 9 (8 mm apart); a marker array behind the axle, or a reversing follower

`channels` must be 8 or 10; anything else raises `ValueError` naming both. The class constants `PINS / POSITIONS_MM` (ten) and `PINS_8 / POSITIONS_MM_8` (eight) carry the layouts, and every array exposes `positions_mm`, its element x coordinates left to right, so a program can name an element by position: `r[qtr.positions_mm.index(12.0)]`.

Steering from elements. The driver reports what each element sees; the discipline is the program's. `ambient()` is 0 black .. 100 white after calibration. The canonical patterns:

- **Right edge** — pick the element the line's right edge should sit under and hold it at half grey: `steer = KP * (50 - r[i].ambient())`. That element reads darker as the robot drifts left (steer right, positive), whiter as it drifts right. On the ten-channel window `i = 7` (+16 mm); on the eight-channel window `i = 5` (+12 mm). Either leaves two mat-side elements beyond the edge (indices 8–9, or 6–7).
- **Left edge** — the mirror: `steer = KP * (r[i].ambient() - 50)` with `i = 2` on either layout (–16 mm on ten channels, –12 mm on eight), elements 0–1 on the mat beyond.
- **Centre** — `steer = KP_MM * r.position()`, the dark-weighted centroid in mm (positive = line right of centre), proportional across the whole window — what you want through sharp corners and after a branch. `position()` is `None` when no element is dark: the line has left the window, so the program keeps the side it was last seen on and steers hard that way.
- **Intersection** — `r.all_dark()`, the whole window on the line.
- **Branch marker** — the far-side elements (the ones beyond the edge you follow) going dark: `r[8].dark()` or `r[9].dark()` for the ten-channel right edge, `r[6].dark()` or `r[7].dark()` on eight.

An edge error is one element's reading, so it is proportional only within about a pitch of that element and rails at ± 50 beyond; the centroid is proportional across the span but blind to which edge is which. Examples: `qtr_line_follow_right.py / _left.py / _center.py`.

Either way the ~20 mm line sits inside the middle of the window with ≥ 2 channels of mat visible on the far side.

Calibration is per array. Each array is ratiometric to its own height and mat, so each one is swept over the line with `calibrate()` and keeps its own file — `"/qtr_front.cal"` and `"/qtr_rear.cal"`, say — loaded with `load_calibration()` at the top of every program. Two arrays with disjoint pins coexist; a pin claimed twice is refused at construction by the pin registry. The claims last for one program: the next run may wire those pins differently without a power cycle.

The ten-channel skip pattern is a palindrome, so if the board is mounted the other way round, only the channel labels swap — GPIO order and geometry stay identical. The same holds for the eight-channel layout.

One board-level note: **GPIO 3 (= element 2, the left-edge element on both layouts) is a strapping pin**, and on the ESP32-S3-COREBOARD V1.4 it optionally carries a 10 k Ω pull-up through the USB-JTAG 0

Ω link. Harmless (per-element calibration absorbs the bias), but if `qtr_calibrate.py` shows element [2] with a conspicuously narrower span than its neighbours, that link is populated — desoldering it is safe if you never use pin-JTAG.

The QTRX board's CTRL (emitter enable) can stay tied high; VCC to 3V3, GND to GND.

2.3.2 If the status LED never lights

The onboard WS2812 speaks a write-only protocol — the firmware can't tell a dead LED from a working one, so a blown pixel looks like software silently failing. Run `examples/led_probe.py` on the hub: it drives full white at both timings across every plausible RGB pin (48/38/47/21) plus a raw `machine.bitstream` write. If nothing lights through all of that, the problem is physical. The decisive check is a **power-off ohmmeter reading from the LED's data pin (or its GPIO) to GND**: a WS2812 whose DI input has failed as an internal short reads $\sim 0\ \Omega$ there and clamps the GPIO — the pin measures millivolts even when driven high, every write “succeeds” into the short, and the symptom history is “worked once, then went dark for good”. Lift the data link to free the pin. Also check the schematic for **population options in the LED circuit**. The ESP32-S3-COREBOARD V1.4 ([docs/datasheets/esp32s3_coreboard_v1.4_schematic.pdf](#)) routes GPIO 48 to the WS2812 through a $0\ \Omega$ RGB link and feeds the LED's VDD from the **5 V rail** through a second optional link — if either link is unpopulated (or the board is powered via 3.3 V only, leaving the 5 V rail dead), the LED never lights while every write succeeds. Bridge the link / feed 5 V, or wire any external WS2812 to 3.3V / GND / GPIO 48 — it becomes the status LED with no firmware change.

2.4 Sensor wiring (I2C)

The colour sensor is the TCS34725. Its I2C address is fixed at $0x29$ — there are no address-select pins — which decides how you wire it:

Sensors	Mode	Extra part
1	<i>Direct</i> — straight onto the ESP32's I2C pins	none
2 or more	<i>Via TCA9548A</i> — one sensor per mux channel	TCA9548A breakout

The IMU in the baseline build is the SPI-wired ICM-45686 (see the GPIO map above) and doesn't share this bus. If you use the I2C BNO055 instead, it sits at $0x28$ and connects straight to GPIO 15/16 in either mode — $0x28$ and $0x29$ coexist fine.

Both modes use the same driver call. `mux[n]` behaves like an I2C bus, so the only difference between the two is what you pass to `TCS34725(...)`.

2.4.1 Mode 1: direct to the ESP32

One colour sensor, no multiplexer:

TCS34725 pin	Connect to	Notes
VIN (or VCC)	3.3V	The breakout has an onboard regulator, but the board's 3.3V is cleanest
GND	GND	Common ground with everything else
SDA	GPIO 15	ESP32-S3 (classic ESP32: GPIO 21)
SCL	GPIO 16	ESP32-S3 (classic ESP32: GPIO 22)
LED / INT	see <i>below</i>	

```

from machine import I2C, Pin
from openbricks.drivers.tcs34725 import TCS34725

i2c = I2C(0, sda=Pin(15), scl=Pin(16), freq=400_000)  # ESP32-S3
sensor = TCS34725(i2c)

```

Check the wiring with `i2c.scan()` — it should list 41 (0x29). A complete program is `examples/read_color.py`.

If you want **exactly two** sensors without a mux, the ESP32's second hardware I2C controller also works: `I2C(1, sda=..., scl=...)` on any two free pins, one sensor per bus. Beyond two, use Mode 2.

2.4.2 Mode 2: via a TCA9548A multiplexer

Two or more colour sensors (a line-follower / zone-detection array). The TCA9548A sits at 0x70 and fans the bus out to eight isolated channels; each sensor lives on its own channel at its own 0x29.

Wire the mux to the ESP32-S3, then each sensor to a mux channel:

TCA9548A pin	Connect to	Notes
VIN	3.3V	
GND	GND	Common ground with everything else
SDA / SCL	GPIO 15 / 16	The main bus (classic ESP32: 21 / 22)
SD0/SC0 ... SD7/SC7	one TCS34725 each	Isolated channels
A0 / A1 / A2	leave open	Default address 0x70; only needed for a second mux

TCS34725 pin	Connect to	Notes
VIN (or VCC)	3.3V	
GND	GND	
SDA / SCL	mux channel SDn / SCn	Not the ESP32 pins
LED / INT	see below	

The Adafruit breakouts include ~10 kΩ SDA/SCL pull-ups, so for a handful of devices you don't need to add your own.

```

from machine import I2C, Pin
from openbricks.drivers.tca9548a import TCA9548A
from openbricks.drivers.tcs34725 import TCS34725

i2c = I2C(0, sda=Pin(15), scl=Pin(16), freq=400_000)  # ESP32-S3
mux = TCA9548A(i2c)                                  # 0x70 by default
sensors = [TCS34725(mux[ch]) for ch in range(3)]     # left, mid, right

```

Check the wiring with `i2c.scan()` — it should list 112 (0x70) for the mux; `mux[n].scan()` should list 41 (0x29) for each channel that has a sensor. For a complete program — a 2-sensor array that combines each sensor's `ambient()` and `rgb()` readings to name the colour under it (red / blue / green / yellow / white / black) — see `examples/color_array.py`.

2.4.3 TCS34725 LED pin

The colour sensor breakout has two extra pins beyond power and I2C:

- **LED** — drives the onboard white illumination LED. On Adafruit boards it defaults **on** (tied to VIN through the ADC-enable trace). To control it, wire it to a spare GPIO and drive high/low; to force it **off**, tie LED to GND. Leave it on for consistent colour readings — ambient light alone is unreliable across environments.
- **INT** — the interrupt output. The driver polls, so leave INT **unconnected**.

2.5 Serial bus servo notes (ST-3032 / ST-3215)

- **Every servo needs a unique bus ID.** Factory default is 1; re-ID one servo at a time with `examples/st3215_reid.py` (same protocol — it works for the ST-3032 too). The bundled drivebase examples assume `left = 1, right = 2`.
- **Speed limits.** Per the STS3032 datasheet (`docs/datasheets/`), no-load top speed at 12V is 148 RPM = 888 °/s; under the rated 3.3 kg·cm load it drops to roughly $\frac{2}{3}$ of that. The driver's default `max_dps=600` clamps requests at the loaded operating point — construct with an explicit `max_dps=900` to chase the no-load number.
- **Mixed fleets:** the ST-3215 tolerates lower voltages, but the ST-3032 browns out below ~9V. If you daisy-chain both on one URT-2, the rail must satisfy the strictest member: 12V.
- **Drivebase:** `ST3032Motor` drops straight into `DriveBase`:

```
from openbricks.drivers.st3032 import ST3032Motor
from openbricks.robotics import DriveBase

left  = ST3032Motor(servo_id=1, uart_id=1, tx=14, rx=41)
right = ST3032Motor(servo_id=2, uart_id=1, tx=14, rx=41, invert=True)
db = DriveBase(left, right, wheel_diameter_mm=65, axle_track_mm=120)
```

Bench-test a fresh build with `examples/st3032_drivebase_test.py`.

2.6 Calibrating the drivebase

`wheel_diameter_mm` and `axle_track_mm` are the two physical parameters that matter for straight-line distance and turn accuracy. Measure them with calipers or a ruler (wheel contact patch to wheel contact patch for axle track, not hub to hub). If `straight(1000)` undershoots, your wheel diameter value is too large; if `turn(360)` overshoots, your axle track is too small.

High-torque 12V servos also deliver the default launch profile much more stiffly than small DC motors — if the chassis pitches or lifts its rear when a move starts, soften the ramp:

```
db.settings(acceleration=180)    # wheel-deg/s2; default 1500
```

2.7 Alternative: DC gear motors with encoders

The original starter build — still fully supported. (Both this and the serial-servo build run in the MuJoCo simulator via the driver shim.)

Part	Qty	Notes
JGB37-520 DC motor with encoder (1:30 gearing)	2	Pick the 12V version; runs fine off 7.4V 2S LiPo
L298N dual H-bridge	1	Cheap and robust. TB6612FNG is a better choice if you can find it module
2S LiPo (7.4V) + buck converter ($\rightarrow 5V, \geq 2A$)	1 each	Don't power the ESP32 from the L298N's onboard 78M05 regulator — it's good for ~ 300 mA and browns out the moment the motors draw current

Wiring topology:



GPIO map (see `examples/esp32_drivebase.py` for the ESP32-S3 pin assignments; the classic-ESP32 equivalents are in git history):

Function	ESP32-S3 GPIO(s)	Devices on this line
Left motor dir	1, 2	L298N / TB6612 IN1, IN2
Left motor PWM	18	L298N / TB6612 ENA
Left encoder A, B	7, 8	JGB37-520 encoder channels
Right motor dir	9, 10	L298N / TB6612 IN3, IN4
Right motor PWM	40	L298N / TB6612 ENB
Right encoder A, B	42, 47	JGB37-520 encoder channels

This build and the QTR line-sensor bar are mutually exclusive. A DC drivebase needs ten GPIOs and the S3 does not have ten free ones outside the ADC1 bank once the other conventions are honoured, so the motor direction and left-encoder lines take six of the bank's pins (GPIO 1, 2, 7, 8, 9, 10) that the `openbricks.drivers.qtr.QTRLineSensor` window otherwise owns. Everything else follows the reference GPIO map unchanged: 11/12/13/17 stay free for the SPI IMU, 15/16 for I2C sensors, 14/41 for a serial-servo arm, and 39/38 for the program / BLE buttons (see `openbricks.hub.ESP32S3DevkitHub` — the launcher polls GPIO 39 as an input, so a motor driver toggling it would read as button presses and stop your program).

2.7.1 Calibrating encoder counts

The default in `jgb37_520.py` is `counts_per_output_rev=1320`, which is $11 \text{ CPR} \times 30:1 \times 4$ (quadrature edges). If you have a different gearbox variant, recompute:

```
counts_per_output_rev = encoder_CPR * gear_ratio * 4
```

Or measure empirically: rotate the output shaft by hand exactly one full turn and read `motor.angle()`. Whatever it reports is what `counts_per_output_rev` should be, scaled so that one turn = 360°.

COMMAND-LINE TOOL

`pipx install openbricks` installs one console script, `openbricks`, which mirrors the `pybricksdev` workflow: flash firmware over USB, then run / upload / stop programs and pull logs over BLE. With the `[sim]` extra installed, `openbricks sim ...` forwards to the *MuJoCo-backed simulator*.

A typical session:

```
$ openbricks flash --name RobotA      # port, chip and newest firmware
↳ auto-detected
$ openbricks list                     # hubs in BLE range
$ openbricks run -n RobotA main.py    # push + stream output
$ openbricks upload -n RobotA main.py # stage; start it with the hub button
$ openbricks stop -n RobotA           # Ctrl-C a running program
$ openbricks log -n RobotA            # dump the most recent run log
$ openbricks docs hardware            # open this manual offline in your
↳ browser
```

3.1 Firmware versions and provenance

`openbricks flash` first reports the firmware already on the chip — version plus an `(official)` / `(customized)` suffix — before it looks up the newest release. Flashing the **same version** again, or an **older** one, asks for confirmation first; pass `--yes` to skip the prompt in scripts. A `--firmware` image's version is read from its file name, so a name without one — the `firmware.bin` a local build produces — cannot be compared with the running version: that flash asks for confirmation too (and a non-interactive flash of it needs `--yes`).

The default output is step-level (probe, download, erase, write, hub name, marker, reboot); pass `--verbose` / `-v` to also echo every underlying `mpremote` / `esptool` command line and the firmware cache paths — useful when reporting a flash problem.

Every firmware image published by CI is signed (Ed25519), and the CLI ships the matching public key. An image whose `.bin.sig` verifies is labeled `(official)`; anything else — a self-built image, a missing or wrong signature — is `(customized)`. Customized firmware flashes normally: the suffix is provenance, not a gate. After each flash the verdict is stored on the hub, which is how the next `openbricks flash` labels the current firmware.

The suffix follows the version everywhere it reaches you: the `firmware 1.79.0 (official)` banner at the top of every `openbricks run`, the `started: header` line in every run log (`openbricks log`), and the flash preflight above. On the hub, `openbricks.firmware_label()` returns the same string.

3.2 Programs are compiled on the host

Since 1.92.0, `openbricks run` and `openbricks upload` cross-compile your script with `mpy-cross` before connecting, and stage compiled bytecode instead of source (like Pybricks). Three things get better:

- **syntax errors surface in milliseconds**, on your terminal, naming your file and line and quoting the offending source line — no BLE scan, no connect, no upload round-trip;
- **programs start faster**: the hub loads bytecode directly and skips its on-device parse/compile step;
- **tracebacks name your real file and line** (File "square.py", line 12) instead of File "<string>".

No flags, nothing to configure. Firmware older than 1.92.0 can't run compiled programs, so the CLI probes the hub's version in-session and sends plain source instead — announced on `stderr`, never silently. `upload --path` (custom boot flows) always stages your file verbatim, uncompiled, at the path you give.

`run` is an **upload-then-run** (since 2.7.0, deliberately different from Pybricks): it stages your script at the button's `/program.mpy` before executing it, so even a run that fails midway leaves the program on the hub — press the start button to rerun it, no BLE round trip needed. The flip side: running a calibration or a one-shot diagnostic replaces the button's program too, so re-upload your mission after such tools (`upload` alone stages without running).

`flash --with-qtr-init` additionally stores a starter QTR line-sensor calibration at `/qtr.cal` (recorded on the reference bench, default pins 1-10), so the line-follow examples work on a fresh hub out of the box. Heights, mats and lighting differ — run `examples/qtr_calibrate.py` once for a calibration measured on your own rig.

3.3 Fast uploads

An upload is one BLE session and, since 4.10.0, one round trip of work on the hub. The CLI connects (discovering only the hub's UART service), interrupts whatever the hub runs and enters the raw REPL in a single write, and pastes **one** program that prints the firmware version, writes the file, syncs the clock, prints the size confirmation and drops straight back into the button's idle loop; the host reads the confirmation and the idle banner off the stream and hangs up. Earlier versions spent four such execs — a version probe, the file, a confirmation, the idle-loop restart — each with its own handshake, plus a fixed settle wait, on every upload.

The CLI remembers each hub's firmware version (in `~/.cache/openbricks/hubs.json`, or `$OPENBRICKS_CACHE_DIR`), so a hub it has met before on firmware 1.92.0 or newer is not probed again. Both directions of a re-flash are caught. A hub remembered on older firmware is probed again rather than trusted, so one upgraded since gets compiled code at once. The staged program still prints its version first and refuses to write compiled code on firmware that cannot run it, so a hub re-flashed to something older is caught in-session: the CLI says so and stages source instead.

On the hub side, firmware 4.10.0 asks for a 512-byte BLE packet size (the stack's default of 256 capped every write at 253 bytes) and advertises every 40 ms instead of 100 ms, so discovery and the connection start sooner.

Every upload ends with a line saying where its time went:

```
staged in 1.42 s (scan 0.31, connect 0.58, subscribe 0.05, raw repl 0.09,
→paste 0.21, confirm 0.12, close 0.06)
```

`scan` and `connect` are the operating system's Bluetooth stack (scanning for the advertisement, then connecting and discovering the service); the rest is the hub. `--debug` adds the packet-level trace.

3.4 One upload at a time

`openbricks run` and `openbricks upload` push a program through the same raw-paste channel, and the operating system shares a single BLE link between every process that connects to the same hub (macOS, Linux and Windows all multiplex). Two transfers started from two terminals used to interleave their bytes on the hub's REPL — a corrupted program, or both terminals reading each other's output — and the hub, which sees one connection, could not tell them apart. Since 3.10.0 the host refuses at once, before any scan:

```
$ openbricks run -n RobotA main.py
error: an upload is ongoing (another openbricks run/upload is transferring
↳to 'RobotA'; wait for it to finish)
```

The guard is a per-hub OS file lock held for the transfer: `upload` holds it until its confirmation returns, `run` releases it the moment the program is staged and started, so a second `run` while the first is only streaming output supersedes it the way a button press would. A CLI that crashes or is killed mid-transfer leaves nothing behind — the kernel releases the lock with the process. Other hubs are unaffected, and so is `openbricks stop`.

The lock is per machine, not per account, because the BLE link is: its file (`openbricks-upload-NAME.lock` in the temp directory, the shared `/tmp` on Linux) is created writable by every account, and a read-only one left by another account still locks. A lock file that cannot be opened at all is an error naming its path — remove that file and retry.

3.5 Reference

The reference below is generated from the CLI's own argument parser, so it always matches the installed version.

Host-side CLI for flashing and running code on openbricks hubs, plus a MuJoCo-backed simulator (`openbricks sim ...`).

```
usage: openbricks [-h] [--version] COMMAND ...
```

3.5.1 Positional Arguments

COMMAND Possible choices: flash, run, upload, stop, list, log, servo-id, paste-probe, docs, doc, bricks, sim

3.5.2 Named Arguments

--version Print the openbricks package version and exit.

3.5.3 Sub-commands

flash

Flash a firmware image onto a hub (via `esptool`) and write the hub's BLE advertising name into NVS (via `mppremote`). `-name` is mandatory so every hub gets a unique identifier — two hubs with the same name can't be individually addressed over BLE.

```
openbricks flash [-h] --name NAME [--port PORT] [--firmware FIRMWARE]
                  [--chip CHIP] [--baud BAUD] [--with-qtr-init] [--skip-
↪erase]
                  [--yes] [--verbose]
```

Named Arguments

--name	Hub identifier for BLE (required, <=20 chars recommended).
--port	Serial port (/dev/ttyUSB0, /dev/cu.usbserial-XXXX, COM5 ...). Omit to auto-detect — works when exactly ONE ESP device is connected (Espressif native USB or a CP210x/CH340/FTDI bridge).
--firmware	Path to firmware.bin produced by scripts/build_firmware.sh or downloaded from the Releases page. Omit to download the newest release automatically for the detected chip (cached under ~/.cache/openbricks/firmware).
--chip	esptool -chip value (esp32, esp32s3, auto). Default: auto. Default: 'auto'
--baud	esptool flash baud rate. Default: 460800. Default: '460800'
--with-qtr-init	After flashing, store a starter QTR line-sensor calibration at /qtr.cal (recorded on the reference bench, default pins 1-10) so line-follow examples work out of the box. Re-run examples/qtr_calibrate.py for a calibration measured on your own mat and lighting. Default: False
--skip-erase	Skip erase_flash (faster dev loop; leaves stale NVS keys). Default: False
--yes	Skip the confirmation prompt when the target firmware is the same version as (or older than) the current one, or when a -firmware file name carries no version to compare (e.g. a local build's firmware.bin). Default: False
--verbose, -v	Echo every subprocess command line (mpremote/esptool) and cache paths. Default output is step-level only. Default: False

run

Connect to the named hub over BLE, push SCRIPT to its REPL (via paste mode), and stream stdout/stderr back to this terminal until the script finishes. Ctrl-C interrupts the remote program. While another `openbricks run / upload` from this machine is still transferring to the same hub, this one refuses at once with `error: an upload is ongoing`.

```
openbricks run [-h] -n NAME [-c CODE] [--scan-timeout SCAN_TIMEOUT] [--
↳debug]
    [SCRIPT]
```

Positional Arguments

SCRIPT Path to the local Python script to run on the hub. Mutually exclusive with `-c`.

Named Arguments

-n, --name Hub name baked in at flash time (`openbricks flash --name`).

-c, --code Inline Python code to run on the hub (analogous to `python -c CODE`). Useful for quick diagnostics — e.g. `openbricks run -n ls -c 'import openbricks; print(openbricks.__version__)'`. Mutually exclusive with the **SCRIPT** positional.

--scan-timeout How long to scan for the named hub before giving up. Default: 5.0 s.
Default: 5.0

--debug Print every BLE notify packet (timestamp + hex + ascii) to stderr as it arrives. Use to diagnose ‘timed out reading from hub’ errors — tells you whether the hub is sending anything at all.
Default: False

upload

Upload **SCRIPT** to the hub’s filesystem (default path `/program.py`). The uploaded code does NOT run automatically — the hub’s frozen `main.py` watches the hub button and `exec`’s the staged script on each short press. Second short-press stops a running program. (Pybricks calls this same operation `download` from the hub’s perspective; we name by direction-of-data-travel — bytes flow *up* to the hub.) While another `openbricks run /upload` from this machine is still transferring to the same hub, this one refuses at once with error: an upload is ongoing.

```
openbricks upload [-h] -n NAME [--path PATH] [--scan-timeout SCAN_TIMEOUT]
    [--debug]
    SCRIPT
```

Positional Arguments

SCRIPT Path to the local Python script to stage.

Named Arguments

-n, --name Hub name baked in at flash time.

--path Destination path on the hub’s filesystem; the file is staged VERBATIM there (no compilation) for custom boot flows. Default: compile with `mpy-cross` and stage `/program.mpy` (which the frozen launcher runs; older firmware gets the source at `/program.py`).

--scan-timeout	BLE scan timeout. Default: 5.0 s. Default: 5.0
--debug	Print every BLE notify packet and the connect timing breakdown to stderr (the per-phase line is printed regardless). Default: False

stop

Connect to the named hub over BLE and send a single Ctrl-C, which MicroPython surfaces as Keyboard-Interrupt. Use when a long-running `openbricks run` has already ended and you just want the hub to idle again.

```
openbricks stop [-h] -n NAME [--scan-timeout SCAN_TIMEOUT]
```

Named Arguments

-n, --name	Hub name.
--scan-timeout	BLE scan timeout. Default: 5.0 s. Default: 5.0

list

Run a BLE scan and print every device found, sorted by RSSI (strongest first). Unnamed devices are shown with a placeholder so you can still spot a hub whose name wasn't flashed.

```
openbricks list [-h] [--timeout TIMEOUT] [--all]
```

Named Arguments

--timeout	Scan duration in seconds. Default: 5.0. Default: 5.0
--all	Show every BLE device, not just those with names. Useful when debugging a hub that came up without a flashed name. Default: False

log

Every program executed via the launcher (button press OR `openbricks run`) gets its stdout / stderr tee'd to a flash file under `/openbricks_logs/`. Ten rotating slots are kept. With no flags this prints the most recent run; `--list` shows the index; `--run N` selects a specific slot. Useful for post-mortem on an untethered run where no live console was attached.

```
openbricks log [-h] -n NAME [--list] [--run RUN] [--scan-timeout_↵
↵SCAN_TIMEOUT]
```

Named Arguments

-n, --name	Hub name baked in at flash time.
--list	List the available run indices + their on-flash size, instead of dumping a run's contents. Default: <code>False</code>
--run	Specific run index to dump. Defaults to the most recent.
--scan-timeout	BLE scan timeout. Default: 5.0 s. Default: <code>5.0</code>

servo-id

Scans the bus (IDs 0..253), rewrites the servo's EEPROM ID register, and verifies the result. With several servos attached, `-old-id` is required so the tool never guesses which one to re-ID. Two transports: by default (or with `-p`) it talks directly to a USB half-duplex adapter's serial port (e.g. the URT-2 board), so a fresh servo can be given its ID before it meets the hub; with `-n NAME` it runs the scan/re-ID through the hub over BLE, with the servo still wired to the robot.

```
openbricks servo-id [-h] [-p PORT] [-n NAME] [--tx TX] [--rx RX]
                    [--scan-timeout SCAN_TIMEOUT] [--scan] [--old-id OLD_ID]
                    [--baudrate BAUDRATE] [--timeout TIMEOUT]
                    [new_id]
```

Positional Arguments

new_id	Bus ID to assign (0..253). Omit with <code>-scan</code> .
---------------	---

Named Arguments

-p, --port	Adapter path only: serial port of the USB adapter, e.g. <code>/dev/cu.usbmodem123</code> . Omitted (and no <code>-n</code>): auto-detected when exactly one USB serial device is connected.
-n, --name	Hub name: run the scan/re-ID THROUGH THE HUB over BLE instead of a USB adapter — the servo stays wired to the robot. Mutually exclusive with <code>-p</code> .
--tx	Hub path only: servo-bus TX pin (default 14). Default: <code>14</code>
--rx	Hub path only: servo-bus RX pin (default 41). Default: <code>41</code>
--scan-timeout	Hub path only: BLE scan timeout. Default: 5.0 s. Default: <code>5.0</code>
--scan	Just list the IDs that answer on the bus; change nothing. Default: <code>False</code>

--old-id	Current ID of the servo to re-ID. Required when more than one servo is attached; otherwise auto-detected.
--baudrate	Adapter path only (refused with -n; the hub's bus runs at 1000000): bus baudrate. Default: 1000000 (Feetech factory). Default: 1000000
--timeout	Adapter path only (refused with -n): per-ping serial read timeout in seconds. Default: 0.02 (a full 254-ID scan takes ~5 s). Default: 0.02

paste-probe

Pastes padded no-op programs of increasing size through the real raw-paste path and reports the largest that completes, plus how each failure presents (truncated vs hung). Use before changing the firmware's `MICROPY_REPL_STDIN_BUFFER_MAX`: two windows may be in flight at once, so that setting is only safe at or below the measured limit.

```
openbricks paste-probe [-h] -n NAME [--scan-timeout SCAN_TIMEOUT] [--max↵
↵MAX]
                        [--timeout TIMEOUT]
```

Named Arguments

-n, --name	Hub BLE name.
--scan-timeout	BLE scan timeout in seconds. Default: 5. Default: 5.0
--max	Largest size to try, bytes. Default: 8192. Default: 8192
--timeout	Per-size timeout in seconds. Default: 15. Default: 15.0

docs (doc)

Opens the full manual in your browser — the same Sphinx build as docs.openbricks.dev, API reference included, bundled and served from disk so no internet is needed. Pass a topic to jump straight to that page.

```
openbricks docs [-h] [topic]
```

Positional Arguments

topic	Page to open (e.g. install, hardware, robotics). Guides and API pages both work. Omit for the index.
--------------	--

bricks

The Assembly Workbench (`openbricks sim`) ships a curated set of LEGO Technic parts with exact LDraw geometry. `bricks fetch` downloads the whole LDraw library into a cache; `bricks convert` turns any part numbers into a bundle file for `openbricks sim workbench --bricks`.

```
openbricks bricks [-h] ACTION ...
```

Positional Arguments

ACTION Possible choices: fetch, convert

Sub-commands

fetch

Fetch parts by number from `ldraw.org` into the sim's library, or (with no numbers) the whole LDraw parts library (145 MB).

```
openbricks bricks fetch [-h] [--dest DIR] [--force] [--no-colors] [NUMBER ..  
↪.]
```

Positional Arguments

NUMBER LDraw part numbers (the LEGO design id, e.g. 2458): each is fetched with the files it needs and kept under the sim's data directory. None: the whole library.

Named Arguments

--dest The LDraw cache: where the whole library is unpacked, and where parts fetched by number keep their files. Default: `$OPENBRICKS_LDRAW_DIR`, else `~/.cache/openbricks/ldraw`.

--force Download again: the whole library even if it is already there; a part's files and Rebrickable's colour tables even if the cache has them.

Default: `False`

--no-colors Fetch parts without the colours Rebrickable lists for them.

Default: `False`

convert

Convert LDraw part numbers into a brick bundle.

```
openbricks bricks convert [-h] [--out FILE] [--weights FILE] [--ldraw DIR]  
NUMBER [NUMBER ...]
```

Positional Arguments

NUMBER LDraw part numbers (the LEGO design id, e.g. 32270).

Named Arguments

--out Write the bundle JSON here (default: stdout).

--weights JSON of {number: {"g": grams}} to record as the parts' masses; unknown parts get a volume estimate.

--ldraw The unpacked LDraw library (default: the cache `bricks fetch` fills).

sim

Forwards all remaining arguments to the MuJoCo-backed simulator's CLI. Use `openbricks sim --help` to see the sim's own subcommand list.

```
openbricks sim
```

`openbricks sim ...` forwards to the simulator's own parser; bare `openbricks sim` launches the sim, the native desktop application:

MuJoCo-backed simulator for openbricks firmware.

```
usage: openbricks sim [-h] [--version] COMMAND ...
```

3.5.4 Positional Arguments

COMMAND Possible choices: app, workbench, preview, run

3.5.5 Named Arguments

--version Print the openbricks package version and exit.

3.5.6 Sub-commands

app

Launches the sim: the Assembly Workbench as a native program with LEGO Technic bricks in exact LDraw geometry, your own STL parts, components, and the robot as the top component. The first run downloads the signed build for this platform into the cache (`~/.cache/openbricks/sim`); `OPENBRICKS_SIM_BIN` points at a build of your own.

```
openbricks sim app [-h] [--bricks FILE] [--bin PATH] [--no-download] [file]
```

Positional Arguments

file A robot.assembly.json to open.

Named Arguments

--bricks	An extra brick bundle from <code>openbricks bricks convert</code> to add to the library (repeatable). Default: []
--bin	Run this sim binary instead of the cached release build.
--no-download	Never download: fail if the build is not cached. Default: <code>False</code>

workbench

Serves the Assembly Workbench on localhost and opens it: LEGO Technic bricks with exact LDraw geometry, STL imports, components, and the robot as the top component with mass, centre of mass and inertia computed at every level. Pass a `robot.assembly.json` to open it.

```
openbricks sim workbench [-h] [--bricks FILE] [--port PORT] [--no-browser]
                        [file]
```

Positional Arguments

file	A <code>robot.assembly.json</code> to open (otherwise the browser's last draft, kept while it fits the browser's storage, else the example).
-------------	--

Named Arguments

--bricks	An extra brick bundle from <code>openbricks bricks convert</code> to add to the library (repeatable). Default: []
--port	Port to serve on. Default: a free one. Default: 0
--no-browser	Print the URL instead of opening a browser. Default: <code>False</code>

preview

Loads the named world (alias or path), splices the default `openbricks-sim` chassis in, and hands off to `mujoco.viewer` for interactive inspection. Pass `--headless` to step N seconds of physics without opening a window (useful for CI smoke tests).

```
openbricks sim preview [-h] [--world WORLD] [--chassis FILE] [--x X] [--y Y]
                      [--yaw YAW] [--headless] [--duration DURATION]
                      [--seed SEED]
```

Named Arguments

--world	World alias or path. Aliases: empty, wro-2026-elementary, wro-2026-junior, wro-2026-senior. Default: empty. Default: 'empty'
--chassis	JSON file of ChassisSpec fields describing YOUR robot: wheel_radius / axle_length (m), body size, line_sensor_x, color_sensor_x/y (sensor placement, m), pos_x / pos_y / yaw_deg (spawn pose). Fields not given keep the default chassis values.
--x	Chassis spawn x (m). Default 0, or the chassis file's pos_x.
--y	Chassis spawn y (m). Default 0, or the chassis file's pos_y.
--yaw	Chassis spawn heading (deg, counter-clockwise from +X). Default 0, or the chassis file's yaw_deg.
--headless	Skip the viewer; step --duration seconds and exit. Default: False
--duration	Headless step duration in seconds. Default: 2.0. Default: 2.0
--seed	Randomization seed for per-round prop permutation. Set to a different integer each round to mimic the WRO judge's pre-round randomization; same seed produces same layout. Has no effect on worlds without a randomization spec.

run

Loads the named world + the default chassis, constructs a `SimRobot` over them, and execs the script with `robot` (plus `drivebase`, `left`, `right` aliases) in its globals. The script drives the sim by calling `robot.run_for` / `run_until` between actions.

```
openbricks sim run [-h] [--world WORLD] [--chassis FILE] [--x X] [--y Y]
                  [--yaw YAW] [--viewer] [--no-shim] [--seed SEED]
                  [--trace FILE] [--max-sim-s MAX_SIM_S]
                  script
```

Positional Arguments

script	Path to the Python script to execute.
---------------	---------------------------------------

Named Arguments

--world	World alias or path (same set as preview). Default: 'empty'
--chassis	JSON file of ChassisSpec fields describing YOUR robot: wheel_radius / axle_length (m), body size, line_sensor_x, color_sensor_x/y (sensor placement, m), pos_x / pos_y / yaw_deg (spawn pose). Fields not given keep the default chassis values.

--x	Chassis spawn x (m). Default 0, or the chassis file's <code>pos_x</code> .
--y	Chassis spawn y (m). Default 0, or the chassis file's <code>pos_y</code> .
--yaw	Chassis spawn heading (deg, counter-clockwise from +X). Default 0, or the chassis file's <code>yaw_deg</code> .
--viewer	Drop into the MuJoCo viewer after the script returns so you can orbit the final scene. Default: <code>False</code>
--no-shim	Skip installing the driver shim. The default behaviour <code>install``s shims for ``machine + openbricks._native</code> and patches <code>time.sleep_ms</code> to advance the sim — disable when your script uses the <code>openbricks_sim</code> API directly. Default: <code>False</code>
--seed	Randomization seed (same semantics as <code>preview --seed</code>).
--trace	Write the chassis pose (<code>t_ms</code> , <code>x_mm</code> , <code>y_mm</code> , <code>yaw_deg</code>) every 50 ms of sim time to this CSV — the run's path, for plotting or asserting on afterwards.
--max-sim-s	Stop the script once this much SIM time has elapsed (exit status 3). A mission loop that never sees its stop condition otherwise runs forever.

SIMULATOR

The `[sim]` extra ships a MuJoCo-backed physics simulator, so you can develop robot programs without a hub on the desk:

```
$ pipx install 'openbricks[sim]'
$ openbricks sim run examples/full_robot.py --viewer
```

The sim runs the **same script you'd push to the hub** — a driver shim maps the `openbricks` API onto simulated motors and sensors, so `ST3032Motor`, `DriveBase`, color sensors, and distance sensors behave like their hardware counterparts.

4.1 Commands

```
$ openbricks sim [robot.assembly.json] [--bricks more.json] [--bin PATH] [--no-download]
```

Launches *the sim*, the native desktop application: the Assembly Workbench with LEGO Technic bricks in exact geometry, your own STL parts, and the robot as the top component. The first run downloads the signed build for your platform (about 15 MB) from the release that matches the installed version into `~/ .cache/openbricks/sim`; it is checked against the same project key that signs firmware images before it runs. `OPENBRICKS_SIM_BIN` points at a build of your own (cargo build `--release` in `tools/sim`). `openbricks sim app` is the explicit form. A file named on the command line opens, or stops the launch: one that cannot be read, is not JSON or is not an assembly is refused with the reason, as is a `--bricks` bundle that will not load — nothing opens on the example, or on a draft of some other session, in its place. (The binary's own `--help` prints its usage and exits 0.)

```
$ openbricks sim workbench [robot.assembly.json] [--bricks more.json] [--port N] [--no-browser]
```

The same workbench as a page in your browser, for a machine without the native build.

```
$ openbricks sim preview [--world WORLD] [--x X] [--y Y] [--headless] [--duration S] [--seed N]
```

Loads the named world (an alias or a path to a map's `map.json`), splices in the default chassis, and opens the MuJoCo viewer so you can inspect the scene. `--headless` steps the physics for `--duration` seconds without opening a window — useful as a smoke test.

```
$ openbricks sim run SCRIPT [--world WORLD] [--chassis FILE] [--x X] [--y Y] [--yaw DEG] [--viewer] [--no-shim] [--seed N]
```

Loads the world plus the chassis and executes `SCRIPT` against the simulated robot. `--viewer` opens the interactive MuJoCo window; without it the sim runs headless (CI-friendly). `--seed` makes randomized worlds reproducible.

Run `openbricks sim --help` for the full, always-current option list.

4.2 The sim

`openbricks sim` opens a window with two tabs. **Workbench** is the editor described in the next section: the library on the left (your components, the LEGO Technic set, other bricks, each with a rendered thumbnail), the 3D view in the middle, the contents of the component you are editing and the inspector on the right. Drag in the view to orbit, right-drag to pan, scroll or pinch to zoom toward the point under the pointer (the brick there, or empty space at the view's depth, stays under it), `F` to fit; drag a brick to move it on the ground plane in 4 mm steps — half a stud, since a part an odd number of studs wide has its centre on a stud and an even one between two; the toolbar's **snap** offers 8, 4, 1 mm or off (shift lifts it). The selection carries handles: three arrows (**Move**, `W`) to slide it along one world axis in grid steps, or three rings (**Rotate**, `E`) to turn it about one axis in 15° steps — hold shift for free movement or rotation; several selected items move and turn together about the first one's origin; **Select all**, beside the contents list's heading, selects everything the component holds (locked items too, which stay put). `R` turns the selection 90°, and the inspector's **Turn 90°**, **Pitch 90°** and **Roll 90°** turn it a quarter about the up (z), y and x axes — the world's axes, as the rings are, and the selection as one piece about the first item's origin, one item or several. With nothing selected inside a component, `R` or **Turn the whole component** turns every part of it (locked ones too: they keep their place among the rest) in place — about the middle of what it fills, and back down so its lowest point is where it was — and every use of the component turns with it; the robot itself is not turned so, since its frame is the way it drives. `S` snaps the selection — one item, or several as one — into the nearest hole, arrows nudge, `Delete` removes, `⌘Z` undoes. `⌘C` copies the selection and `⌘V` pastes it: back into the same component two modules over, into another component as it was, or into another window (the clipboard carries every part and component definition it needs). `⌘L` locks the selection so nothing moves, turns or removes it until `⌘⇧L` unlocks it; locked bricks draw faded and show  in the contents list. **Import STL...** in the library brings in a part from a mesh file: choose the file's units, where its origin goes (as in the file, the bounding-box centre or the bottom centre), and a weighed mass or a density; the part gets its exact volume, centre of mass and inertia from the closed mesh (an open mesh gets a box's inertia and needs a mass), plus any 4.8 mm pin holes found on it, and lands in the library and the view. A component can start from nothing: type a name beside the library's **Components** and press **New** (or Enter), and an empty component of that name joins the library and opens for building; add bricks to it, then **+** **add** it to the robot or any other component. Double-click a component to edit its definition in place; every use follows. Open and save `robot.assembly.json` from the toolbar. **Import...** brings a saved build's components and bricks into this library, to add from: what is here already, the same, is left, and a clash comes in under a new id; what the file has wrong (a brick whose number this library does not carry) is listed as it is on Open, and a build with a component inside itself is refused. A component's **save** in the library (or **Save as build...** on its page) writes it as a build of its own — a file whose robot is that component, with the bricks and components it needs — to open, import or put on a map elsewhere. Unsaved work is kept: a couple of seconds after a change settles (a drag, a run of nudges or of typing in the route's **Definitions** box, an undo, once it ends — never part way through), the build and the route are written as drafts under the data directory (`~/.local/share/openbricks/drafts`, or `$OPENBRICKS_DATA_DIR`), and the next `openbricks sim` started without a file brings them back — the status line says so, and **Save** writes the build to the file it came from. Saving drops the draft. A draft whose note (the file it belongs to, when it was kept) cannot be read comes back all the same, and the status line says so, naming the file, rather than guess. A route (a draft's, or one opened) planned on a map this machine has not is shown on the first shipped map, the status line naming both; the route keeps its own map's name until you pick a map. (The map editor's props are saved by **Save map** and are not drafted.)

Simulate runs your program on a map with the chassis you assembled. Pick the map (a shipped world, or one you saved from the Map tab) and it appears in the view at once, as does the chassis (a `robot.assembly.json`, or the build open in the Workbench tab once it is saved) when you choose it; pick the program (`main.py`), then Run. The view is a plan: the whole map seen straight from above, north up, with no perspective, fitted edge to edge and never panned or zoomed — whatever changed the map while you were on another tab (the Workbench keeps its own 3D camera). Pause, Resume and Stop do what they say; the speed slider runs the physics slower or faster than wall time; the program's prints and errors appear in the log panel below the view, and while the map is not there yet the view says what the run server is doing. The loaded chassis stands on the map at true scale: drag it to put it where a run should start — a 70 % transparent copy follows the pointer with its axle centre under it, and letting go places the robot there (shift turns it) — or type the pose in the **Route** panel. That panel plans a route as actions placed on the map. Click a tool — **→ Straight**, **⌒ Curve**, **⤿ Turn**, **■ Stop** or **f Custom** — then click the map: a straight takes its start and its end (the start snaps to where the previous action ends, or to the chassis, so paths chain; an end clicked within 7.5° of the way the robot arrives lands exactly on that heading, so chained straights run on with no turn between them — shift keeps any angle); a curve takes its start and its end — it enters the way the robot arrives at its start and is the one arc from there through its end, one `curve` command, so the heading it ends facing follows from the geometry (an end dead ahead is a straight run; one straight behind, or within 2° of it — where the one arc would loop nearly the whole circle on a radius of metres — is not drawn, and a click or a handle drag there is refused: “no single arc reaches a point straight behind the start, or within 2° of it: click elsewhere”); as the first action of a route, a curve also takes a point to face at its start between the two, the way the robot starts; a turn takes where it turns and a point to face; a stop or a custom call takes one point. Every click leaves a marker; after the first click the line (or the arc, or the turn's arrow) follows the pointer with its length, radius and angle, or heading written beside it, and once placed each action keeps that label next to its path, and an arrowhead at the end of every path shows the way the robot faces there (a turn's arrow shows its own). Paths are drawn three pixels wide at any zoom, over everything on the map (markers sit between the map and the paths), in the kind's colour — blue straights, green curves, orange turns, stops and calls — or a colour of your own, picked in the popup or the panel (“default” goes back to the kind's). A popup then asks for the parameters: the speed (the drive base's default, shown in mm/s for your wheels), **continuous** for a move that flows into the next one without slowing (`then=Stop.NONE`) or otherwise the end state (coast, brake, hold) — a curve asks for nothing more: the popup shows the headings it enters and ends with and the arc's radius and angle, all set by the clicks — a turn's heading and rate, a stop's wait, a custom action's call (the picker lists what the **Definitions** box defines, such as `def line_follow(): ...`) and whether it moves the robot, in which case one more click says where it ends. Where an action does not start where the previous one ends, a dashed line shows the drive the program inserts to get there; the first action placed on an empty route starts it, so the chassis goes to where that action begins, facing its way (drag the chassis afterwards for another start; shift-drag turns it, and a first curve follows). Every action is an object on the map: click its path to select it and edit its parameters in the panel, drag its handles (a straight's ends; a curve's start, which carries its end along, and its end, which bends the arc; a turn's heading arrow) or the path itself to move it, `⌘C` / `⌘V` to copy and paste it (the copy lands a little to the side), `⌘L` / `⌘⇧L` to lock and unlock it (a locked action shows  and cannot be moved, edited or deleted), Delete to remove it, `⌘Z` to undo, Esc to cancel a placement. The numbers on the map are the order the program runs them in; rows drag by their `≡` grip (or `↑` `↓`) to reorder. **Markers** are named points you add to a map — a corner of the mat, a mission object, a line junction: click **● Marker**, click the map, name it in the popup. They show as flags with their names, drag to move, rename or remove in the panel, and route clicks snap to them, so a path can start or end exactly on a marker. They are kept with the map on your machine (under `~/local/share/openbricks/markers/`, or `$OPENBRICKS_DATA_DIR`), not in the route file, and come back whenever that map loads.

Routes save and load as `*.route.json` (the map, the start pose, the actions and the definitions). Older route formats are converted on load; a move that has no place in the current one is refused with its action number rather than placed as a different move: a backward straight or a backward curve (a negative mm

or radius) from an `openbricks-route/2` file, a full-circle curve from one (it ends where it starts), an `openbricks-route/3` curve without a numeric `radius_mm`, and a curve of any format whose end lies exactly straight behind its start. The 2° band is for clicks and drags only: a curve in a file that loops 358° loads and runs as written. ► **Run route** writes the route as a hub-style program (`ST3032Motor` wheels and a `DriveBase` sized from the chassis; edit the motor lines for other wiring) and runs it from the route's start — the chassis is put there first, so a second run repeats the route — and “show the program” prints it. Under the hood the sim starts the MuJoCo runtime as a child process — the same runtime, driver shim and C cores `openbricks sim run` uses — and draws the run from the poses it streams, with every brick of the chassis in its exact geometry and the map's mesh props (the WRO senior mosaic frame) as MuJoCo has them. The chassis is built from the assembly: the wheel, caster and sensor roles place the physics skeleton, the brick-by-brick mass properties become the body's inertia, and each brick rides along as a visual geom.

4.3 The map editor

Map edits the map itself in a 3D view of its own — drag to orbit, shift-drag (or right-drag) to pan, scroll or pinch to zoom toward the point of the map under the pointer, **Fit** or `F` to frame the map, `Iso / Top / Side / Front` as on the Workbench. Once a map is framed the view is yours: moving, adding, removing or sticking a prop rebuilds the map on the run server, but the camera stays where you put it, and so it does when you save the map under a name of your own; only loading another map, or **Fit**, frames it again. The toolbar on this tab names the map shown — the assembly's `Open / Save / Save as...` buttons and its component path belong to the Workbench tab. The props on the map are the LEGO-built objects a mission puts on the mat (each an LDraw model among the map's `props`) and whatever you add: drag a prop to move it (its outline lights under the pointer, the selected one carries its name), shift-drag to turn it by hand, `R` or **Turn 90°** for a quarter turn, or set the **heading** field outright (degrees counter-clockwise from the map's x axis). **Pitch 90°** and **Roll 90°** tip it a quarter about the map's y and x axes, and the **pitch** and **roll** fields set those angles outright (the Workbench's order: roll about x, then pitch about y, then yaw about z). Every such turn is in place: about the middle of the prop, and up or down so its lowest point stays where it was — a prop standing on the mat stays standing on it, however it is tipped — and it is refused, as a drag is, when turned it would overlap another prop or the robot. The map keeps the angles as the prop's `yaw`, `pitch` and `roll`. Click one, or its row in the panel, to select it; `⌘D` or **Duplicate** puts another like it a little to the side, `Del` or **Remove** takes it away, **Add...** lists the kinds of prop the map has and puts one of that kind at the map's origin.

Adding what you built. The panel lists the components of a build from the Workbench — the build open there, or any saved build: **Open a build...** takes an `.assembly.json`, and the builds opened are listed under the Workbench's; click one to list its components. The whole build comes first, then each component that holds bricks, each with a **+** button that puts it on the map (the whole build is named after its file). Or add one brick from the library (search it by number or name, then **+ to map**). Either lands at the origin, standing on the map — its lowest brick on the floor, whatever the build's own origin is (a brick's is its top face) — as a prop of its own whose `file` is an `openbricks-assembly/1` document, kept under the data directory until the map is saved — drawn with the exact bricks and colliding as their boxes with their catalogue masses. Nothing sits under the map: a prop a map file places so low that a brick of it would be below the floor is lifted onto the floor when the map loads.

No two props overlap. A prop dragged or turned into another prop, or into the robot, is not taken there: it stays where it was last clear and the panel says what it would overlap (its exact bricks against theirs, as on the Workbench; faces that only touch are fine, so props stack and stand side by side). A prop added onto another — a copy lands a little to the side of its original, a build or brick at the origin — is moved along the map's x axis, a module at a time, until it is clear, and says whom it was placed beside.

Free or stuck. A prop is free by default: it has a free joint, so the physics settles it and the robot can push it. Tick **stuck to the map** on the selected prop to weld it there — no joint, nothing but the editor moves it;

a stuck prop wears a pin through its centre, and its row says so. Every move, add, remove, stick and unstick is sent to the run server, which moves the live body at once (a chassis place, which resets the physics, keeps the prop where it was put) and rewrites the prop in the map it holds, so the physics, the picture and the map agree; adding, removing, sticking and unsticking rebuild the world with the chassis where it stands. Nothing moves while a program runs.

Save as a new map writes the map as it stands — every prop where it is, the ones added included, with the map's artwork and the props' models (documents added since the load copied into the map's `props/`) — to `worlds/<name>/` under the data directory (`$OPENBRICKS_DATA_DIR`, else `$XDG_DATA_HOME/openbricks`, else `~/.local/share/openbricks`, the same place the markers live). The run server lists your maps beside the shipped ones (marked “yours”), the tab switches to the new map, and its markers come along; saving again under the same name replaces it, and a shipped map's name is refused so it is never shadowed. Routes remember the map they were planned on by that name. On a map of your own the panel also offers **Save changes**, which writes the map shown (the one the view and the props list are of — never a map picked whose load was refused) in place under its own name — its artwork and models stay, only `map.json` is rewritten. A shipped map has no such button: its layout is kept as shipped, and your changes go into a map of your own.

Share a map. Export map... writes the map as it stands — every prop where it is, the ones added included — to one JSON file you name, with everything it needs inside: the mat's artwork and meshes (as base64), the props' LDraw models (as text) and builds (as JSON). **Import map...** takes such a file and makes it a map of your own under its name (the next free `-2`, `-3`... when that name is a shipped map's or taken), listed with the rest and shown at once, the status line saying so (“imported as ...”, as a save says “saved as ...”); like saving and exporting, importing waits for a running program to be stopped. A file that is not a whole export — a bare `map.json`, or one lacking a file its map names — is refused by name, and nothing is left behind; an export the sim refuses (a model gone since it was added) leaves the file already at that path as it was. Components travel the same way on the Workbench: a component's **save** writes it as a build of its own, and **Import...** brings a build's components in.

Maps are JSON. A map is a folder: `map.json` (format `openbricks-map/1`) and the files it names. `map.json` holds the physics settings, the textures, materials and meshes, the lights, the fixed geoms (the mat, the walls), the props and the cameras, each with MuJoCo's own attribute names and units — metres, vectors as arrays — and any of them may carry a note; a prop is `{"name", "ldr", "pos", "mass", "color"}` for an LDraw model or `{"name", "file", "pos"}` for a build, with `yaw`, `pitch`, `roll` and `fixed` when it has them. MuJoCo itself reads no JSON (its models are XML or its own binary), so the sim makes the model from `map.json` in memory as it loads the map; no XML is ever written. A map of your own saved before 4.32.0 (a `world.xml`) becomes `map.json` the first time the sim lists your maps; the old file is left where it was. One that cannot be converted (its `world.xml` truncated, or not a map's MJCF) is listed all the same — its row in the map picker cannot be chosen, and shows the file and the reason when hovered — never hides the maps beside it, and is refused by name, with the reason, should a route or a draft name it.

4.4 The Assembly Workbench

The sim and `openbricks sim workbench` (the same editor as a page in the browser) read and write one file. The editor is a 3D view of the robot as a tree of components:

- **Bricks** are recorded once, with their geometry, mass and provenance (`measured`, `datasheet`, `vendor` or `placeholder`). The library that ships in the wheel holds a curated set of popular LEGO Technic parts converted from the [LDraw parts library](#) (CC BY 2.0 / 4.0): beams in every common length, bent and L beams, frames, Technic bricks and plates, pins, axles, bushes, connectors, gears, a few rims and tyres, and fairing panels, with BrickLink catalogue weights where known — and every part of the two WRO sets, 45811 (the Brick Set: the mission bricks, tiles, hoses and balls) and

45819 (the Expansion Set). Type a set number, or “WRO”, into the library’s search to see a set’s bricks, each with how many the set holds. A name matches however its spaces run and whatever its case: “plate 4 x 8” finds LDraw’s “Plate 4 x 8”. Type a part number the library lacks — the LEGO design id printed on the part, 2458 — and the library says so and offers **Fetch 2458 from LDraw**: the part’s own file and the few subparts and primitives it references come from ldraw.org (a few dozen kilobytes, not the whole 145 MB library; the first fetch also brings Rebrickable’s colour tables, 1.1 MB, once), are converted like the shipped parts, get the colours Rebrickable lists for the part (or a note that it lists none), and the part joins the library at once and for every later launch, kept under `~/ .local/share/openbricks/bricks` (or `$OPENBRICKS_DATA_DIR`; on Windows `~\is %USERPROFILE%`). One fetch runs at a time, and **Cancel** stops it. A build that uses a fetched part carries its record along — mesh, mass properties and connectors, the stud sockets derived on arrival — so it opens, and stacks, on a machine whose library never fetched it. Fetching needs the sim to have been started by `openbricks sim` (it runs the package’s Python); `openbricks bricks fetch 2458 3005` does the same from a terminal, and a fetched part is in the MuJoCo runtime’s library too, so a fetched wheel drives. Servos, boards and wheels are recorded as boxes, cylinders and spheres (a cylinder runs along *x*, *y* or *z* — *z* when the file leaves *axis* out; any other value is listed with the file’s errors, naming the brick, and the MuJoCo runtime refuses the file the same way), and any part you have as a mesh comes in through **Import a part from an STL file** (binary or ASCII; mm, cm, inch or m; a weighed mass or a density such as PLA 1.24 g/cm³). A mesh that reaches beyond 327.67 mm from its origin — more than the library’s 0.01 mm steps hold — is packed at 0.1 mm (ten times coarser for every ten it is over), the record carries the step and the import summary says so; choose the bounding-box-centre origin to bring a part exported at its assembly position back to the fine step. A corner of the mesh that is not a number is refused.

- **Components** are lists of bricks and other components, each placed by a position and a roll / pitch / yaw. Drag bricks from the library into the view, move and rotate them with the gizmos, select what you built and *Group* it — or, in the sim’s library, name a component and press *New* to start it empty and build it from bricks. Either way the new component joins the library and can be dropped anywhere, as many times as you like. Double-click an instance to edit its definition in place; every use follows. Every brick is placed in a LEGO colour: the library lists, for each part, the colours it really comes in (from Rebrickable’s tables), a row’s colour combo picks the one the next **+** places it in, and a placed brick’s page (or a selection’s, for the colours they all share) changes it. A part in a given colour is its own LEGO *element* with the number set inventories print, and the page names it; typing an element number into the library’s search finds the part and picks that colour. “—” leaves a brick in its category’s colour. The file records the colour as its LDraw id (`color` on the instance), and the Map and Simulate tabs draw builds in their colours too.
- **Connections.** Pins, axles and studs are real features of the LDraw parts, and 4.8 mm bores are recognised as pin holes on every mesh, imported STL files included; a pin LDraw draws from plain cylinders (the WRO set’s “Type 2” pins) is recognised on its mesh the same way, and so is an axle hole drawn from plain walls (the 24-tooth gear 3648’s); the rounded ends of an axle hole’s four arms are not taken for a pin hole. Let go of a part near a hole and it snaps: the pin axis aligns to the hole, a pin half centres in its module, an axle keeps its position along the hole. Bricks and plates stack: under every stud a part has a socket, and a tile, which has none, has one under every place of its footprint a stud fits, so a plate or a tile let go over a brick lands on its stud grid, squared up, with every stud it covers mated — the turn and shift that mate the most features win, which also seats a two-pin connector in both holes. Dragging a part near a hole or a grid pulls it on (it lets go again as the pointer moves away); a stud also seats in a Technic hole or, for the half-stud offset, in a tube — sitting at the hole’s mouth, as a stud does. Several selected parts — an assembled hinge, a stack — are pulled and seated as one, keeping their own fit: their features together against everything else, shifted alike. The magnet never turns a part: a heading is yours to set (**R**, the rotate handles), and a part let go askew of a hole stays askew. Of the seats within reach it takes the one the most features

agree on; one that would land the part on a neighbour is passed over for the next as good (a pin let go nearer a taken hole than a free one lands in the free one), and when none as good fits, the drop is refused. A part let go lands with its height on the stud module (1.6 mm: a plate is two, a beam five, a brick six) and its plan position on the 4 mm half-stud grid, and the magnet takes it from there. The inspector lists what each part is mated to. So that a stack stays legible, the sim draws every brick with its edges — the seams between stacked bricks, the rims of studs and holes — in a darker shade of its own colour (lighter on a black brick); **edges** in the toolbar turns them off.

- **No two bricks overlap.** A brick let go, nudged, turned or given a pose where its material would pass through a neighbour's is put back where it was, and a red banner across the top of the view names both (the status line too), a brick inside a component by its path (`aa/lego_2`), until the next change is kept or the status line says something else — while it is dragged it flushes red wherever it would overlap, and Escape abandons the drag. Touching is not overlapping, and at a joint — a feature seated in the other part's (a pin all the way in its hole, a plate down on its studs, an axle anywhere along a hole it runs through) — the two may cross a little — within a millimetre of the seated feature, as a brick's tube pushed onto a pin presses on the pin's collar: that is the press fit ABS allows, a feature pushed in firmly, and it is not counted. A brick on a pin turns about it at any angle: the rotate rings step 15° (45° is three steps) and the inspector's **roll**, **pitch** and **yaw** take any angle. Turn it about the pin's own axis — the ring of the pin's colour (red x, green y, blue z): a pin standing up turns the brick in **yaw**, a level pin along y in **pitch**; about any other axis the brick would twist off the pin, and the banner says so. Only there, though: the rest of the two parts' surfaces are checked as ever, so a plate let go turned on one stud, its walls through the others, is refused. A new brick lands beside what is already at the origin; a duplicate or a paste goes as far along -y as it needs to be clear. A file that already holds overlaps opens and says how many, and stays editable: an old overlap may be moved out of, not further in.
- **Mass properties** are never typed in above the brick level. Volume, centre of mass and the inertia tensor of every LDraw and STL part come from its closed mesh, so a recorded weight becomes a full inertia tensor; components and the robot roll their children up with the parallel-axis theorem. Weight divided by exact volume is shown as a density on every part, which catches a wrong weight or a wrong part at a glance (ABS is about 1.05 g/cm^3).
- **Roles** name the parts the simulator binds: the two drive wheels, the caster, the reflectance arrays, the colour sensor, the range sensor and the IMU. From them the page derives the flat `ChassisSpec` fields (what the simulator receives) with the axle midpoint as the origin, so a build can be run today with `openbricks sim run --chassis`.

The file the editor reads and writes, `robot.assembly.json`, stores recorded facts only: bricks, poses, roles, spawn pose. Everything computed is recomputed on load. Open one with `openbricks sim robot.assembly.json` (or `openbricks sim workbench robot.assembly.json` in the browser, which also keeps your last draft between visits while it fits the browser's storage, and says so on the page when it does not; a file it is given that does not validate is not opened, and the page says why rather than open the draft instead).

4.5 The brick library

```
$ openbricks bricks fetch NUMBER [NUMBER ...] [--dest DIR] [--force] [--no-colors]
$ openbricks bricks fetch [--dest DIR] [--force]
$ openbricks bricks convert NUMBER [NUMBER ...] [--out FILE] [--weights FILE] [--ldraw DIR]
$ openbricks sim workbench --bricks FILE
```

The wheel ships the curated Technic set and every part of the WRO sets 45811 and 45819 (the library says which sets hold a brick, and how many). Any other part is a number away: `bricks fetch 2458` gets the part's files from `ldraw.org` one by one (the same `~/ .cache/openbricks/ldraw` layout the whole library unpacks into, so a full library is used as it is and a sparse one grows), converts it, adds the colours Rebrickable lists for it, and keeps it under the sim's data directory, where `openbricks sim` finds it on every launch — what the sim's own library does from its search box. A number `ldraw.org` has not, official or on the tracker, is said so; a file the part needs that `ldraw.org` has not is an error naming the part and the file, never a part with a hole in it; every pause `ldraw.org` asks for (sixty requests a minute; a part of two hundred files takes a few) is waited out, a connection that stalls for thirty seconds is an error, and a page served in place of a file is refused, as is a file name that is not one of the library's (a . . ., an absolute path) before anything is asked for. A part too long for 0.01 mm steps (a 47-stud hose) is packed at 0.1 mm, and the fetch says so. A number the library ships is refused, and so is an inventory number it ships under LDraw's (41250, the WRO inventory's 52 mm ball, is the library's 22119) — the shipped record, weighed and in its sets, wins over a fetched copy, and one left under the data directory by an earlier fetch is ignored and noted at launch, by both numbers, by the sim and the runtime alike. A `bricks` directory that is there but cannot be listed (a plain file in its place, a permission lost) is noted at the sim's launch too, naming it and the reason, rather than taken for nothing fetched. `--force` fetches a part's files (its own and its subparts, not the primitives) and Rebrickable's tables again; `--dest` is the LDraw cache for both forms. The whole LDraw library (every LEGO part ever catalogued, 145 MB to download, about 600 MB unpacked) is `bricks fetch` alone, and `bricks convert` turns any part numbers — the LEGO design ids printed on the parts, 3648 for the 24-tooth gear — from it into a bundle file that `openbricks sim --bricks` (and `openbricks sim workbench --bricks`) adds to the library. Converted parts without a weight carry a volume estimate at 1.05 g/cm³ and are flagged until you weigh them; pass `--weights` with a JSON of `{"3648": {"g": 1.62}}` to record real ones.

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4.6 Describing your robot

The default chassis is a 60 mm-wheel, 150 mm-axle box with every down-facing sensor 60 mm ahead of the axle. A real robot differs, and those differences decide whether a mission script's numbers work: `--chassis FILE` loads a JSON object of `ChassisSpec` fields (metres, kilograms, degrees) that describe the robot the script was written for. Fields not given keep the defaults.

```
{
  "wheel_radius": 0.0432,    "axle_length": 0.135,
  "body_length": 0.16,      "body_width": 0.12,
  "line_sensor_x": 0.06,
  "line_sensor_2_x": -0.03,  "line_sensor_2_y": 0.0,
  "color_sensor_x": 0.06,    "color_sensor_y": 0.184,
  "pos_x": -0.547,          "pos_y": -0.15,          "yaw_deg": 90
}
```

- `wheel_radius` / `axle_length` size the chassis at load time. The `DriveBase(wheel_diameter_mm=..., axle_track_mm=...)` in the script resizes it again at adoption, so the script's geometry always wins — set them here so a preview shows the same robot.
- `line_sensor_x` places the first reflectance-array site (`chassis_line`) ahead of the axle; `line_sensor_2_x` / `_y` place the second (`chassis_line2`, default 30 mm behind the axle on the centre line, the same height). Reflectance arrays bind these sites in **construction order** within

one run: the first `QTRArray` / `QTRLineSensor` / `QTRChannel` the script constructs reads `chassis_line`, the second reads `chassis_line2`, and a third raises `RuntimeError` (two sites is the chassis's limit). The counter resets when the shim is installed for a run, so every `sim run` starts with both sites free. `color_sensor_x` / `_y` / `_z` place the centre colour camera (`chassis_cam_down`, the no-mux TCS34725) in the chassis frame (the floor is at $-(\text{wheel_radius} + 0.005)$); `color_sensor_yaw` / `_pitch` aim it (default straight down; a sensor on the robot's left flank reading bricks beside the line is yaw 90, pitch 0 at brick height); `color_sensor_fov` is the cone it integrates (degrees, 0 = one ray) and `color_sensor_range` how far it sees. The left/right down pair rides 18 mm either side of (`color_sensor_x`, `color_sensor_y`).

- `pos_x` / `pos_y` / `yaw_deg` are the spawn pose; `--x` / `--y` / `--yaw` on the command line override them one at a time. `yaw_deg` is counter-clockwise from +X seen from above (0 = facing +X).

4.7 What the shim simulates

Firmwa class	Sim binding
ST3032 / ST3215	The first two servo ids become the chassis wheels, the third and fourth kinematic task shafts (a gripper motor that turns but pushes nothing). A <code>DriveBase</code> always gets the physical wheels for the pair it adopts, whatever order the script constructed its motors in, and re-constructing a motor for a servo id yields the same motor — both firmware rules.
Drive- Base	The firmware engine over an emulated <code>st_bus</code> ; <code>use_gyro(True)</code> reads the chassis's true yaw.
ICM456 / BNO055	Ground-truth chassis heading; the ICM's bias estimator reports calibrated at once.
TCS347	The firmware driver class over a synthesised raw read: the centre camera (no mux) or the left/right pair (mux channels 1 / 0) casts along its own axis — optionally a cone, with a range — and the first geom hit (a mat texel; a LEGO brick's colour: an LDraw prop's material, or a Workbench build brick's own colour from the palette, as the view draws it) gives the reflectance; <code>rgb()</code> / <code>ambient()</code> are the driver's channel-over-clear arithmetic, so white reads about (85, 85, 85) and a blue brick has the largest b, as on the robot.
QTR- Li- ne- Sen- sor / QTRAr- ray / QTR- Chan- nel	The firmware driver over a reflectance model: one element per array position (<code>QTRLineSensor(channels=8)</code> gives the eight-channel front layout, exactly as on the hub), spread left-to-right from the site the array bound at construction — the first array a run constructs reads <code>chassis_line</code> , the second <code>chassis_line2</code> , a third raises <code>RuntimeError</code> — each element averaging the floor over a 3 mm spot so an edge reads as a gradient, which is what makes <code>50 - reading[i].ambient()</code> proportional. <code>load_calibration("/ray / qtr_front.cal")</code> and <code>calibrate()</code> need no file — the sim's reflectance is born normalised.
Dis- tance sen- sors	A forward ray from the <code>chassis_dist</code> site.

Nothing above has a load: task motors don't grip, and a prop is only pushed when the chassis body drives into it.

4.8 Notes

- The sim needs the `[sim] extra (mujoco, numpy)`. Without it, `openbricks sim ...` prints an install hint instead of crashing.
- The wheel carries the firmware package (`openbricks.drivers.*`, `openbricks.parameters`, ...) since 3.6.0, so a plain `pipx install 'openbricks[sim]'` runs hub-style scripts; earlier releases needed a repo checkout for that.
- Firmware-only users never need the simulator — it's strictly host-side tooling.

EXAMPLES

The repo ships runnable example programs in `examples/`. Push any of them to a hub with `openbricks run -n <name> <script>` or try them in the simulator with `openbricks sim run <script>`.

Two representative ones are reproduced below.

5.1 Drive a square (ST-3032 drivebase)

```
# SPDX-License-Identifier: MIT
"""
Drive a square with the ST-3032 drivebase.

Four sides of ``SIDE_MM`` straight + ``+90°`` turn each. Demonstrates
``DriveBase.straight`` / ``turn`` composing into a closed loop – if
the chassis geometry (``WHEEL_DIAMETER_MM`` / ``AXLE_TRACK_MM``) is
calibrated correctly, the robot returns to within a few cm of its
starting pose after one lap.

Uses the new ``then=Stop.COAST`` default end-state (1.6.7), so the
wheels free-wheel briefly between segments. If your bench has
significant momentum carryover and you'd rather pin the wheels
between sides, pass ``then=Stop.BRAKE`` to the ``straight`` / ``turn``
calls (or ``then=Stop.HOLD`` for active position lock – ST-3032
supports it).

Hardware: identical to ``examples/st3032_drivebase_test.py``.

Run with:
    openbricks run -n ls examples/st3032_drivebase_square.py
"""

from openbricks.drivers.st3032 import ST3032Motor
from openbricks.robotics import DriveBase
from openbricks.parameters import Stop

LEFT_ID, RIGHT_ID = 2, 1
UART_ID, TX, RX   = 1, 14, 41
```

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```

WHEEL_DIAMETER_MM = 88
AXLE_TRACK_MM     = 136

SIDE_MM    = 200
NUM_LAPS   = 1

STRAIGHT_SPEED_MM = 150
TURN_RATE_DPS     = 200

def main():
    print("--- ST-3032 drivebase square (%d mm sides × %d laps) ---" %
          (SIDE_MM, NUM_LAPS))

    left  = ST3032Motor(servo_id=LEFT_ID,  uart_id=UART_ID, tx=TX, rx=RX,
→invert=True)
    right = ST3032Motor(servo_id=RIGHT_ID, uart_id=UART_ID, tx=TX, rx=RX)

    db = DriveBase(left, right,
                    wheel_diameter_mm=WHEEL_DIAMETER_MM,
                    axle_track_mm=AXLE_TRACK_MM)
    db.settings(straight_speed=STRAIGHT_SPEED_MM, turn_rate=TURN_RATE_DPS)

    for lap in range(NUM_LAPS):
        for side in range(4):
            print(" lap %d side %d: straight(%d) → turn(+90)" %
                  (lap + 1, side + 1, SIDE_MM))
            db.straight(SIDE_MM)
            db.turn(90)

        print("--- done ---")

main()

```

5.2 Gyro-corrected square (ICM-45686)

With an ICM-45686 attached, `use_gyro(True)` moves heading control off the encoders and onto measured body rotation — corrected every millisecond inside the firmware's 1 kHz control tick, with no Python in the loop. This script drives the same square twice, encoders-only and gyro-corrected, and prints each pass's heading drift so the difference is a number, not an impression. On the reference bench the gyro pass returns within $\sim 0.6^\circ$ over all four turns; wheel slip that would bend the encoder pass simply gets steered back out.

```

# SPDX-License-Identifier: MIT
"""
Gyro-corrected square — ICM-45686 on the hard tick.

The ICM-45686 is read INSIDE the 1 kHz control tick over SPI, so

```

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with ``use\_gyro(True)`` the DriveBase corrects heading every millisecond in C – no Python in the loop. This script drives a square twice, encoder-only and then gyro-corrected, and prints how far each pass drifts from its starting heading. Bench 2026-08-10: the gyro pass returned within +0.6 degrees over all four turns.

The learned gyro bias is persisted to NVS after the stillness lock, so later boots start corrected without the ~0.5 s wait.

Wiring: breakout on 3V3/GND plus the four SPI pins below; INT stays unwired (the tick polls). Needs ~0.5 m of clear floor.

Run:

```
openbricks run -n <hub> examples/icm45686_square.py
"""
```

```
import time
```

```
from openbricks.drivers.icm45686 import ICM45686
from openbricks.drivers.st3032 import ST3032Motor
from openbricks.robotics import DriveBase
from openbricks.parameters import Stop
```

```
LEFT_ID, RIGHT_ID = 2, 1
UART_ID, TX, RX = 1, 14, 41
SCK, MOSI, MISO, CS = 12, 13, 11, 17
```

```
WHEEL_DIAMETER_MM = 88
AXLE_TRACK_MM = 136
```

```
STRAIGHT_SPEED = 300
TURN_RATE = 200
SIDE_MM = 300
```

```
def square_drift(db, imu):
    start = imu.heading()
    for side in range(4):
        db.straight(SIDE_MM)
        db.turn(90)
        print("corner %d: heading %.1f" % (side + 1, imu.heading() - start))
    return imu.heading() - start - 360
```

```
print("hold still for gyro bias lock ...")
imu = ICM45686(sck=SCK, mosi=MOSI, miso=MISO, cs=CS)
while not imu.calibrated():
    time.sleep_ms(100)
imu.save_calibration()
```

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```

left = ST3032Motor(servo_id=LEFT_ID, uart_id=UART_ID, tx=TX, rx=RX,
                   invert=True)
right = ST3032Motor(servo_id=RIGHT_ID, uart_id=UART_ID, tx=TX, rx=RX)
db = DriveBase(left, right, wheel_diameter_mm=WHEEL_DIAMETER_MM,
               axle_track_mm=AXLE_TRACK_MM, imu=imu)
db.settings(straight_speed=STRAIGHT_SPEED, turn_rate=TURN_RATE)

print("pass 1: encoders only")
print("drift: %+.1f deg" % square_drift(db, imu))

db.use_gyro(True)
print("pass 2: gyro-corrected at 1 kHz")
print("drift: %+.1f deg" % square_drift(db, imu))
db.stop(then=Stop.BRAKE)

```

Wiring for the IMU is four SPI pins plus power — see [Hardware guide](#). The first still half-second learns the gyro bias; `save_calibration()` persists it so later boots skip the wait.

5.3 Rounded square (DriveBase.curve)

`curve(radius, angle)` follows the Pybricks contract — positional order, parameter names, and sign semantics: positive angle arcs right (clockwise), a negative radius drives the arc backward, and `curve(0, angle)` degrades to a turn in place. The forward and turn profiles run with proportionally scaled speed *and* acceleration, so the path is a true circle even through the ramps, and the outer wheel is automatically capped at the `straight_speed` setting. (One deviation: then defaults to `Stop.COAST` like every openbricks move; pass `then=Stop.HOLD` for the Pybricks end state.)

```

# SPDX-License-Identifier: MIT
"""Drive a rounded square with DriveBase.curve().

Four straights joined by four quarter-circle arcs – the robot never
stops to pivot, so the lap is faster and smoother than the
straight+turn square. curve(radius, angle) follows Pybricks: positive
angle arcs right, the radius sign picks forward/backward, and the
outer wheel is automatically capped at the straight_speed setting.

Run with:
    openbricks run -n ls examples/st3032_drivebase_curve.py
"""

from openbricks.drivers.st3032 import ST3032Motor
from openbricks.robotics import DriveBase

LEFT_ID, RIGHT_ID = 2, 1
UART_ID, TX, RX = 1, 14, 41

WHEEL_DIAMETER_MM = 88
AXLE_TRACK_MM = 136

```

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```

SIDE_MM = 150
RADIUS_MM = 60
NUM_LAPS = 1

left = ST3032Motor(servo_id=LEFT_ID, uart_id=UART_ID, tx=TX, rx=RX,
                   invert=True)
right = ST3032Motor(servo_id=RIGHT_ID, uart_id=UART_ID, tx=TX, rx=RX)
db = DriveBase(left, right,
               wheel_diameter_mm=WHEEL_DIAMETER_MM,
               axle_track_mm=AXLE_TRACK_MM)
db.settings(straight_speed=150, turn_rate=200)

print("rounded square: %d mm sides, %d mm corner radius" %
      (SIDE_MM, RADIUS_MM))
for lap in range(NUM_LAPS):
    for side in range(4):
        db.straight(SIDE_MM)
        db.curve(radius=RADIUS_MM, angle=90)
print("done")

```

5.4 Colour sensor, direct to the ESP32 (one TCS34725)

Mode 1 from the *hardware guide*: a single TCS34725 on GPIO 15/16, no multiplexer. The driver is handed the bus itself.

```

# SPDX-License-Identifier: MIT
"""
Example: read RGB from a TCS34725 over I2C and classify the color.

Hardware:
    * ESP32-S3 (or classic ESP32)
    * TCS34725 breakout on I2C bus 0 (3.3V, GND)
      SDA=15, SCL=16 on ESP32-S3; SDA=21, SCL=22 on classic ESP32
"""

from machine import I2C, Pin

from openbricks.drivers.tcs34725 import TCS34725
from openbricks.tools import wait

i2c = I2C(0, sda=Pin(15), scl=Pin(16), freq=400_000)
color = TCS34725(i2c, integration_ms=50, gain=4)

def classify(r, g, b):
    """Very rough color classifier. Good enough to tell red from green."""
    if r > g and r > b:

```

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```

        return "red"
    if g > r and g > b:
        return "green"
    if b > r and b > g:
        return "blue"
    if r > 200 and g > 200 and b > 200:
        return "white"
    if r < 40 and g < 40 and b < 40:
        return "black"
    return "unknown"

while True:
    r, g, b = color.rgb()
    print("rgb=({:3d},{:3d},{:3d})  ambient={:3d}  -> {}".format(
        r, g, b, color.ambient(), classify(r, g, b)
    ))
    wait(200)

```

5.5 Colour sensor array via a TCA9548A (two TCS34725s)

Mode 2: the TCS34725's address is fixed at 0x29, so two or more go through a TCA9548A multiplexer, one per channel. `mux[n]` behaves like an I2C bus, so the driver call is the same as above — only the bus argument changes. Each loop combines `ambient()` and `rgb()` to name the colour under every sensor.

```

# SPDX-License-Identifier: MIT
"""
Demo: name the colour under each of 2 TCS34725s through a TCA9548A mux.

The TCS34725's I2C address is fixed at 0x29, so two of them can't
share a bare bus — this is exactly the job the TCA9548A multiplexer
exists for. Each sensor hangs off its own mux channel, and ``mux[n]``
behaves like a normal I2C bus, so the driver is constructed unchanged.

Each loop reads both signals from every sensor and combines them:

* ``ambient()`` (0..100) — brightness. Separates black (too dark to
  trust hue at all) and, together with the channel spread, white.
* ``rgb()`` (0..255 each, clear-normalised) — hue. Splits the
  chromatic colours: red, yellow, green, blue.

``classify()`` reduces one sensor's (ambient, rgb) pair to one of
``red / blue / green / yellow / white / black``.

Hardware:
* ESP32-S3 (or classic ESP32)
* TCA9548A breakout on I2C bus 0 (3.3V, GND)
  SDA=15, SCL=16 on ESP32-S3; SDA=21, SCL=22 on classic ESP32
* 2x TCS34725, one per mux channel 0..1, facing the surface

```

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Calibration matters: TCS34725 readings shift with illumination LED, distance, and surface gloss. The thresholds below are starting points – print ``sensor.ambient()`` and ``sensor.rgb()`` over each of your own surfaces and adjust.

```
"""
```

```
BLACK_AMBIENT = 12
```

```
NEUTRAL_SPREAD = 45
```

```
WHITE_AMBIENT = 35
```

```
YELLOW_G_OVER_R = 0.6
```

```
def classify(ambient, rgb):
```

```
    """Reduce one sensor's (ambient 0..100, (r, g, b) 0..255) reading
    to ``"red" / "blue" / "green" / "yellow" / "white" / "black"``.
```

```
    Decision order: black by darkness first (hue is noise down
    there), then white/black by neutral spread + brightness, then the
    chromatic colours by channel dominance.
```

```
    """
```

```
    r, g, b = rgb
```

```
    if ambient < BLACK_AMBIENT:
```

```
        return "black"
```

```
    spread = max(r, g, b) - min(r, g, b)
```

```
    if spread < NEUTRAL_SPREAD:
```

```
        return "white" if ambient >= WHITE_AMBIENT else "black"
```

```
    if b >= r and b >= g:
```

```
        return "blue"
```

```
    if g >= r:
```

```
        return "green"
```

```
    if g >= r * YELLOW_G_OVER_R:
```

```
        return "yellow"
```

```
    return "red"
```

```
def main():
```

```
    from machine import I2C, Pin
```

```
    from openbricks.drivers.tca9548a import TCA9548A
```

```
    from openbricks.drivers.tcs34725 import TCS34725
```

```
    from openbricks.tools import wait
```

```
    i2c = I2C(0, sda=Pin(15), scl=Pin(16), freq=400_000)
```

```
    mux = TCA9548A(i2c)
```

```
    sensors = [TCS34725(mux[ch]) for ch in range(2)]
```

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```

print("Hold surfaces under both sensors (Ctrl-C to stop)...")
while True:
    parts = []
    for i, s in enumerate(sensors):
        ambient = s.ambient()
        rgb = s.rgb()
        parts.append("s%d: ambient=%3d rgb=(%3d,%3d,%3d) -> %s" % (
            i, ambient, rgb[0], rgb[1], rgb[2],
            classify(ambient, rgb)))
    print(" | ".join(parts))
    wait(200)

if __name__ == "__main__":
    main()

```

5.6 Line following (QTR sensor bar)

Three followers on the default ten-channel QTRLineSensor window, each a short control-law block over the element readings — the driver reports what every element sees, the program picks the discipline (see *Hardware guide* for the element tables and the eight-channel variant):

- `examples/qtr_line_follow_right.py` holds the line's RIGHT edge under element 7 (+16 mm, GPIO 8): `steer = KP * (50 - r[7].ambient())`. Elements 8 and 9 sit on the mat beyond the edge; either going dark is a branch marker.
- `examples/qtr_line_follow_left.py` is the mirror: LEFT edge under element 2 (-16 mm, GPIO 3), `steer = KP * (r[2].ambient() - 50)`, elements 0 and 1 as the branch watch.
- `examples/qtr_line_follow_center.py` (below) steers on `reading.position()`, the dark-weighted centroid in mm, so the error is proportional across the whole 56 mm window. When `position()` is `None` the line has left the window and the robot turns hard toward the side it was last seen on — the program keeps that side itself. The two outermost elements on each side are the branch watch.

In all three the whole window going dark (`all_dark()`) is the intersection that ends the run.

```

# SPDX-License-Identifier: MIT
"""Line following on the QTRLineSensor window - centre.

Run ``examples/qtr_calibrate.py`` once first. Holds the line's
CENTRE under the middle of the window, steering on
``reading.position()`` - the dark-weighted centroid in mm, positive
when the line sits right of centre - so the error stays
proportional across the whole 56 mm span. When no element is dark
the line has left the window and the robot turns hard toward the
side it was last seen on. The two outermost elements on each side
are the branch watch. The whole window going dark ends the run.
Edge disciplines: ``examples/qtr_line_follow_left.py`` / ``_right.py``.
"""

```

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```

import time

from openbricks.drivers.qtr import QTRLineSensor
from openbricks.drivers.st3032 import ST3032Motor
from openbricks.robotics import DriveBase

# --- control law (pure logic, unit-tested in
↳tests/test_qtr_line_follow.py) ---

CRUISE_DPS = 200
KP_MM = 6.0
MAX_DPS = 400
FLAG_COUNT = 2

last_side = 1

def clamp(dps):
    return max(0, min(MAX_DPS, int(dps)))

def get_wheel_speeds(reading):
    global last_side
    if reading.all_dark():
        return None
    position = reading.position()
    if position is None:
        steer = last_side * CRUISE_DPS
    else:
        if position > 0:
            last_side = 1
        elif position < 0:
            last_side = -1
        steer = KP_MM * position
    return (clamp(CRUISE_DPS + steer),
            clamp(CRUISE_DPS - steer))

def branch_seen(reading):
    for e in reading.elements[:FLAG_COUNT] + reading.elements[-FLAG_COUNT:]:
        if e.dark():
            return True
    return False

# --- end control law ---

qtr = QTRLineSensor()
qtr.load_calibration("/qtr.cal")

```

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```

left_motor = ST3032Motor(servo_id=2, uart_id=1, tx=14, rx=41,
                        invert=True)
right_motor = ST3032Motor(servo_id=1, uart_id=1, tx=14, rx=41)
db = DriveBase(left_motor, right_motor,
               wheel_diameter_mm=88, axle_track_mm=136)

print("following. Full-window dark stops the run.")
while True:
    reading = qtr.read()
    speeds = get_wheel_speeds(reading)
    if speeds is None:
        db.stop()
        print("intersection - stopped")
        break
    if branch_seen(reading):
        print("branch marker")
    db.move_wheels(speeds[0], speeds[1])
    time.sleep_ms(5)

```

5.7 Square up on a line (QTR sensor bar)

The classic align move on the QTRLineSensor window: each half of the ten-element bar acts as one virtual corner sensor, in two passes. Seek: drive slowly toward the line — the wheel whose half reaches it first stops while the other keeps rolling, pivoting the chassis square. Edge: servo each wheel proportionally — the follower's KP discipline — until its half reads ambient of about 50, the elements straddling the black/white boundary, parked right ON the line's edge. Calibrate once with `examples/qtr_calibrate.py` first; mount the bar ahead of the wheels.

```

# SPDX-License-Identifier: MIT
"""Square up on the edge of a perpendicular line, QTRLineSensor.

The classic FLL/WRO align move, on one sensor bar instead of two
corner sensors: each half of the window is one virtual corner
sensor, and each wheel servos proportionally until its half's mean
ambient reads 50 - the elements straddling the black/white
boundary. The wheel whose half arrives first holds while the other
pivots the chassis on; overshoot backs up. Both halves end ON the
edge, so the bar - and the chassis - is square right at it.

Run ``examples/qtr_calibrate.py`` once first. The bar must be
mounted ahead of the wheels; the farther ahead, the finer the final
heading.
"""

import time

from openbricks.drivers.qtr import QTRLineSensor
from openbricks.drivers.st3032 import ST3032Motor

```

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```

from openbricks.robotics import DriveBase
from openbricks.parameters import Stop

# --- control law (pure logic, unit-tested in tests/test_qtr_align.py) ---

KP = 1.3
EDGE_TOLERANCE = 8
SIDE_COUNT = 5

def side_ambient(elements):
    total = 0
    for e in elements:
        total += e.ambient()
    return total // len(elements)

def edge_dps(elements, target):
    error = side_ambient(elements) - target
    if abs(error) <= EDGE_TOLERANCE:
        return 0
    return int(KP * error)

def edge_wheel_speeds(reading):
    left = edge_dps(reading.elements[:SIDE_COUNT], 50)
    right = edge_dps(reading.elements[-SIDE_COUNT:], 50)
    if left == 0 and right == 0:
        return None
    return (left, right)

# --- end control law ---

qtr = QTRLineSensor()
qtr.load_calibration("/qtr.cal")

left_motor = ST3032Motor(servo_id=2, uart_id=1, tx=14, rx=41,
                        invert=True)
right_motor = ST3032Motor(servo_id=1, uart_id=1, tx=14, rx=41)
db = DriveBase(left_motor, right_motor,
               wheel_diameter_mm=88, axle_track_mm=136)

print("aligning on the line ...")
while True:
    speeds = edge_wheel_speeds(qtr.read())
    if speeds is None:
        break
    db.move_wheels(speeds[0], speeds[1])

```

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```

    time.sleep_ms(5)
db.stop(then=Stop.BRAKE)
print("aligned - square on the edge")

```

5.8 Square up on a line (two color sensors)

The same maneuver with a corner-mounted color sensor per side, for rigs without the QTR bar.

```

# SPDX-License-Identifier: MIT
"""
Demo: square the robot up on a dark line using two colour sensors.

The classic FLL/WRO "align on a line" move: drive slowly toward a
black line with one colour sensor mounted near each front corner,
ahead of the wheels. The moment a sensor crosses onto the line its
wheel brakes - the other wheel keeps rolling, pivoting the chassis
until its sensor reaches the line too. When both have stopped, the
sensor pair (and therefore the chassis) is parallel to the line, no
matter how crooked the approach was.

Geometry matters: the sensors must sit ahead of the axle and be
mounted symmetrically. The farther apart they are, the more accurate
the final heading.

Line detection reuses the idea from ``color_array.py``: a black line
on a light mat is simply "too dark to be the mat" - ``ambient()``
below a threshold. Print ``sensor.ambient()`` over your own mat and
line and put ``LINE_AMBIENT`` between the two readings.

Hardware (same bus layout as ``color_array.py`` / ``full_robot.py``):
    * ESP32-S3 (I2C on 15/16; serial bus UART on 14/6)
    * 2x ST-3032 wheel servos, IDs 1 (left) / 2 (right)
    * TCA9548A mux, one TCS34725 per channel: 0 = left, 1 = right,
      both facing the mat at the front of the chassis
"""

from machine import I2C, Pin
from openbricks.tools import wait

from openbricks.drivers.st3032 import ST3032Motor
from openbricks.drivers.tca9548a import TCA9548A
from openbricks.drivers.tcs34725 import TCS34725
from openbricks.robotics import DriveBase
from openbricks.parameters import Stop

left_motor = ST3032Motor(servo_id=2, uart_id=1, tx=14, rx=41)
right_motor = ST3032Motor(servo_id=1, uart_id=1, tx=14, rx=41, invert=True)
db = DriveBase(left_motor, right_motor,

```

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```

        wheel_diameter_mm=88, axle_track_mm=136)

i2c = I2C(0, sda=Pin(15), scl=Pin(16), freq=400_000)
mux = TCA9548A(i2c)
left_sensor = TCS34725(mux[1])
right_sensor = TCS34725(mux[0])

LINE_AMBIENT = 20

def align_on_line():

    approach_dps = 100
    poll_ms = 10
    timeout_ms = 8000

    left_done = False
    right_done = False
    left_ambient = None
    right_ambient = None
    speeds = [approach_dps, approach_dps]
    db.move_wheels(speeds[0], speeds[1])
    try:
        for _ in range(max(1, timeout_ms // poll_ms)):
            if not left_done:
                left_ambient = left_sensor.ambient()
                if left_ambient < LINE_AMBIENT:
                    speeds[0] = 0
                    db.move_wheels(speeds[0], speeds[1])
                    left_done = True
                    print('left wheel stopped (ambient=%d)' % left_ambient)
            if not right_done:
                right_ambient = right_sensor.ambient()
                if right_ambient < LINE_AMBIENT:
                    speeds[1] = 0
                    db.move_wheels(speeds[0], speeds[1])
                    right_done = True
                    print('right wheel stopped (ambient=%d)' %
↪right_ambient)

            if left_done and right_done:
                return
        wait(poll_ms)
    finally:
        db.stop(then=Stop.BRAKE)
    raise RuntimeError(
        "no line found within %d ms (last ambient: left=%r right=%r) - "
        "is the line in reach, and is LINE_AMBIENT calibrated for "

```

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```

        "your mat?" % (timeout_ms, left_ambient, right_ambient))

def main():

    print("aligning on the line ...")
    align_on_line()
    print("aligned – square to the line.")
    wait(500)
    left_motor.coast()
    right_motor.coast()

if __name__ == "__main__":
    main()

```

5.9 Full robot (ST-3032 drivebase + IMU + colour sensor + arm)

Everything from the reference build on one bus map — the wheels and the optional ST-3215 arm share the serial bus (IDs 1, 2 and 3), the IMU is on SPI, the colour sensor on I2C, and the QTR bank (GPIO 1-10) is left untouched.

```

# SPDX-License-Identifier: MIT
"""
Example: a small robot that rolls forward until its colour sensor sees red,
then demos a servo wave.

Hardware – the reference build from docs/hardware.md, wired to its
GPIO map so nothing collides with the QTR bank (GPIO 1-10):
    * ESP32-S3 DevKitC-1
    * 2x ST-3032 serial bus servos (wheel mode, IDs 1 and 2) on one
      URT-2: GPIO 14 -> URT-2 RX, GPIO 41 -> URT-2 TX
    * 1x ICM-45686 IMU on SPI: SCK 12, MOSI 13, MISO 11, CS 17
    * 1x TCS34725 RGB colour sensor on I2C: SDA 15, SCL 16
    * 1x ST-3215 serial bus servo (ID 3) daisy-chained on the SAME
      bus as the wheels (optional – the servo demo at the bottom
      just prints a message if it isn't attached)
"""

import time

from machine import I2C, Pin

from openbricks.drivers.icm45686 import ICM45686
from openbricks.drivers.st3032 import ST3032Motor
from openbricks.drivers.st3215 import ST3215
from openbricks.drivers.tcs34725 import TCS34725
from openbricks.robotics import DriveBase

```

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```

I2C_SDA, I2C_SCL = 15, 16
SCK, MOSI, MISO, CS = 12, 13, 11, 17
UART_ID, TX, RX = 1, 14, 41
LEFT_ID, RIGHT_ID, ARM_ID = 1, 2, 3

WHEEL_DIAMETER_MM = 88
AXLE_TRACK_MM      = 138

i2c  = I2C(0, sda=Pin(I2C_SDA), scl=Pin(I2C_SCL), freq=400_000)
imu   = ICM45686(sck=SCK, mosi=MOSI, miso=MISO, cs=CS)
color = TCS34725(i2c)

left  = ST3032Motor(servo_id=LEFT_ID, uart_id=UART_ID, tx=TX, rx=RX,
                    invert=True)
right = ST3032Motor(servo_id=RIGHT_ID, uart_id=UART_ID, tx=TX, rx=RX)

drivebase = DriveBase(left, right,
                       wheel_diameter_mm=WHEEL_DIAMETER_MM,
                       axle_track_mm=AXLE_TRACK_MM)

try:
    arm = ST3215(servo_id=ARM_ID, uart_id=UART_ID, tx=TX, rx=RX)
    if not arm.ping():
        raise OSError("servo %d did not answer ping" % ARM_ID)
except Exception as e:
    print("no arm servo attached:", e)
    arm = None

while True:
    r, g, b = color.rgb()
    heading = imu.heading()

    print("rgb=({:3d},{:3d},{:3d}) heading={:6.1f}".format(r, g, b,
↵heading))

    if r > g and r > b and r > 120:
        print("Red detected - stopping.")
        drivebase.stop()
        break

    drivebase.drive(speed_mm_s=150, turn_rate_dps=0)
    time.sleep_ms(50)

if arm is not None:
    arm.move_to(180, speed=500)
    time.sleep_ms(500)

```

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```
arm.move_to(0, speed=500)
```

5.10 Servo dynamometer (two ST-3032s, coupled)

Measures one servo's real no-load speed, stall-torque estimate and gear play against a second servo used as the load — the per-unit numbers to compare across your spares. Procedure and how to read the summary: *Servo health & characterisation*.

```
# SPDX-License-Identifier: MIT
"""ST-3032 dynamometer - measure what YOUR servo does, unit by unit.
```

Couple two ST-3032s horn to horn (a rigid coupler; the pair bolted down). The servo under test drives at a fixed duty while the second servo loads it with an opposing duty in steps; the speed and supply current of the servo under test give a torque-speed line through the motor constant (torque = K_t x current above no-load), and a gentle reversal against the held load servo gives the pair's combined gear play. Run it on every spare and keep the summary lines: the unit whose no-load speed or stall estimate sits well off the others is the one to leave out of the drive pair.

```
openbricks upload -n <hub> examples/st3032_dyno.py
openbricks log -n <hub>
```

The sweep stops at 30 % duty on both servos; the fitted line is extrapolated to stall (a DC motor behind a gearbox droops linearly).

```
"""
```

```
import time
```

```
from openbricks.drivers.st3032 import ST3032Motor
```

```
UART_ID, TX, RX = 1, 14, 41
DUT_ID, LOAD_ID = 1, 2
DUTY = 30
LOAD_STEPS = (0, 5, 10, 15, 20, 25, 30)
SETTLE_MS = 800
SAMPLES = 10
PLAY_DUTY = 8
```

```
# --- dyno summary ---
```

```
def torque_mnm(current_a, no_load_current_a, kt_mnm_per_a):
    """Shaft torque from supply current above the no-load current,
    through the motor constant (ST-3032 datasheet: 6.3 kg·cm/A)."""
    extra = current_a - no_load_current_a
    if extra < 0.0:
        extra = 0.0
    return extra * kt_mnm_per_a
```

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```

def fit_torque_speed(points):
    """Least-squares line through (torque_mnm, speed_dps) points:
    (no_load_dps, stall_mnm, slope_dps_per_mnm) - the speed at zero
    torque, the torque at zero speed, and the droop between."""
    n = len(points)
    mean_t = sum(t for t, _ in points) / n
    mean_s = sum(s for _, s in points) / n
    sxx = sum((t - mean_t) ** 2 for t, _ in points)
    if sxx == 0.0:
        raise ValueError("torque did not change across the sweep - "
                          "is the load servo coupled and powered?")
    slope = sum((t - mean_t) * (s - mean_s) for t, s in points) / sxx
    no_load = mean_s - slope * mean_t
    if slope >= 0.0:
        raise ValueError("speed rose with torque - the load servo is "
                          "helping, not opposing; flip its sign")
    return (no_load, -no_load / slope, slope)

def play_deg(plus_angles, minus_angles):
    """Combined gear play of the coupled pair: the mean rest angle
    after pushing gently one way minus the mean after pushing the
    other way."""
    return abs(sum(plus_angles) / len(plus_angles)
               - sum(minus_angles) / len(minus_angles))

# --- end dyno summary ---

dut = ST3032Motor(servo_id=DUT_ID, uart_id=UART_ID, tx=TX, rx=RX)
load = ST3032Motor(servo_id=LOAD_ID, uart_id=UART_ID, tx=TX, rx=RX)

print("load_duty_pct,speed_dps,current_a,torque_mnm")
points = []
no_load_current = None
for step in LOAD_STEPS:
    dut.dc(DUTY)
    load.dc(-step)
    time.sleep_ms(SETTLE_MS)
    speed = 0.0
    current = 0.0
    for _ in range(SAMPLES):
        s = dut.speed()
        if s is None:
            raise OSError("servo %d went silent mid-sweep" % DUT_ID)
        speed += s
        current += dut.health().current
    time.sleep_ms(20)

```

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```

speed /= SAMPLES
current /= SAMPLES
if no_load_current is None:
    no_load_current = current
torque = torque_mnm(current, no_load_current,
                    ST3032Motor.KT_MNM_PER_A)
points.append((torque, abs(speed)))
print("%d,%.1f,%.3f,%.1f" % (step, speed, current, torque))
dut.coast()
load.coast()
time.sleep_ms(500)

load.hold()
plus = []
minus = []
for _ in range(3):
    dut.dc(PLAY_DUTY)
    time.sleep_ms(400)
    a = dut.angle()
    dut.dc(-PLAY_DUTY)
    time.sleep_ms(400)
    b = dut.angle()
    if a is None or b is None:
        raise OSError("servo %d went silent during the play check"
                        % DUT_ID)

    plus.append(a)
    minus.append(b)
dut.coast()
load.coast()

no_load, stall, slope = fit_torque_speed(points)
print("# --- dyno summary ---")
print("servo %d at %d %% duty: no-load %.0f dps, stall %.0f mNm "
      "(%.2f kg.cm), droop %.3f dps per mNm"
      % (DUT_ID, DUTY, no_load, stall, stall / 98.07, slope))
print("scaled to 100 %% duty: stall ~%.0f mNm, datasheet %.0f; "
      "no-load ~%.0f dps, datasheet %.0f"
      % (stall * 100.0 / DUTY, ST3032Motor.STALL_TORQUE_MNM,
         no_load * 100.0 / DUTY, ST3032Motor.max_dps_default()))
print("combined gear play of the pair: %.2f deg (datasheet <= 1.0 "
      "per servo)" % play_deg(plus, minus))

```

5.11 Servo soak test (one ST-3032)

An hour of swings with voltage, temperature, current and gear play logged along the way, and the drift summarised at the end — the pre-competition check for a servo you are not sure about.

```

# SPDX-License-Identifier: MIT
"""ST-3032 soak test - run a servo for an hour and watch it age.

```

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Before a competition, or after a crash: one servo on the bus swings +/-90 deg at a gentle speed for the chosen duration, and every REPORT_EVERY cycles it logs supply voltage, temperature, current and its own gear play (one target, approached from below and from above under hold). The summary at the end is the drift: a temperature that keeps climbing, or play that grows through the run, is a servo to retire before it retires you on the mat.

```

    openbricks upload -n <hub> examples/st3032_soak_test.py
    openbricks log -n <hub>
"""

import time

from openbricks.drivers.st3032 import ST3032Motor
from openbricks.parameters import Stop

UART_ID, TX, RX = 1, 14, 41
SERVO_ID = 3
DURATION_MIN = 60
SPEED_DPS = 120
SWING_DEG = 90
PLAY_APPROACH_DEG = 20
REPORT_EVERY = 20

# --- soak summary ---
def play_deg(from_below, from_above):
    """Gear play: the rest angle reached approaching one target from
    above minus the one reached approaching it from below."""
    return abs(from_above - from_below)

def soak_summary(rows):
    """``rows`` = (t_min, voltage, temperature, current, play_deg) in
    time order. The drift from the first report to the last, the
    peak temperature and the lowest supply voltage seen."""
    first = rows[0]
    last = rows[-1]
    return {
        "temp_rise": last[2] - first[2],
        "temp_peak": max(r[2] for r in rows),
        "current_change": last[3] - first[3],
        "play_change": last[4] - first[4],
        "voltage_min": min(r[1] for r in rows),
    }
# --- end soak summary ---

```

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```

m = ST3032Motor(servo_id=SERVO_ID, uart_id=UART_ID, tx=TX, rx=RX)

print("t_min,voltage_v,temperature_c,current_a,play_deg,cycles")
rows = []
start = time.ticks_ms()
cycles = 0
while time.ticks_diff(time.ticks_ms(), start) < DURATION_MIN * 60000:
    m.run_angle(SPEED_DPS, SWING_DEG)
    m.run_angle(SPEED_DPS, -SWING_DEG)
    cycles += 1
    if cycles % REPORT_EVERY:
        continue
    m.run_angle(SPEED_DPS, -PLAY_APPROACH_DEG)
    m.run_angle(SPEED_DPS, PLAY_APPROACH_DEG, then=Stop.HOLD)
    time.sleep_ms(300)
    below = m.angle()
    m.run_angle(SPEED_DPS, PLAY_APPROACH_DEG)
    m.run_angle(SPEED_DPS, -PLAY_APPROACH_DEG, then=Stop.HOLD)
    time.sleep_ms(300)
    above = m.angle()
    if below is None or above is None:
        raise OSError("servo %d went silent during the play check"
                       % SERVO_ID)
    h = m.health()
    t_min = time.ticks_diff(time.ticks_ms(), start) / 60000.0
    rows.append((t_min, h.voltage, h.temperature, h.current,
                 play_deg(below, above)))
    print("%.1f,%.1f,%.0f,%.3f,%.2f,%d"
          % (t_min, h.voltage, h.temperature, h.current,
             rows[-1][4], cycles))
    if h.flags:
        print("protection flags: %s" % ", ".join(h.flags))
m.coast()

s = soak_summary(rows)
print("# --- soak summary ---")
print("%d cycles: temperature %+.0f C (peak %.0f C, shutdown at 80), "
      "current %+.3f A, play %+.2f deg, supply dipped to %.1f V"
      % (cycles, s["temp_rise"], s["temp_peak"], s["current_change"],
         s["play_change"], s["voltage_min"]))

```

MEASURING WHEEL DIAMETER & AXLE TRACK

DriveBase converts your commands from millimeters and body-degrees into wheel rotations using exactly two numbers:

```
db = DriveBase(left, right, wheel_diameter_mm=88, axle_track_mm=138)
```

- **wheel_diameter_mm** — how far one wheel rotation moves the robot. Every `straight()` distance scales linearly with it.
- **axle_track_mm** — the distance between the two wheels' *contact points* with the floor. Every `turn()` angle scales linearly with it.

Getting these two right is worth more than any controller tuning: a 2 % diameter error is 20 mm of error on a 1 m drive, and a 2 % track error is $\sim 7^\circ$ of error on a full spin. A ruler gets you close; the two test drives below get you to a few tenths of a percent.

Calibrate **wheel diameter first, then axle track** — the turn calculation uses the wheel diameter, so a diameter error contaminates the track measurement.

6.1 Step 1 — wheel diameter, from a straight drive

Start from the nominal value (caliper across the tire, or the manufacturer's spec). Then:

1. Put a strip of masking tape on the floor and align a marked point of the robot (e.g. the axle center) with its edge.
2. Run a long straight — the longer the drive, the better the resolution:

```
db.settings(straight_speed=150)
db.straight(1000)
```

3. Measure the distance actually traveled, from tape edge to the same marked point, in millimeters.
4. Scale the diameter by how far the robot *really* went:

```
new_diameter = old_diameter * measured_mm / 1000
```

Traveled 1023 mm with `wheel_diameter_mm=88`? Then $88 \times 1023 / 1000 = 90.0$ is your real diameter.

Repeat once with the new value: the measured distance should now land within a few millimeters of the command. Soft tires compress under the robot's weight, so the *effective* diameter is often $\sim 1\text{-}3\%$ smaller than the caliper says — the test drive measures reality, load included.

6.2 Step 2 — axle track, from an in-place spin

With the wheel diameter calibrated, the fastest way is to let an attached ICM-45686 measure the spin for you: `examples/icm45686_axle_track.py` commands ten encoder-only turns (the drivebase's gyro stays off — the point is to measure what the *encoders* produce), reads the true rotation off the IMU, and prints the corrected `axle_track_mm` to paste into your `DriveBase(...)` call. Ten turns make each 0.1° of gyro error only 0.003 % of track.

No IMU on the robot? The manual version:

1. Align the robot against a straightedge (a wall or ruler touching both wheels works well) and note its exact heading.
2. Command several full spins in place — more turns amplify the error so it's easier to measure:

```
db.settings(turn_rate=120)
db.turn(3600)           # ten full clockwise spins
```

3. Measure the final heading error `err_deg` against your straightedge: positive if the robot rotated *past* the start alignment, negative if it stopped short. A protractor or a phone compass app is plenty — over ten spins, each degree of final error is only 0.1 % of track.
4. Scale the track by how far the robot really rotated:

```
new_track = old_track * 3600 / (3600 + err_deg)
```

Overshot by 18° with `axle_track_mm=138`? Then $138 \times 3600 / 3618 = 137.3$ is your real track.

Note the direction: if the robot turns **too far**, the real track is **smaller** than configured (each wheel-degree of travel produced *more* body rotation than the math assumed), so the correction *decreases* the configured value. Stopping short means the opposite — increase it.

The contact *points* matter, not the wheel centers: wide, soft tires effectively touch the ground inboard of their centerline, so the real track is usually a few millimeters less than what you measure center-to-center with a ruler.

6.3 Checking the result

Drive a square and see how close the robot returns to its start pose:

```
for _ in range(4):
    db.straight(300)
    db.turn(90)
```

With both values calibrated, the return error on a 300 mm square is typically under 1 cm and a few degrees. Do the calibration on the same surface the robot will compete on — carpet, foam mats, and wood all load the tires differently.

6.4 What about the gyro?

`use_gyro(True)` (with an `imu=` attached) makes turns terminate on *measured* body rotation, so heading no longer depends on the axle track being exact — wheel slip included. Calibrate the track anyway: the

controller still uses it to shape the commanded wheel speeds, and encoder-only operation (gyro off, or no IMU on the robot) depends on it entirely.

How much the gyro buys depends on which IMU: an ICM-45686 read inside the 1 kHz control tick returned a four-turn square within **+0.6° total** on the reference bench, versus +0.5° to +1.8° *per turn* for a Python-pumped BNO055 and a few degrees encoder-only. `examples/icm45686_square.py` measures all of this on your own robot — run it and read the two drift numbers.

SERVO HEALTH & CHARACTERISATION

Every Feetech STS servo reports its own supply voltage, case temperature, supply current and a byte of protection flags, next to the position/speed/load block the drivers already read. Since 3.3.0 that block is one call away, and two bundled programs turn it into the two questions a team actually asks about a servo: *what does this particular unit do, and is it still doing it.*

7.1 `health()`

```
from openbricks.drivers.st3032 import ST3032Motor

m = ST3032Motor(servo_id=3, uart_id=1, tx=14, rx=41)
h = m.health()
print(h.voltage, "V", h.temperature, "C", h.current, "A", h.flags)
```

`health()` returns a `ServoHealth` tuple:

Field	Meaning
<code>voltage</code>	supply rail at the servo, volts (0.1 V resolution)
<code>temperature</code>	case temperature, °C
<code>current</code>	supply current, amps (6.5 mA resolution)
<code>flags</code>	the protection flags currently set, as names — <code>voltage</code> , <code>sensor</code> , <code>temperature</code> , <code>current</code> , <code>angle</code> , <code>overload</code> ; empty when healthy
<code>status</code>	the raw status byte

It works on position servos (ST3215 / ST3032) and wheel-mode motors alike, adopted by a `DriveBase` or on their own bus. It **raises** `OSError` when the bus is silent rather than returning nothing — a health check that can't reach the servo is the failure it exists to catch. On an adopted motor the four registers are staged through the native bus pump, which takes a few milliseconds: read it for a log line, not inside a control loop.

The ST-3032's own protections, from its datasheet, are the numbers to read the fields against: it cuts torque above 80 °C, flags **over-voltage** outside 9–14 V, and latches **overload** after 2 s above 80 % of stall (re-issuing a command clears it). Nominal current is 100 mA free-running, 500 mA at rated load, 1.6 A stalled.

7.2 Dynamometer: what does *this* servo do?

`examples/st3032_dyno.py` characterises one servo against a second one, the way actuator labs do it with a load motor — minus the load cells. Couple two ST-3032s horn to horn with a rigid coupler and bolt the pair down. The servo under test drives at a fixed duty; the load servo opposes it with a duty that steps up from 0 to 30 %; at each step the program averages the driven servo's speed and supply current. The current above the unloaded baseline, times the motor constant (6.3 kg·cm/A, `ST3032Motor.KT_MNM_PER_A`), is the shaft torque, so the sweep yields a torque–speed line, and a DC motor behind a gearbox droops linearly — the fit's intercepts are the unit's **no-load speed** and **stall torque** at that duty, scaled to 100 % in the summary. A gentle reversal against the held load servo at the end measures the pair's combined **gear play**.

```
load_duty_pct, speed_dps, current_a, torque_mnm
0, 262.4, 0.118, 0.0
5, 251.0, 0.151, 20.4
...
# --- dyno summary ---
servo 1 at 30 % duty: no-load 263 dps, stall 226 mNm (2.30 kg.cm), droop -
↪1.16 dps per mNm
scaled to 100 % duty: stall ~753 mNm, datasheet 980; no-load ~877 dps, ↪
↪datasheet 888
combined gear play of the pair: 0.95 deg (datasheet <= 1.0 per servo)
```

Run it on every servo you own and keep the summary lines. The numbers are not lab-grade — the torque rides on a datasheet constant, and the stall figure is an extrapolation from a 30 % sweep (the bench cap; raise `DUTY` and `LOAD_STEPS` only with the pair bolted down) — but they are *consistent*, which is what matters: the unit whose no-load speed or stall sits well off its siblings is the one to keep out of the drive pair, and a pair that measures alike will track alike under `DriveBase`. The simulator's servos are ideal velocity loops, so there is nothing to feed these numbers into; they are for choosing and comparing hardware.

7.3 Soak test: is it still good?

`examples/st3032_soak_test.py` runs one servo through $\pm 90^\circ$ swings for an hour (`DURATION_MIN`) and every `REPORT_EVERY` cycles logs `health()` plus the servo's gear play — one target approached from below and from above under hold, the difference between the two rest angles. The summary is the drift from the first report to the last:

```
# --- soak summary ---
1780 cycles: temperature +14 C (peak 46 C, shutdown at 80), current +0.021 ↪
↪A, play +0.08 deg, supply dipped to 11.4 V
```

What to look for: a temperature that keeps climbing instead of levelling off, play that grows through the run, current that rises at the same speed and load, or any protection flag at all. Each is a servo to retire — or a wiring/battery problem to fix — before it shows up on the mat. Feetech's own life test is 50 000 cycles at a fifth of stall torque; an hour is a couple of thousand, enough to expose a unit that is already going.

API REFERENCE

Everything below ships frozen into the firmware image — `import openbricks` on the hub gives you all of it. The pages are generated from the package’s docstrings.

The API is layered:

- **Drivers** (`openbricks.drivers.*`) — one module per supported component. Construct these with your wiring’s pins / IDs.
- **Interfaces** (`openbricks.interfaces`, `openbricks.distance`) — the abstract contracts drivers implement. Higher layers depend only on these, which is what makes the system plug-and-play.
- **Robotics** (`openbricks.robotics`) — building blocks composed from interfaces, like the two-wheel `DriveBase`.
- **Parameters** (`openbricks.parameters`) — the enums every option argument takes (`Stop`, `DriveMode`).
- **Hub & runtime** — board-level peripherals, BLE, program launching, and log capture.

8.1 robotics — DriveBase

Drive a robot, not two motors: `DriveBase` couples a left and a right motor into one chassis with moves in millimeters and body-degrees.

```
from openbricks.drivers.st3032 import ST3032Motor
from openbricks.robotics import DriveBase

left  = ST3032Motor(servo_id=1, uart_id=1, tx=14, rx=41)
right = ST3032Motor(servo_id=2, uart_id=1, tx=14, rx=41, invert=True)

db = DriveBase(left, right, wheel_diameter_mm=88, axle_track_mm=138)
db.settings(straight_speed=250, turn_rate=200,
            acceleration=1000, turn_acceleration=800)

for _ in range(4):           # a 300 mm square
    db.straight(300)
    db.turn(90)
```

Moves block by default. Pass `wait=False` to return immediately and poll `done()` — the Pybricks pattern for driving while reading sensors; any new move command supersedes the pending one:

```
db.straight(600, wait=False)
while not db.done():
    if bumper_pressed():
        db.stop()
        break
    time.sleep_ms(10)
```

For direct control of each wheel — line-following, tank-style teleop, or any controller that computes its own per-wheel outputs — `move_wheels` takes two speeds in wheel-deg/s:

```
db.move_wheels(200, 120)      # gentle right-hand arc
time.sleep_ms(500)
db.stop()
```

Both setpoints leave in a single sync-write packet on serial-bus motors, so the wheels change speed at the same packet boundary. Reach for this rather than a `SyncServoGroup` over the wheels: a `DriveBase` hands their UART to the native bus driver when it adopts them, so a `SyncServoGroup` can't drive them at all.

Moves take a `then=` end state — a `Stop` member, never a string (3.0.0): `Stop.COAST` (the default) decelerates to rest and free-wheels; `Stop.BRAKE` / `Stop.HOLD` end actively. `then=Stop.NONE` on straight and `curve` does NOT decelerate at the end: the move finishes at cruise speed and the wheels keep it until the next command, so chained segments flow through their seams. A segment too short to reach cruise under `settings.acceleration` hands over at the speed it does reach on its target (the acceleration is never raised to make it):

```
from openbricks.parameters import Stop

db.straight(300, then=Stop.NONE)    # ends AT cruise
db.curve(150, 90, then=Stop.NONE)  # picks the speed up
db.straight(300)                   # decelerates to rest
```

Move endings are SHAPED all the way down (2.6.0): the controller runs position integral action (pbio's integrator rules — the same control law Pybricks uses) so tracking error is squeezed out near the target, and any residual left when a profile expires is closed by a small landing trajectory under the same acceleration limit as every other motion — never a raw feedback step. A robot that ends a mission simply comes to rest on its mark; a genuinely stuck robot still refuses to report `done()` and the stall watchdog raises.

`stop()` is Pybricks parity: it coasts and returns immediately. `then` picks the end state (`Stop.COAST`, `Stop.BRAKE`, `Stop.HOLD`). `Stop.BRAKE` and `Stop.HOLD` decelerate at the `acceleration` setting as a move of the coupled controller, with the heading loop closed all the way down — with `use_gyro(True)` the IMU corrects any yaw the brake induces (one wheel gripping harder than the other), so the robot stops on the heading it had (3.2.0). Coast cannot: it releases torque at once, and nothing can steer wheels that carry no torque — so to end a line-follow on heading, brake:

```
while True:
    r = qtr.read()
    if r.all_dark():                # the intersection
        break
    db.drive(SPEED, KP * (50 - r[7].ambient()))  # right edge under +16 mm
db.stop(then=Stop.BRAKE)           # gyro-held deceleration
db.straight(200)                   # holds the heading the follow reached
```

A program that ends by itself — returns, or dies on an exception — gets the same brake for free (3.4.0): the launcher decelerates an adopted drive base to rest under the controller before the torque-off that ends every run, so a mission that finishes mid-move no longer rolls out on friction (or away, on a ramp). The stop button keeps its instant torque-off — an e-stop must not spend half a second driving under power. The run log records the outcome (`brake: wheels to rest in 412 ms`, or the time-out that hands a blocked robot to the torque-off).

Short moves armed while the robot is already fast raise their own deceleration to land at rest exactly on target, so you rarely need more — but `wait=True` is available to block until the stop has LANDED: both wheels' measured speeds read ~ 0 and, on an adopted drive base, the engine reports the stop finished (the decel ramp for brake/hold, the physical freewheel decay for coast). A brake or hold lands the moment the wheels are measured at rest after its ramp, whatever the position residual — a stop is done when the robot has stopped (3.10.1; before, duty-mode stiction could keep it “active” for the 400 ms settle window, and a `reset()` in that window raised). It raises `ValueError` on open-loop pairs (no measured speed) and, if the stop never lands within 5 s, `RuntimeError` naming the measured speeds — a stopped robot that is still moving is a fault, not a detail to hide.

A wheel that stops answering the bus — no power, a knocked-loose TX/RX wire, the wrong `servo_id` — raises instead of quietly doing nothing, and the error names the motor:

```
OSError: motor is not responding on the bus: right wheel
(servo id 1, slot 1) on UART1 tx=14 rx=6 - 0 replies, 137 failed
reads (137 in a row). Check the servo's power and TX/RX wiring,
and that it really has that bus id - `openbricks servo-id --scan`
lists the ids actually answering on the bus.
```

(`openbricks servo-id` talks through the URT-2's USB port, or, with the servo already wired to the hub, through the hub: `openbricks servo-id -n NAME --scan` lists the ids answering on the hub's bus and `openbricks servo-id -n NAME 3` re-IDs the one servo wired to it — the same safety contract either way. The hub route runs as a program, so it gets the launcher's program-boundary reset, a run log and the stop button.)

Both wheels are verified when the `DriveBase` is constructed, and on every move afterwards. If one goes silent mid-move the controller halts immediately rather than winding that wheel's command to the rail — a frozen odometry reading would otherwise look like “infinite error” to the heading loop. `db.check_motors()` runs the same check on demand.

With an IMU attached, `use_gyro(True)` steers by measured body rotation instead of the encoder differential — immune to wheel slip. The preferred IMU is the [ICM45686](#): it is read *inside* the 1 kHz control tick over SPI, so the heading correction runs every millisecond in C with no Python in the loop (bench: +0.6° total drift over a four-turn square):

```
from openbricks.drivers.icm45686 import ICM45686

imu = ICM45686(sck=12, mosi=13, miso=11, cs=17)
db = DriveBase(left, right, wheel_diameter_mm=88,
               axle_track_mm=138, imu=imu)
db.use_gyro(True)
```

(A legacy [BNO055](#) on I2C still works — its fused heading is pumped from Python between ticks, which corrects noticeably slower, typically +0.5° to +1.8° per turn. New builds should use the ICM-45686.)

The heading target is absolute: each move steers to where the plan says the robot should be, so a turn's arrival residual is corrected by the next move instead of accumulating. `drive() / move_wheels` (a line-

follow) sit outside that plan — they rotate the chassis by whatever your controller decides — so the next coupled command after one (a move, or a brake/hold stop) takes the heading the follow *reached* as its target rather than steering back to the pre-follow one (3.2.0; earlier releases needed `reset()` at that hand-off).

To re-zero the heading frame mid-mission (say, after squaring up on a line), call `db.reset()` between moves — afterwards the robot’s **CURRENT** pose is heading zero for both the drive base and `imu.heading()`, atomically. `imu.heading()` re-zeroes on an IMU that can reset its heading (the ICM-45686); a BNO055 keeps reporting its absolute fused heading, and the drive base re-bases against it:

```
db.straight(100)
db.turn(-90)
db.reset()           # here, now = heading zero
db.straight(130)     # drives straight along the NEW zero
```

`imu.reset_heading()` refuses (`OSError`) while a drive base steers by the gyro — same rule as Pybricks (“can’t reset heading while gyro in use”): zeroing the integrator under an armed heading controller shifts the measurement out from under the held target, and the next move veers chasing the old frame. Use `db.reset()`, or `use_gyro(False)` first. `db.reset()` itself raises while a **MOVE** is in progress (a straight / turn / curve armed with `wait=False` and not yet done) — wait for it or stop first. A brake/hold stop still on its ramp is **NOT** a move to `reset()`: it waits for the stop to land (up to 1.5 s, pumping a soft gyro meanwhile) and lands it itself at the bound, so `stop()` followed by `reset()` never raises, with or without `wait=True` (3.10.1 — a 3.2.0–3.10.0 firmware raised here when duty-mode stiction kept the brake “active” after the wheels had stopped; a mission died on it mid-competition).

Accurate `wheel_diameter_mm` / `axle_track_mm` values matter more than any tuning — calibrate both with two short test drives: [Measuring wheel diameter & axle track](#). Two-wheel differential drivebase.

Thin Python wrapper over `_openbricks_native.DriveBase` — the C implementation at `native/user_c_modules/openbricks/drivebase.c` that runs 2-DOF coupled control at 1 kHz. Both motors are driven by a single forward-progress trajectory and a heading-hold trajectory; a heading-error feedback term keeps them in sync even when one wheel has more friction than the other.

Public API matches the M1 Python version so existing code and tests don’t need to change:

```
db = DriveBase(left, right, wheel_diameter_mm=56, axle_track_mm=114)
db.settings(straight_speed=200, turn_rate=180) # deg/s at wheels
db.straight(500) # mm, blocking
db.turn(90) # deg body heading, blocking
db.drive(100, 0) # non-blocking kinematic mapping
```

Serial-bus motors (ST-3215 / ST-3032) are adopted transparently onto the hard-tick engine (firmware) or the emulated bus (sim) — same class, same code, one controller. There is no Python control loop: motor pairs with neither a native servo nor a serial-bus adoption path get open-loop `drive()/stop()` only, and `straight()/turn()` raise.

Open-loop `drive()` bypasses the coupled controller; it just maps (`speed_mm_s`, `turn_rate_dps`) → (`left_dps`, `right_dps`) and hands them to each servo’s `run_speed`. Useful for interactive control where profile-based moves would feel sluggish.

```
class openbricks.robotics.drivebase.DriveBase(left, right, wheel_diameter_mm,
                                              axle_track_mm, imu=None,
                                              drive=DriveMode.DUTY)
```

Bases: `object`

A two-wheel differential drive robot: two motors, one chassis.

Pybricks-compatible surface: `straight(distance_mm)`, `turn(angle_deg)`,

`drive(speed_mm_s, turn_rate_dps), stop(then=...), settings(...), use_gyro(True)` and non-blocking moves via `wait=False + done()`. Positive turn is right/clockwise viewed from above.

Give it any two closed-loop motors and it picks the right controller automatically:

- **Encoder servos** (`JGB37Motor`, `MG370Motor`) — the native C 2-DOF coupled controller at 1 kHz.
- **Serial-bus servos** (`ST3032Motor`, `ST3215Motor`) — the motors are *adopted* onto the hard-tick native bus engine (~220 Hz odometry per wheel, immune to Python stalls). Their wheel-mode motor API keeps working after adoption.
- **Open-loop motors** (`L298NMotor`) — kinematic `drive()` / `stop()` only; moves by distance need feedback and raise.

Example:

```
from openbricks.drivers.st3032 import ST3032Motor
from openbricks.robotics import DriveBase

left  = ST3032Motor(servo_id=1, uart_id=1, tx=14, rx=41)
right = ST3032Motor(servo_id=2, uart_id=1, tx=14, rx=41,
                    invert=True)
db = DriveBase(left, right, wheel_diameter_mm=88,
               axle_track_mm=138)
db.straight(300)      # forward 300 mm
db.turn(90)           # turn right 90 degrees
```

Accurate `wheel_diameter_mm` / `axle_track_mm` values matter more than any controller gain — see [Measuring wheel diameter & axle track](#) for how to calibrate both in two short test drives.

settings (`straight_speed=None`, `turn_rate=None`, `acceleration=None`, `turn_acceleration=None`)

Tune cruise + ramp parameters for subsequent moves.

Parameters

- **straight_speed** – cruise speed for `straight()`, wheel-deg/s.
- **turn_rate** – cruise rate for `turn()`, wheel-deg/s.
- **acceleration** – STRAIGHT (and curve) trajectory acceleration, wheel-deg/s² — Pybricks' `straight_acceleration`. Serial-engine default 1500 (Pybricks parity: their 2000 dps² motor accel × 3/4) — lower it if the robot pitches or lifts its rear on launch. In mm/s² that's `acceleration * wheel_circumference / 360`. Applies on both paths: the native (encoder-servo) controller arms its C trajectory with it, and the serial-bus engine forwards it to the hard-tick controller.
- **turn_acceleration** – `turn()` ramps' own acceleration, wheel-deg/s², independent of `acceleration` — Pybricks parity (serial default 1500). Serial-bus drivebase only; passing it on an encoder/DC pair raises.

use_gyro (*enable*)

Switch the heading feedback source between encoder-diff (default) and the attached IMU (when True). Pybricks-style.

Requires an `imu=` argument to the constructor. With the gyro, heading is slip-immune — wheel slip or wildly asymmetric friction won't throw the robot off course, because the IMU sees actual body rotation regardless of what the wheels did. Works on both the native (encoder-servo) path and the serial-bus engine.

reset()

Re-zero the heading frame: after `reset()`, the robot's CURRENT pose is heading zero — for the drive base's controller and `imu.heading()` together (Pybricks `DriveBase.reset()`). Call it between moves; it raises `RuntimeError` while a move is active, with or without the gyro. `imu.heading()` re-zeroes on an IMU that can reset its heading (the ICM-45686); a BNO055 keeps reporting its absolute fused heading, and the controller re-bases against it.

This is the supported way to re-zero mid-mission. `imu.reset_heading()` refuses while a drive base steers by the gyro, because zeroing the integrator under an armed controller shifts the measurement out from under the held target — the next `straight()` then veers chasing the old frame.

drive(speed_mm_s, turn_rate_dps)

Start driving at a given forward speed + body turn rate.

Kinematic one-shot — no coupled feedback. Call again (or `stop()`) to change. Positive turn rate = right turn (clockwise viewed from above), Pybricks convention.

Speed changes ramp at `settings(acceleration=...)` (the uniform-accel rule, 1.94.0) — proportionally across the two wheels, so an arc keeps its radius through the ramp.

check_motors()

Raise if a wheel has stopped answering the bus.

Serial-bus wheels are checked at construction and on every move, so you rarely need this directly — reach for it in a long-running open-loop control loop (`move_wheels` in a `while` loop already calls it for you), or to verify the chassis before a run. The error names the motor: side, bus id, slot, UART and pins.

move_wheels(left_wheel_speed, right_wheel_speed)

Drive the two wheels at independent speeds, in wheel-deg/s.

Positive is forward on both sides (each motor's `invert` is already applied), so `move_wheels(200, 200)` drives straight and `move_wheels(200, -200)` spins in place.

Non-blocking and continuous, like `drive()`: the wheels hold these speeds until you call it again, issue another move, or `stop()`. It supersedes any move in flight.

Use this instead of building a `SyncServoGroup` over the wheels. On serial-bus motors both setpoints leave in a single sync-write packet, so the wheels change speed at the same packet boundary — and a `SyncServoGroup` could not drive them anyway, because adopting them into a `DriveBase` hands their UART to the native driver. On encoder servos both targets are set and both servos subscribed inside one native call.

Where `drive(speed_mm_s, turn_rate_dps)` speaks chassis kinematics, this speaks wheels directly — the right tool for line-following, tank-style teleop, or any controller that computes per-wheel outputs itself.

Example:

```
db.move_wheels(200, 120)      # gentle right-hand arc
time.sleep_ms(500)
db.stop()
```

Open-loop motor pairs (no encoder, no serial bus) are supported but cannot batch: the two speeds are written one after the other.

stop (*then=Stop.COAST, wait=False*)

Halt both wheels. Also clears any pending `wait=False` move (new command supersedes, pybricks-style). `then` selects the end-state:

- `Stop.COAST` (default) — both motors free-wheel.
- `Stop.BRAKE` — both motors actively resist motion at zero velocity.
- `Stop.HOLD` — both motors actively hold their current angle. Requires motors that implement `hold()` (e.g. `ST3215Motor`); open-loop drivers raise `NotImplementedError`.

Both wheels are always commanded together, never one motor at a time. On serial-bus (adopted) motors the whole stop is staged atomically in the C engine and reaches the wheels at the same bus-packet boundary. `brake` and `hold` DECELERATE at `settings(acceleration=...)` first (the uniform-accel rule) as a move of the coupled controller, so the heading loop stays closed all the way down: with `use_gyro(True)` the IMU corrects any yaw the brake induces (one wheel gripping harder than the other) exactly as it does mid-straight, and the robot stops on the heading it had. `Hold` anchors where the robot actually stops. `coast` releases torque immediately — a freewheel has no controlled deceleration, and nothing can steer wheels that carry no torque. After `drive()` / `move_wheels` (a line-follow), a brake/hold — like the next move — takes the heading the follow REACHED as its target rather than steering back to the pre-follow one. On encoder servos `coast` / `brake` likewise apply to both bridges inside one native call, so the second wheel's 1 kHz control tick can't keep driving while the first is already released.

Pybricks parity by default: the call returns immediately (their stop/brake do too — verified from their source). Since 2.4.0 short moves armed at speed raise their own deceleration to land at rest on target, so waiting is rarely needed; pass `wait=True` to `BLOCK` until both wheels' MEASURED speeds read ~0 — for brake/hold the decel ramp finishing plus settle, for `coast` the physical freewheel decay. `wait=True` raises `ValueError` on open-loop pairs (no measured speed to wait on) and `RuntimeError` if the wheels never settle within the timeout.

done ()

Pybricks-style status check for in-flight `straight(wait=False)` / `turn(wait=False)`. Returns `True` if no move is pending or the active move has reached its target (and `stop(then=...)` has run). Returns `False` while the move is still progressing.

The controller runs the trajectory independently on the hard tick (native path: 1 kHz C scheduler; serial path: the `st_bus` pump); `done()` checks a flag — plus, on the serial path with the gyro enabled, feeds the IMU heading into the hard-tick heading hold. The natural polling cadence is `time.sleep_ms(10)`.

straight (*distance_mm, then=Stop.COAST, wait=True*)

Drive forward by `distance_mm`. 2-DOF coupled.

`then` is forwarded to `stop()` — see its docstring for coast/brake/hold semantics.

`wait=True` (default) blocks until the move completes. `wait=False` returns immediately after arming the move; the caller polls `done()` to check completion, and the `then=dispatch` is deferred until `done()` reports the target was reached. Concurrent use with another `wait=False` move on a separate `DriveBase` (or with `motor.run_angle(wait=False)` calls) is the intended pattern.

Any subsequent move command supersedes the previous pending `wait=False` move (pybricks “new command wins”).

`then=Stop.NONE` (Pybricks `Stop.NONE`) does not decelerate at the end: the move finishes AT cruise speed and the wheels keep it until the next command — chain `straight/curve` segments without stopping between them. A segment too short to reach cruise under `settings.acceleration` hands over at the speed it does reach on its target (the acceleration is never raised to make it).

Raises `RuntimeError` for open-loop motor pairs — moves by distance need feedback; use `drive()/stop()`.

turn (*angle_deg*, *then=Stop.COAST*, *wait=True*)

Turn in place by `angle_deg` body heading (positive = right/clockwise viewed from above, Pybricks convention).

Same `then / wait` semantics as `straight()` — see its docstring — except `then=Stop.NONE`: a turn in place ends facing its target heading, so there is no speed worth carrying.

curve (*radius*, *angle*, *then=Stop.COAST*, *wait=True*)

Drive an arc along a circle of `|radius|` mm, changing heading by `angle` degrees — Pybricks `DriveBase.curve()`, including the parameter names, so Pybricks-style keyword calls (`curve(radius=150, angle=90)`) work verbatim. The one deviation: our `then` defaults to `Stop.COAST` like every openbricks move (Pybricks defaults to hold) — pass `then=Stop.HOLD` for the Pybricks end state.

Positive `angle` turns right (clockwise from above, the system-wide sign convention, same as `turn()`); the SIGN of `radius` picks the travel direction along the arc (positive = forward, negative = backward). `curve(150, 90)` sweeps a forward quarter-circle to the right around a centre 150 mm to the robot’s right; `curve(150, -90)` the mirror to the left.

The forward and turn profiles run simultaneously with proportional speed AND acceleration, so heading stays proportional to distance at every instant — the path is a true circle through the accel/decel ramps, not just at the endpoints. The centre speed is the `straight_speed` setting scaled by $|R| / (|R| + \text{track}/2)$ so the OUTER wheel never exceeds `straight_speed`. `curve(0, angle)` degrades to a turn in place.

Same `then / wait` semantics as `straight()`, including `then=Stop.NONE` — the arc hands its full speed to the next command.

8.2 Parameters

Enumerated options — the Pybricks `pybricks.parameters` pattern. Every API argument that selects one of a fixed set of behaviours takes one of these members, never a string:

```
from openbricks.parameters import Stop, DriveMode

db.straight(300, then=Stop.BRAKE)
db = DriveBase(left, right, 88, 136, drive=DriveMode.WHEEL)
```

A string is rejected at the call with a `TypeError` that names the members, so a typo or a drifted spelling can never silently select a different behaviour. Enumerated parameters — the Pybricks `pybricks.parameters` pattern.

Every API argument that selects one of a fixed set of behaviours takes a member of one of these classes, never a string. A string compares equal to nothing here, so a typo (`then="cost"`) or a drifted spelling (`"continue"` vs `"none"`) fails at the call with a message naming the members, instead of silently selecting a default somewhere downstream. Members print as `Stop.COAST` like Pybricks' `_PybricksEnum`, and `Stop` carries the Pybricks numbering (`COAST=0`, `BRAKE=1`, `HOLD=2`, `NONE=3`), which the native drivebase stop codes share.

Members compare by (enum name, member name) rather than by object identity: the simulator drops the firmware modules from `sys.modules` between runs and re-imports them, so a `Stop.COAST` bound before the reload must still equal the one created after it. A string compares equal to no member either way.

MicroPython ships no `enum` module, so this is the smallest class that gives the same surface: `.name`, `.value`, `repr`, and a class-level member list.

Example:

```
from openbricks.parameters import Stop
db.straight(300, then=Stop.BRAKE)
```

```
class openbricks.parameters.Stop(name, value)
```

Bases: `_Enum`

What a motor or drivebase does when a move ends — Pybricks `Stop`.

- `COAST` — cut torque; the wheel free-wheels.
- `BRAKE` — actively hold zero speed (passively resist on open-loop motors).
- `HOLD` — keep controlling the motor to hold the reached position.
- `NONE` — do not decelerate; the move hands its end speed to the next command (concatenate maneuvers without stopping).

```
BRAKE = Stop.BRAKE
```

```
COAST = Stop.COAST
```

```
HOLD = Stop.HOLD
```

```
NONE = Stop.NONE
```

```
class openbricks.parameters.DriveMode(name, value)
```

Bases: `_Enum`

How a serial-bus wheel servo is driven by the drivebase engine.

- `DUTY` — the servo runs open-loop and the engine's FF+PI loop is the speed controller (the default since 1.89.0).
- `WHEEL` — the servo's own wheel-mode velocity loop closes the speed; the engine commands speed setpoints.

```
DUTY = DriveMode.DUTY
```

```
WHEEL = DriveMode.WHEEL
```

`openbricks.parameters.check` (*enum_cls*, *value*, *param*, *allowed=None*)

Raise `TypeError` unless *value* is a member of *enum_cls* (and of *allowed*, when the call accepts only a subset). The message names the accepted members; a string argument is called out explicitly, since that is the mistake this module exists to catch.

8.3 Motor drivers

Every motor class implements the same `Motor` contract (*Interfaces*), so higher layers — and your code — swap motor types without changes. Speeds are output-shaft degrees per second, angles are degrees, `dc()` duty is -100..100.

```
from openbricks.drivers.st3032 import ST3032Motor

m = ST3032Motor(servo_id=1, uart_id=1, tx=14, rx=6)
m.run_speed(120)           # wheel mode: spin at 120 deg/s
print(m.angle())           # multi-turn accumulated degrees
m.coast()

from openbricks.drivers.jgb37_520 import JGB37Motor

e = JGB37Motor(in1=12, in2=14, pwm=27, encoder_a=18, encoder_b=19)
e.run_angle(360, 720)      # two turns at 360 deg/s, blocking
```

8.3.1 Serial bus servos (recommended)

ST-3032

ST-3032 (Feetech STS3032) serial bus servo.

Smaller sibling of the ST-3215 — same SCS protocol, same 4096-count 12-bit magnetic encoder, same operating modes (0 = position, 1 = wheel), same default 1 Mbps bus. The only differences are mechanical / electrical.

Full datasheet: docs/datasheets/feetech_sts3032.pdf (model ST-3032-C062, Edition A/0 2025-11-30). Headline specs at typical 12 V operation, all $\pm 10\%$ per Feetech:

Working voltage	9–14 V (typ 12 V)	ST-3215 typ 6–12.6 V
No-load speed	0.067 s/60° = 148 RPM = 888 °/s	
	(``ST3032Motor`` defaults ``max_dps`` to this — the ST-3215's protective 600 default capped the wire command well below what the servo can do)	
Stall torque	10 kg·cm (139 oz·in)	ST-3215 ~30 kg·cm
Stall current	1.6 A	
Rated torque	3.3 kg·cm (1/3 of stall)	
Rated current	500 mA	
No-load current	100 mA	
Idle current	20 mA	
Kt	6.3 kg·cm/A	
Size	23 × 12 × 27.5 mm	ST-3215 larger

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Weight	20.6 g
Gear	1/205, coreless motor
Backlash	$\leq 1.0^\circ$
Over-temp shutdown	80 °C
Over-current	> 80 % stall for 2 s
Max position update	1 ms

Because the wire protocol is identical, `ST3032` and `ST3032Motor` are thin marker subclasses of `ST3215` / `ST3215Motor` with no behavioural override. Use them when your bench actually carries an ST-3032 — the typed name documents the hardware and gives us a place to specialise defaults later (lower `max_dps`, torque caps) if hardware testing surfaces a real difference.

ST-3032 and ST-3215 instances on the same UART share the bus registry, so mixing them on one daisy chain is fine — each servo is addressed by its 1-byte ID regardless of model.

△ Voltage rail: an ST-3032 will brown out below ~9 V. Don't share a 6 V bus with ST-3215s; budget a separate 12 V rail.

```
class openbricks.drivers.st3032.ST3032(servo_id, uart_id=1, tx=14, rx=41,
                                         baud=1_000_000, dir_pin=None, min_raw=0,
                                         max_raw=4095, range_deg=360)
```

Bases: `ST3215`

One ST-3032 in position-servo mode (Servo interface).

Behaves identically to `ST3215` — see that class for the full API. Subclassed only to let user code spell the actual hardware in place.

```
class openbricks.drivers.st3032.ST3032Motor(servo_id, uart_id=1, tx=14, rx=41,
                                              baud=1_000_000, dir_pin=None,
                                              invert=False,
                                              steps_per_dps=_DEFAULT_STEPS_PER_DPS,
                                              max_dps=ST3032_NO_LOAD_DPS,
                                              accel_dps2=1500.0, raise_on_stall=False,
                                              stall_idle_ms=1000)
```

Bases: `ST3215Motor`

One ST-3032 in wheel/continuous-rotation mode (Motor interface).

Behaves identically to `ST3215Motor` — see that class for `run_speed` / `angle` / `run_angle` semantics — except for ONE specialised default: `max_dps` matches the ST-3032's datasheet no-load speed (888 °/s) instead of the ST-3215's 600, so the driver clamp can't cap the servo below its own spec.

```
STALL_TORQUE_MNM = 980.0
```

```
KT_MNM_PER_A = 617.9
```

```
NO_LOAD_CURRENT_A = 0.1
```

```
STALL_CURRENT_A = 1.6
```

```
classmethod max_dps_default()
```

The datasheet no-load speed this class defaults `max_dps` to — the reference a measured no-load speed is compared against.

ST-3215

ST-3215 (Waveshare/FeeTech) serial bus servo.

These servos daisy-chain on a half-duplex UART bus: one TX line shared by host and all servos, each servo addressed by a 1-byte ID. Packet format (as in the FeeTech SCServo / Dynamixel-style protocol):

0xFF 0xFF ID LEN INSTR PARAM... CHECKSUM

CHECKSUM = ~(ID + LEN + INSTR + sum(PARAM)) & 0xFF
LEN = number of params + 2

Common instructions:

```

0x01 PING      -> probe the servo
0x02 READ      -> READ reg len
0x03 WRITE     -> WRITE reg value...

```

Key registers (ST-3215):

```

0x21 Operation Mode - 0 = position, 1 = wheel/continuous
0x28 Torque Switch - 1 = enable, 0 = coast
0x2A Goal position (low, high) - int16, 0..4095 over ~360°
0x2E Goal speed (low, high) - sign-magnitude; in wheel mode this
    is the velocity setpoint (bit 15 of high byte = direction)
0x38 Present position (low, high) - read only. 12-bit absolute
    angle within one revolution: 0..4095 over 360°. Wraps to 0
    at every full turn; we accumulate multi-turn revolutions in
    software via a wrap heuristic in ``ST3215Motor.angle``.

```

Half-duplex wiring: most ST-3215 boards use a single data line driven by a TX/RX switching circuit, but MicroPython UART pins are usually separate. If your adapter exposes TX and RX separately, just wire them normally. If you have a true half-duplex bus, you'll need a driver chip or a direction-enable GPIO — set `dir_pin` below.

Two classes here:

- `ST3215` — position-mode (Servo interface), for grippers / lifts / sensor turrets. `move_to(angle)` blocks until reached.
- `ST3215Motor` — continuous-rotation mode (Motor interface), for drivebase wheels. Switches the servo into `mode=1` at construction and exposes `run_speed(dps)` / `angle()` / `brake()` so it drops into `DriveBase` the same way `MG370Motor` does.

The ST-3032 (smaller sibling — 12 V, ~10 kg·cm, same SCS protocol) ships as marker subclasses in `openbricks.drivers.st3032`.

Only a minimal subset of the protocol is implemented here. PR welcome.

```
class openbricks.drivers.st3215.ServoHealth(voltage, temperature, current, flags, status)
```

Bases: `tuple`

current

Alias for field number 2

flags

Alias for field number 3

status

Alias for field number 4

temperature

Alias for field number 1

voltage

Alias for field number 0

```
class openbricks.drivers.st3215.ST3215(servo_id, uart_id=1, tx=14, rx=41,
                                         baud=1_000_000, dir_pin=None, min_raw=0,
                                         max_raw=4095, range_deg=360)
```

Bases: *Servo*

One ST-3215 servo on a shared bus.

move_to(*angle_deg*, *speed*=None, *wait*=True)

Move to *angle_deg* within the configured position range.

speed (raw goal-speed register units) is optional; with *wait*=True polls until within 2% of target or a 3 s timeout.

angle()

Current shaft angle in degrees within the position range, or None if the bus read timed out.

health()

Supply voltage (V), case temperature (°C), supply current (A) and the protection flags, as a *ServoHealth* tuple — see *ST3215Motor.health*. Raises *OSError* if the bus is silent.

ping()

True if the servo answers on the bus.

```
class openbricks.drivers.st3215.ST3215Motor(servo_id, uart_id=1, tx=14, rx=41,
                                             baud=1_000_000, dir_pin=None,
                                             invert=False,
                                             steps_per_dps=_DEFAULT_STEPS_PER_DPS,
                                             max_dps=600.0, accel_dps2=1500.0,
                                             raise_on_stall=False, stall_idle_ms=1000)
```

Bases: *Motor*

One ST-3215 in wheel/continuous-rotation mode.

Implements the openbricks *Motor* interface so it drops directly into *DriveBase*. The servo's internal velocity loop handles closed-loop speed tracking — we just write the setpoint and read the multi-turn accumulated angle.

angle() accumulates in software because the servo's Present-Position register is a 16-bit signed counter that wraps every ~3 turns (4096 counts/rev × ~8 turns to wrap at ±32767). Same wrap-correction shape as *PCNTEncoder*.

STALL_TORQUE_MNM = 2940.0

STALL_LOAD_PCT = 80

STALL_SPEED_DPS = 20.0

dc (*duty*)

True `Pybricks Motor.dc()`: raw duty, no speed regulation. Switches the servo to its open-loop mode (2) and writes duty straight to the output stage — speed sags under load, exactly like a plain DC motor at fixed voltage.

On a motor adopted by the native engine the bus belongs to the hard tick, so duty still maps onto `run_speed` scaled to `max_dps` there (transitional until the engine grows its own duty drive).

speed ()

Measured shaft speed in deg/s from the present-speed register (sign-magnitude, bit 15; steps/s scaled by `steps_per_dps`). Returns `None` if the bus is silent, matching `angle()`. On an adopted motor (native DriveBase) the value comes from the hard-tick pump's widened feedback read — no extra bus traffic.

load ()

Estimated shaft torque in mNm — `Pybricks Motor.load()` shape. The present-load register reports 0.1 %-of-stall units (sign in bit 10, per the Feetech SCServo SDK); scaled by the model's datasheet stall torque (`STALL_TORQUE_MNM`), so treat it as an estimate, not a measurement. `None` if the bus is silent. On an adopted motor the value comes from the hard-tick pump's widened feedback read (user frame — the slot carries this motor's invert).

health ()

Supply voltage (V), case temperature (°C), supply current (A) and the servo's protection flags, as a `ServoHealth` tuple: `flags` holds the set names among `voltage`, `sensor`, `temperature`, `current`, `angle` and `overload` (empty when healthy), `status` the raw byte. From the present-voltage / temperature / status / current registers (0.1 V, °C, 6.5 mA per LSB). Raises `OSError` when the bus is silent — a health check that returns nothing is the failure it exists to catch. On an adopted motor the four reads are staged through the native pump (a few ms — for a log line, not a control loop).

stalled ()

True when the servo is pushing hard (load magnitude at least `STALL_LOAD_PCT` percent of stall) but barely moving (speed magnitude at most `STALL_SPEED_DPS`) — the `Pybricks Motor.stalled()` contract, from the servo's own feedback registers. Raises `OSError` if the bus is silent (a silent bus must not read as “not stalled”).

run_speed (*deg_per_s*)

Set continuous wheel velocity in degrees per second.

brake ()

Ramp to zero velocity and hold (servo's internal loop).

Deliberately RAMPED at `accel_dps2` like every other speed change (user decision, reverting 1.18.1's instant bypass): one uniform rule — the acceleration default governs all commanded speed transitions, brake included. The SAFETY stop is unaffected: the e-stop and `coast()` cut the torque register, which the ramp does not govern, and remain instant. A hard local stop without torque-off is `accel_dps2=0` at construction.

coast ()

Disable torque — wheel free-wheels.

classmethod migrate_bus_to_native (*bus*, *skip=()*)

Move every motor still on `bus` onto native slots.

Called by DriveBase adoption right after it takes the UART: the wheels become slots 0/1 by their own path, and anything else that was sharing the MicroPython bus has to come across too or it is left talking into a closed UART.

hold()

Actively hold the current shaft angle so the position PID resists rotation. Subsequent `run_speed / brake / coast` calls transparently restore wheel mode.

On an adopted motor (native DriveBase) the hold is a per-slot position lock on the hard tick (`st_move_core`) — it corrects disturbances continuously at the bus feedback rate.

Holding never crosses a turn boundary (the shaft is meant to stay put), so the mechanism is chosen to avoid disturbing the servo's turn model:

- If this motor has already been switched to multi-turn step mode (a prior `run_angle`), hold with a zero-count step — step mode keeps holding its current target.
- Otherwise (e.g. a DriveBase wheel that has only ever run in velocity mode), hold in single-turn position mode with `goal=present`. This deliberately does NOT zero the angle-limit registers, so a wheel motor keeps its stock single-turn present-position reads and its odometry stays intact.

done()

Pybricks-style status check for an in-flight `run_angle(wait=False)` move. Returns `True` if no move is in flight (the normal case) or the active move has parked (its remaining-to-target register has counted down to within tolerance). Returns `False` while the move is still running.

Calling `done()` is what advances the move: each call reads the step-mode remaining register once. For a move larger than 7 turns (issued as back-to-back steps), `done()` writes the next step once the current one parks, and on the final step runs the end-of-move `then=dispatch` and clears the pending state. So polling cadence matters for >7-turn moves — the wheel sits idle at each step boundary until the next `done()` advances it. With a typical `time.sleep_ms(10)` poll that's a single-tick gap; if you never poll, a multi-step move stalls after the first step.

On an adopted motor the C controller advances the move on the hard tick AND applies the `then= end-state` itself the tick the move arrives (3.9.0) — whether or not `done()` is ever called; `done()` just checks the arrival flag.

angle()

Return shaft angle in degrees, multi-turn accumulated.

In STEP mode the present-position register reads remaining-to- target rather than absolute position, so we can't derive the angle from it — return the software accumulator, which `run_angle` advances by the executed step counts as each step parks. In wheel/position mode, rebuild the accumulator from the encoder via the wrap heuristic.

reset_angle(angle=0)

Set the current shaft angle to `angle` (degrees).

run_angle(deg_per_s, target_angle, wait=True, tolerance_deg=0.5, kp=None, poll_ms=None, debug=False, then=Stop.COAST)

Rotate by `target_angle` degrees at up to `deg_per_s`, ending within `tolerance_deg` of the target.

On an adopted motor (native DriveBase) the move runs as a per-slot trapezoid position move on the hard tick (`st_move_core`) — same trajectory + arrival semantics as the drivebase's own moves; `tolerance_deg` is fixed at the C core's arrival tolerance there.

`target_angle` is **RELATIVE** and **UNBOUNDED** — `run_angle(200, 360)` rotates one full turn forward, `run_angle(200, 1080)` three turns, `run_angle(200, -540)` one and a half turns back. Direction is the sign of `target_angle`.

Implementation: the servo is driven in **step mode** (`op_mode=3`) for the move. In step mode the goal-position register is a *signed relative step* — write N counts and the shaft advances N counts from where it is, with no single-turn 0/4095 wrap to cross (the prerequisite is `angle_limits = 0`, set once on the first call by `_ensure_step_limits`). This is what makes moves past 180° / past one full turn work: the older single-turn position mode (`op_mode=0`) clamps to 0..4095 and a target across the boundary was executed the *wrong way round*, capping real motion at roughly half a turn.

A single step write covers up to ± 7 turns (the STS multi-turn envelope); larger moves are issued as back-to-back ± 7 -turn steps, each parking before the next — still no boundary to cross. The servo's internal PID handles convergence ($\approx 0.088^\circ$ per encoder count); completion is detected by reading the step-mode *remaining-to-target* register and waiting for it to count down to ~ 0 (see `_read_step_remaining`). The shaft-angle accumulator is advanced by the counts actually travelled so `angle()` / `reset_angle` stay correct across the move.

then selects the end-state, pybricks-style:

- `Stop.COAST` (default) — cut torque; wheel free-wheels. The next `run_speed` / `brake` / `run_angle` transparently re-enables torque and restores the mode it needs.
- `Stop.BRAKE` — restore wheel mode and write `goal_speed=0` so the servo's velocity loop actively holds zero rotation rate.
- `Stop.HOLD` — leave the servo in step mode; its position PID is already holding the target and resisting rotation.

`wait=False` kicks off the move and returns immediately without blocking. Use it for concurrent multi-motor moves, pybricks-style:

```
left.run_angle(60, 720, wait=False)
right.run_angle(60, 720, wait=False)
while not (left.done() and right.done()):
    time.sleep_ms(10)
```

Multi-revolution targets are supported in `wait=False` mode too. For a move within ± 7 turns the whole thing is one step write and `done()` simply reports convergence; for a larger move `done()` issues each subsequent ± 7 -turn step once the previous one parks, so you must keep polling. On this (Python-driven) path the end-state `then= dispatch` is deferred until `done()` reports the final step has converged. On an **ADOPTED** motor (the wheels and task shafts of a native drive base) the C controller applies `then=` the tick the move arrives, polled or not — a `run_angle(..., wait=False)` your program fires and forgets ends in the state it asked for instead of holding under power for the rest of the run (3.9.0).

Any subsequent motion command (`run`, `run_speed`, `brake`, `coast`, `hold`, `run_angle`) supersedes a pending `wait=False` move — the new command takes over and the pending state is dropped (pybricks “new command wins”).

The legacy `kp` / `poll_ms` / `debug` arguments are accepted for back-compat with the velocity-mode implementation but no longer apply — the PID lives on the servo, not in Python.

ping()

True if the servo answers on the bus.

class openbricks.drivers.st3215.**SyncServoGroup**(servos)

Bases: `object`

Coordinated multi-servo writes via SCServo SYNC WRITE.

All servos must share one `_SCServoBus` (same UART). Mixed servo types (`ST3215`, `ST3215Motor`, `ST3032`, `ST3032Motor`) are fine since they all speak the same SCS protocol — SYNC WRITE just blasts the same register on all listed IDs.

Use this whenever you have multiple servos that should apply a setpoint at the same packet boundary (a multi-finger gripper, a multi-axis arm) — each servo receives its byte slot of the broadcast packet at the same instant, instead of N serialised individual writes.

NOT for drivebase wheels: use `DriveBase.move_wheels(left, right)` instead. It gives the same one-packet guarantee, and a `SyncServoGroup` cannot drive adopted wheels at all — a `DriveBase` hands their UART to the native bus driver, so the MicroPython bus this class writes through is closed.

Example

```
from openbricks.drivers.st3215 import ST3215Motor, SyncServoGroup

thumb = ST3215Motor(servo_id=1)
index = ST3215Motor(servo_id=2)
group = SyncServoGroup([thumb, index])

# Both fingers start moving at the same packet boundary -
# one SYNC WRITE instead of two individual writes.
group.set_goal_speeds([200, 200])
```

set_goal_speeds(speeds_dps)

Write goal-speed on every servo in one SYNC WRITE packet.

`speeds_dps` is a list parallel to the servos given at construction. Each servo's own `_encode_goal_speed` is used, so per-servo `invert` / `steps_per_dps` / `max_dps` are respected.

Servos that don't expose `_encode_goal_speed` (i.e. the position-mode `ST3215` class) raise `TypeError`.

8.3.2 DC gear motors with encoders

JGB37-520

JGB37-520 geared DC motor with magnetic quadrature encoder.

This is a 6-wire motor: two power leads (into an H-bridge, typically L298N), Vcc and GND for the hall sensors, and two encoder channels A and B.

Typical spec sheet values (varies by gear ratio variant):

- Encoder CPR at motor shaft: 11
- Gear ratio (example): 1:30 -> output shaft CPR = $11 * 30 * 4 = 1320$ edges (multiply by 4 because we count both edges on both channels)

This class is a thin Python wrapper over the C `Servo` type in `_openbricks_native` (see `native/user_c_modules/openbricks/servo.c`). The wrapper's job is to build the hardware handles (Pin /

PWM / encoder) from pin numbers and pass them to the native servo — the closed-loop control tick runs in C at the `motor_process` tick rate (1 kHz default).

```
class openbricks.drivers.jgb37_520.JGB37Motor(in1, in2, pwm, encoder_a, encoder_b,
                                             counts_per_output_rev=1320,
                                             invert=False, encoder_invert=False,
                                             kp=_DEFAULT_KP)
```

Bases: *Motor*

JGB37-520 DC gear motor with quadrature encoder, closed-loop.

Runs the full *Motor* contract on the native 1 kHz servo core: `run_speed(deg_per_s)`, `run_angle(deg_per_s, target_angle)`, `dc(duty)`, `brake()` / `coast()`, `angle()` / `reset_angle()`, `speed()`. Pair two of them in a *DriveBase* for the coupled 2-DOF chassis controller.

Parameters

- **in1** – H-bridge pins (e.g. L298N IN1/IN2/EN).
- **in2** – H-bridge pins (e.g. L298N IN1/IN2/EN).
- **pwm** – H-bridge pins (e.g. L298N IN1/IN2/EN).
- **encoder_a** – quadrature encoder channel pins.
- **encoder_b** – quadrature encoder channel pins.
- **counts_per_output_rev** – encoder edges per output-shaft revolution ($4 \times \text{PPR} \times \text{gear ratio}$; 1320 for the common 11-PPR 1:30 variant).
- **invert** – flip motor command AND encoder together — for a motor wired backwards end-to-end (typically the mirrored side of a drivebase).
- **encoder_invert** – flip ONLY the encoder reading — for mirror-mounted encoders that count down on motor-forward.
- **kp** – speed-loop proportional gain (native servo core).

Example:

```
from openbricks.drivers.jgb37_520 import JGB37Motor

m = JGB37Motor(in1=12, in2=14, pwm=27,
               encoder_a=18, encoder_b=19)
m.run_angle(360, 720)      # two turns at 360 deg/s, blocking
print(m.angle())
```

dc(*duty*)

Run at a fixed raw duty cycle (-100..100), open loop — `PybricksMotor.dc()`. Non-blocking. This is the pre-1.21.0 `run()`.

speed()

Measured shaft speed in degrees per second — `PybricksMotor.speed()`. Closed-loop drivers implement it (encoder observer / servo present-speed register).

brake()

Stop with active braking (both terminals shorted).

coast ()

Stop by cutting drive power (motor free-wheels).

angle ()

Return the current shaft angle in degrees.

reset_angle (*angle=0*)

Set the current angle to *angle* degrees.

run_speed (*deg_per_s*)

Enter closed-loop speed control with the given target (deg/s).

run_angle (*deg_per_s, target_angle, wait=True, accel_dps2=1500.0*)

Rotate by *target_angle* degrees, following a trapezoidal profile at up to *deg_per_s*.

The native servo builds the profile once and the scheduler samples it at 1 kHz, so acceleration / cruise / deceleration phases are smooth and the move stops cleanly at the target without overshoot (apart from small integration error).

With *wait=False* the scheduler keeps running the profile; the caller can poll `self._servo.is_done()` or eventually call `brake()` / `coast()`.

MG370

MG370 geared DC motor with GMR (giant magnetoresistance) quadrature encoder.

The GMR variant has a 500-PPR encoder on the motor shaft, giving massively more resolution than a standard Hall-effect encoder — at a 1:34 gearbox the output shaft sees $500 * 4 * 34.014 \approx 68028$ CPR. That edge rate is too fast for the software `QuadratureEncoder` (`Pin.irq()` tops out around 5-10 kHz), so this driver uses the native `PCNTEncoder` — a C wrapper over the ESP32 PCNT peripheral — baked into the firmware image.

Every `MG370Motor` instance needs its own PCNT unit — ESP32 has 8, ESP32-S3 has 4. Pick `unit=0` for the first motor, `unit=1` for the second, etc.

Apart from the encoder layer, `MG370Motor` is identical to `JGB37Motor`: same native `Servo` underneath, same control tick, same closed-loop API.

```
class openbricks.drivers.mg370.MG370Motor (in1, in2, pwm, encoder_a, encoder_b,
                                           pcnt_unit=0, pcnt_filter=1023,
                                           counts_per_output_rev=_DEFAULT_CPR,
                                           invert=False, encoder_invert=False,
                                           kp=_DEFAULT_KP)
```

Bases: `Motor`

MG370 DC gear motor with high-resolution GMR encoder, closed-loop.

Same `Motor` contract and native 1 kHz servo core as `JGB37Motor`; the difference is the encoder path — the 500-PPR GMR encoder is counted by the ESP32 PCNT hardware peripheral (`pcnt_unit`), not GPIO interrupts, because its edge rate exceeds what `Pin.irq` can service.

Parameters

- **in1** – H-bridge pins.
- **in2** – H-bridge pins.
- **pwm** – H-bridge pins.

- **encoder_a** – encoder channel pins (into PCNT).
- **encoder_b** – encoder channel pins (into PCNT).
- **pcnt_unit** – PCNT peripheral unit, unique per motor (ESP32 has 8, ESP32-S3 has 4 — first motor 0, second 1, ...).
- **pcnt_filter** – PCNT glitch filter in APB cycles (default 1023 $\approx$ 12.8 μ s; lower it only for encoders faster than ~35 kHz edge rate).
- **counts_per_output_rev** – encoder edges per output-shaft revolution (default matches the 1:34 GMR variant).
- **kp** (*invert / encoder_invert /*) – as in JGB37Motor.

dc (*duty*)

Run at a fixed raw duty cycle (-100..100), open loop — `PybricksMotor.dc()`. Non-blocking. This is the pre-1.21.0 `run()`.

speed ()

Measured shaft speed in degrees per second — `PybricksMotor.speed()`. Closed-loop drivers implement it (encoder observer / servo present-speed register).

brake ()

Stop with active braking (both terminals shorted).

coast ()

Stop by cutting drive power (motor free-wheels).

angle ()

Return the current shaft angle in degrees.

reset_angle (*angle=0*)

Set the current angle to *angle* degrees.

run_speed (*deg_per_s*)

Hold a target speed (closed loop).

run_angle (*deg_per_s, target_angle, wait=True, accel_dps2=1500.0*)

Rotate by *target_angle* degrees at *deg_per_s*. Blocks if *wait*; otherwise returns immediately and the caller polls `done()` to advance the move and detect completion.

8.3.3 H-bridge drivers (open loop)

L298N

L298N H-bridge motor driver.

The L298N drives a brushed DC motor via two direction pins (IN1/IN2) and one PWM pin (EN-A or EN-B). It's open-loop: the class knows nothing about the physical motor speed or position. For closed-loop use, wrap one of these in `jgb37_520.JGB37Motor` (which adds an encoder).

Pinout recap for one channel:

```
IN1  = direction bit A
IN2  = direction bit B
PWM  = enable / speed (tie to VCC for 100%, PWM for speed control)
```

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IN1	IN2	result
---	---	-----
0	0	coast
0	1	reverse
1	0	forward
1	1	brake

```
class openbricks.drivers.l298n.L298NMotor(in1, in2, pwm, invert=False,
                                          pwm_freq=_PWM_FREQ_HZ)
```

Bases: *Motor*

Open-loop brushed DC motor on an L298N H-bridge channel.

No encoder, so only the open-loop subset of the *Motor* contract works: `dc(duty)`, `brake()`, `coast()`. Closed-loop methods (`run_speed`, `run_angle`, `angle`, `hold`) raise `NotImplementedError` — wrap the same H-bridge channel in *JGB37Motor* when the motor has an encoder.

dc(*duty*)

Duty is -100..100 — Pybricks `Motor.dc()`. (Open-loop H-bridge: there is no closed-loop `run(speed)` here; the inherited `run` surfaces `run_speed`'s `NotImplementedError`.)

brake()

Short both terminals — active brake.

coast()

Cut drive entirely — motor spins freely.

TB6612FNG

TB6612FNG dual MOSFET H-bridge — re-exposes `L298NMotor` under the TB6612 name.

Both chips drive each motor channel with the same three signals:

IN1 — direction bit 1 IN2 — direction bit 2 PWM — speed (PWM duty cycle)

...so the openbricks driver is identical. TB6612 is generally the better commodity choice — MOSFETs instead of Darlingtons, ~0.3 V drop instead of ~1.8 V, 3.3 V-logic compatible directly from an ESP32 GPIO, and 3.2 A peak per channel vs L298N's 2 A.

Wiring difference to be aware of: TB6612 has a chip-level **STBY** pin that must be held high for either channel to operate. Tie it to 3.3 V on your breakout, or drive a spare GPIO high at boot — openbricks doesn't model STBY because most breakout modules already pull it up.

8.4 Sensor drivers

All sensors construct against a `machine.I2C` bus (or a *TCA9548A* mux channel, which quacks the same) except two: the GPIO-driven HC-SR04, and the ICM-45686 IMU, which wires over SPI so the firmware can read it inside the 1 kHz control tick.

```
from machine import I2C, Pin
from openbricks.drivers.icm45686 import ICM45686
```

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```

from openbricks.drivers.tcs34725 import TCS34725
from openbricks.drivers.tca9548a import TCA9548A

i2c = I2C(0, sda=Pin(15), scl=Pin(16), freq=400_000) # ESP32-S3 pins
mux = TCA9548A(i2c)

color = TCS34725(mux[0]) # two same-address sensors ...
color2 = TCS34725(mux[1]) # ... on separate mux channels
imu = ICM45686(sck=12, mosi=13, miso=11, cs=17) # SPI, not I2C

print(color.rgb()) # (r, g, b) each 0-255
print(color.ambient()) # clear channel, 0-100
print(imu.heading()) # degrees, CW-positive

```

8.4.1 ICM-45686 (raw IMU, hard-tick heading)

TDK ICM-45686 raw 6-axis IMU — the hard-tick heading source.

Unlike the BNO055 (fused heading over I2C, pumped from Python at ~50-100 Hz during moves), this part is read **INSIDE** the 1 kHz hard tick over SPI (~13 μ s per burst): gyro-Z integrates into a continuous heading in C (`imu_yaw_core` — stationarity-gated bias learning, Pybricks-Prime architecture), and a gyro DriveBase consumes it every millisecond with no Python in the loop.

Wiring: 4 free GPIOs to the breakout's SPI pins (SCLK/MOSI(SDI)/ MISO(SDO)/CS) — the I2C bus, mux and color sensors are untouched. INT/FSYNC/CLKIN stay unwired: the hard tick polls, nothing interrupts.

Example:

```

from openbricks.drivers.icm45686 import ICM45686
from openbricks.drivers.st3032 import ST3032Motor
from openbricks.robotics import DriveBase

imu = ICM45686(sck=12, mosi=13, miso=11, cs=17)
left = ST3032Motor(servo_id=2, uart_id=1, tx=14, rx=41, invert=True)
right = ST3032Motor(servo_id=1, uart_id=1, tx=14, rx=41)
db = DriveBase(left, right, wheel_diameter_mm=88,
               axle_track_mm=138, imu=imu)
db.use_gyro(True) # heading now corrects at 1 kHz in C

```

Calibration: gyro bias learns automatically during stillness (fast at boot rest, slow tracking after). `save_calibration()` persists the learned bias to NVS; the next construction seeds from it (the pbio trick), so boot-and-immediately-run starts corrected.

```

class openbricks.drivers.icm45686.ICM45686(sck, mosi, miso, cs, hz=8_000_000, mode=3,
                                           scale=-1.0)

```

Bases: `object`

heading()

Continuous body heading in degrees, CW-positive — the hard-tick integrator's value. Same contract as `BNO055.heading()` except unwrapped (multi-turn) rather than `[-180, 180)`; DriveBase accepts both.

reset_heading()

Zero the heading frame (calibration is kept).

Refused (`OSError` from the yaw binding) while a `DriveBase` steers by this gyro — Pybricks parity: “Can’t reset heading while gyro in use”. The controller’s measured frame would shift under its held target and the next move veers chasing the difference (bench 2026-08-13: `straight()` after `turn(-90) + reset_heading()` pivoted left). Use `db.reset()` — it re-bases the controller and the heading together — or `use_gyro(False)` first.

gyro()

(x, y, z) rates in dps from the last hard-tick sample.

acceleration()

(x, y, z) in g from the last hard-tick sample.

calibrated()

True once the bias estimator has locked (robot has been still for ~0.5 s since boot, or a saved calibration was loaded).

save_calibration()

Persist the learned gyro bias to NVS so the next boot starts corrected instead of waiting for stillness.

stats()

(reads_ok, read_errors, configured) — hard-tick health.

8.4.2 BNO055 (IMU)

BNO055 — re-export of the native BNO055 C type.

The implementation lives in `native/user_c_modules/openbricks/bno055.c` so the drivebase tick can read `imu.heading()` every ms (when `use_gyro=True`) without a Python frame on the hot path. This module exists so existing user imports stay the same:

```
from openbricks.drivers.bno055 import BNO055
```

The class itself is implemented in C (so the drivebase can read the heading on its 1 kHz tick); its Python-facing API:

```
class openbricks.drivers.bno055.BNO055 (i2c, address=0x28)
```

Bosch BNO055 9-axis IMU in 6-DOF fusion mode (accelerometer + gyro; the magnetometer is deliberately unused — motor magnets and steel in floors bend the local field, and a drive robot only needs *relative* heading). Heading zeroes where the robot points at construction.

Parameters

- **i2c** – `machine.I2C` (or a mux channel).
- **address** – 0x28, or 0x29 on breakouts whose ADR pin straps high.

heading()

Body heading in degrees, wrapped to [-180, 180). CW-positive: turning right (clockwise viewed from above) increases it — compass and Pybricks convention. This is what `DriveBase(imu=...)` reads.

euler()

(heading, roll, pitch) tuple in degrees.

```
angular_velocity()
    (x, y, z) gyro rates in deg/s.

acceleration()
    (x, y, z) accelerometer in m/s2.
```

8.4.3 TCS34725 (color)

AMS TCS34725 RGB + clear light-to-digital sensor.

The TCS34725 returns four 16-bit channels (clear, red, green, blue) over I2C at address 0x29. There's an onboard LED that we leave under user control — some breakout boards wire it to the LED pin on reset, others require GPIO control.

Reference: TCS34725 datasheet (AMS / ams-OSRAM), sections 2.4 and 3.

I2C command byte format (from datasheet):

bit 7 (CMD) = 1 (always for command byte) bits 6:5 (TYPE) = 01 (auto-increment) or 00 (single)
bits 4:0 (ADDR) = register address

```
class openbricks.drivers.tcs34725.TCS34725(i2c, address=_ADDR, integration_ms=2.4,
                                           gain=16)
```

Bases: `ColorSensor`

RGB + clear color sensor, fixed at I2C address 0x29.

Implements the `ColorSensor` contract: `rgb()` raw 16-bit channels, `reflection()` and the calibrated helpers built on it. Two or more on one robot need a [TCA9548A](#) mux (the address is not configurable).

```
raw()
    Return the raw (clear, red, green, blue) 16-bit readings.
```

```
rgb()
    Return (r, g, b) scaled to 0..255 using the clear channel.
```

Dividing by the clear channel normalizes for ambient brightness, so a white object reports roughly (255, 255, 255) at any light level within the sensor's range.

```
ambient()
    Return clear-channel brightness scaled to 0..100.
```

100 means the clear ADC is saturated *for the configured integration time* — 1024 counts per 2.4 ms cycle, capped at 65535 (datasheet “MAX COUNT”). Scale non-linearly would be nicer but keep it simple.

8.4.4 QTR / QTRX (reflectance array)

Pololu QTR / QTRX reflectance sensor arrays (analog outputs).

A row of IR emitter/phototransistor pairs a few millimetres above the mat: dark line = high reading, light mat = low. Unlike a pair of colour sensors, the array gives a CONTINUOUS line position — a weighted centroid across all elements — so a follower steers on a real analog error instead of edge-crossings.

Wiring (QTRX-HD-15A on ESP32-S3): the analog outputs go to ADC1 pins (GPIO 1..10 — ADC2 fights the radios). Any subset of the array's channels works; pass the pins left-to-right as mounted, and set `pitch_mm` to the spacing of the channels you actually wired (4 mm for adjacent QTRX-HD channels, 8 mm if every other one). CTRL (emitter enable) may be tied high or given a pin.

Readings are ratiometric to whatever height and mat you mounted over, so the array **MUST** be calibrated once per session: sweep it across the line while `calibrate()` runs. Reading before calibration raises — an uncalibrated centroid is a plausible-looking wrong number.

Steering is the program's job, from the element readings. The three canonical patterns (`ambient()` is 0 black .. 100 white after calibration):

- Edge following — pick the element the line's edge should sit under: `steer = KP * (50 - r[i].ambient())` for a **RIGHT** edge (element `i` just inside the right edge reads darker as the robot drifts left), `steer = KP * (r[i].ambient() - 50)` for a **LEFT** edge. The error is proportional within about one pitch of that element and rails beyond.
- Centre following — `steer = KP_MM * r.position()` (mm, positive = line right of centre); `position()` is `None` when the line is outside the window, so the program keeps the side it was last seen on and steers that way.
- Intersection — `r.all_dark()`. Branch marker — the far-side elements (the ones beyond the edge you follow) going dark.

Example (the front window on GPIO 1..8, a two-element rear array on GPIO 9/10, each with its own calibration file):

```
from openbricks.drivers.qtr import QTRArray, QTRLineSensor

front = QTRLineSensor(channels=8)          # -28..+28 mm, 8 mm pitch
rear = QTRArray(pins=(9, 10), pitch_mm=8.0)
front.load_calibration("/qtr_front.cal")
rear.load_calibration("/qtr_rear.cal")
while True:
    r = front.read()
    steer = KP * (50 - r[5].ambient())      # right edge under +12 mm
    if r.all_dark():
        break                               # intersection
    if r[6].dark() or r[7].dark():
        pass                                # branch on the far side
    if rear.read().all_dark():
        pass                                # the rear crossed it too
```

class openbricks.drivers.qtr.QTRElement (*value, threshold*)

Bases: `object`

One calibrated element reading, as returned by `QTRArray.read()`.

`value` is the 0 (mat) .. 1000 (line) int; `dark()` / `white()` answer per element exactly like `QTRChannel.dark()`. Deliberately NOT an int subclass: MicroPython cannot reflect-compare int against one, so `max()` over such readings raises on the hub while passing on the desktop — numeric code uses `.value` instead (`max(e.value for e in readings)`).

dark()

True when this element is over the line.

white()

True when this element is over the mat.

ambient()

Reflected brightness as 0 (black) .. 100 (white) — the Pybricks scale, inverted from `value`.

The edge follower's error: `50 - ambient()` is zero exactly when the element straddles the black/white boundary.

class `openbricks.drivers.qtr.QTRReading` (*elements, array*)

Bases: `object`

One snapshot of the whole array, as returned by `QTRArray.read()`.

Behaves like the list of `QTRElement` it holds (`index`, `iterate`, `len`) and carries every aggregate view of exactly this snapshot — so a control tick reads once and derives everything from the same coherent sample:

```
r = qtr.read()
r[0].dark()           # per element
r[7].ambient()        # 0 black .. 100 white (edge error)
r.position()          # global centroid, mm
r.left_edge_position() # the line's left edge, mm
r.right_edge_position() # the line's right edge, mm
r.leftmost_position()  # fork clusters
r.rightmost_position()
r.dark_count()         # how many elements on the line
r.all_dark()           # whole array on the line
r.max()               # brightest value (phantom gate)
```

values()

The plain 0..1000 ints, left to right.

max()

The brightest calibrated value in this snapshot — the follower's off-mat phantom gate.

dark_count()

all_dark()

Every element over the line — with the branch flag dark too, the whole crossing is under the robot (the follower's stop condition).

position()

left_edge_position()

right_edge_position()

leftmost_position()

rightmost_position()

class `openbricks.drivers.qtr.QTRArray` (*pins, pitch_mm=4.0, ctrl=None, dark_threshold=300, positions_mm=None*)

Bases: `object`

Analog QTR/QTRX reflectance array on ESP32 ADC pins.

Parameters

- **pins** – ADC-capable GPIO numbers, LEFT to RIGHT as mounted.
- **pitch_mm** – physical spacing between the wired channels.

- **ctrl** – optional emitter-control GPIO (QTRX CTRL). Driven high at construction (emitters on). `None` = tied high.
- **dark_threshold** – calibrated value (0..1000) above which an element counts as “over the line” for `position()` / `dark_count()`.

Each array owns its pins for the run: a second array (or *QTRChannel*) naming a pin this one already holds is refused at construction, by GPIO number and both arrays. The claims end with the program, so the next run may wire those pins differently. Two arrays on disjoint pins coexist — the front window on GPIO 1..8 and a rear pair on GPIO 9/10 — and each calibrates and stores its own file (`save_calibration()` records the wiring, so the files cannot be swapped by accident).

property **positions_mm**

The element x coordinates in mm, left to right — the same frame as `position()`. Name elements by where they sit: `r[qtr.positions_mm.index(12.0)]` is the element 12 mm right of centre.

calibrate (*duration_ms=3000, poll_ms=5*)

Learn each element’s mat/line extremes.

Sweep the array across the line while this runs (rotate the robot, or slide it by hand). Extends any previous calibration rather than replacing it, so repeated calls refine.

save_calibration (*path*)

Persist the calibration to a file on the hub filesystem, so one sweep (examples/`qtr_calibrate.py`) serves every later run. The wiring is stored with it: a calibration is per- element min/max, so loading it onto different pins would silently mis-scale every reading. One file per array — the front window and a rear pair each keep their own ("`/qtr_front.cal`", "`/qtr_rear.cal`").

load_calibration (*path*)

Load a calibration saved by `save_calibration()`.

Raises with the remedy when the file is missing (run the calibrate script), corrupt, or was recorded for DIFFERENT wiring. Calibration is height- and mat-dependent — resweep after remounting the array or changing mats.

read ()

One calibrated snapshot: a *QTRReading* holding a *QTRElement* per array element, left to right — `r[i].value` 0 (mat) .. 1000 (line), `r[i].dark()` / `r[i].white()` / `r[i].ambient()`, and the aggregate views (`r.position()`, `r.dark_count()`, ...) computed from exactly this sample.

dark_count (*readings=None*)

How many elements are over the line — the intersection / stop-bar signal (a full-width bar darkens most of the array, a branch stub only one side).

position (*readings=None*)

Line centre in mm relative to the array centre; positive = line is to the RIGHT. `None` when no element sees the line — it is outside the window, and the program steers toward the side it last saw the line on (keep that yourself from the sign of the last non-`None` value).

left_edge_position (*readings=None*)

Millimetre position of the line’s LEFT edge — the white→black boundary on the left side of the leftmost dark cluster, in the same frame as `position()` (positive = right of the array centre). `None` when nothing is dark.

The crossing is interpolated between the last white element and the first dark one: the point where the calibrated value passes `dark_threshold`. An edge follower keeps THIS at 0, so the robot straddles the boundary — half the array over mat, half over line — instead of centring on the line itself. That also makes line width irrelevant to steering.

When the leftmost element is itself dark the true edge is off-array to the left; the estimate saturates half a pitch beyond that element, so the error keeps its sign and magnitude instead of vanishing.

right_edge_position (*readings=None*)

Mirror of `left_edge_position()`: the line's RIGHT edge — the black→white boundary on the right side of the rightmost dark cluster, in the same mm frame. `None` when nothing is dark; a dark rightmost element saturates half a pitch off-array to the right.

A right-edge follower keeps THIS at 0 — the mirror-image track discipline of the left-edge follower, same sign convention (the P law is symmetric between the two).

leftmost_position (*readings=None*)

Centroid of the LEFTMOST contiguous dark cluster, in mm (same frame as `position()`), or `None` when nothing is dark.

At a branch the array sees TWO dark regions and the global centroid lands between them — steering into the gap. The leftmost cluster is the left line's own centre: a follower that must take the left fork steers on this while its branch flag says a second line is present. Computed the same way every tick, so switching between the two is jump-free.

rightmost_position (*readings=None*)

Mirror of `leftmost_position()`: the RIGHTMOST contiguous dark cluster's centre — the right fork, for a route policy that takes it.

emitters (*on*)

Drive the CTRL pin (no-op when CTRL is tied high).

class `openbricks.drivers.qtr.QTRChannel` (*pin, dark_threshold=300*)

Bases: `QTRArray`

One reflectance element with the array's calibrate/read contract — for a DETECTOR channel wired apart from the line cluster (a branch / marker flag).

Kept out of `QTRArray` on purpose: a flag element folded into the steering centroid would yank the position toward every marker it passes. Steer on the array; DECIDE on this.

Example (GPIO 9, one of the two ADC1 pins the 8-channel front window leaves free):

```
branch = QTRChannel(pin=9)
branch.calibrate(duration_ms=3000)    # same sweep as the array
if branch.dark():
    ...    # marker under the flag channel
```

value ()

Calibrated reading, 0 (mat) .. 1000 (marker/line).

dark ()

True when the element is over a marker/line.

white()

True when the element is over the mat.

class openbricks.drivers.qtr.QTRLineSensor(*channels=10, dark_threshold=300*)

Bases: [QTRArray](#)

THE bench line sensor: one QTRX-HD-15A window with its geometry pre-wired — construct it, pick an element, follow:

```
qtr = QTRLineSensor()                # ten channels, GPIO 1..10
r = qtr.read()
steer = KP * (50 - r[7].ambient())    # right edge under +16 mm
```

Two layouts (detailed tables in docs/hardware.md), selected with `channels`:

- `channels=10` (default): QTRX channels 1, 3, 4, 5, 7, 9, 11, 12, 13, 15 left-to-right onto GPIO 1..10 in order — a 56 mm window at spacings 8/4/4/8/8/8/4/4/8 mm (the pattern is a palindrome, so board orientation only changes the channel LABELS, never the geometry). Right edge under index 7 (+16 mm), left edge under index 2 (-16 mm).
- `channels=8`: QTRX channels 1, 3, 5, 7, 9, 11, 13, 15 (every other one, 8 mm pitch) onto GPIO 1..8 — the same 56 mm window on eight pins, leaving GPIO 9 and 10 (the last two ADC1 pins on the ESP32-S3) for a SECOND array. Right edge under index 5 (+12 mm), left edge under index 2 (-12 mm).

Both leave two mat-side elements beyond the followed edge for the branch watch. Different wiring: use [QTRArray](#) directly with your own pins/positions.

PINS = (1, 2, 3, 4, 5, 6, 7, 8, 9, 10)

POSITIONS_MM = (-28.0, -20.0, -16.0, -12.0, -4.0, 4.0, 12.0, 16.0, 20.0, 28.0)

PINS_8 = (1, 2, 3, 4, 5, 6, 7, 8)

POSITIONS_MM_8 = (-28.0, -20.0, -12.0, -4.0, 4.0, 12.0, 20.0, 28.0)

8.4.5 HC-SR04 (ultrasonic distance)

HC-SR04 ultrasonic distance sensor.

The HC-SR04 has two pins: `trig` (input — we drive it) and `echo` (output — it pulls high for as long as the round-trip echo took). Sequence:

1. Drive `trig` high for ~10 μ s.
2. Read the duration of the resulting pulse on `echo`.
3. `distance_mm = pulse_us × speed_of_sound_mm_per_us / 2`.

Speed of sound in air at room temperature is ~0.343 mm/ μ s, so a 60 cm target gives ~3.5 ms of echo pulse — well within the 30 ms default timeout.

This driver is pure Python — there's no closed-loop control on a range sensor, so the 1 kHz hot-path concern doesn't apply. The `machine.time_pulse_us` builtin does the actual measurement; typical call latency is ~1 ms (pulse round-trip) which is fine for the cold-path use cases (line-following, wall avoidance, mission “approach until N mm” loops at <100 Hz).

```
class openbricks.drivers.hcsr04.HCSR04 (trig, echo, timeout_us=_DEFAULT_TIMEOUT_US)
```

Bases: *DistanceSensor*

HC-SR04 ultrasonic distance sensor.

Parameters

- **trig** – GPIO pin number wired to the sensor’s TRIG pin.
- **echo** – GPIO pin number wired to ECHO.
- **timeout_us** – how long to wait for the echo. `time_pulse_us` returns -1 past this; we map that to -1 from `distance_mm()` to mean “no return”.

```
distance_mm()
```

Return the round-trip distance to the nearest reflector ahead, in millimetres, or -1 if no echo arrived inside `timeout_us`.

8.4.6 VL53L0X (laser distance)

ST VL53L0X laser time-of-flight distance sensor.

The VL53L0X talks I2C at default address 0x29. A measurement cycle:

1. Write 0x01 to `SYSRANGE_START` (register 0x00).
2. Poll `RESULT_INTERRUPT_STATUS` (0x13) bit 0 until set (typical 33 ms at 33 Hz default rate).
3. Read `RESULT_RANGE_STATUS + 10` (0x14 + 10 = 0x1E) as a 16-bit big-endian integer — the raw distance in millimetres.
4. Write 0x01 to `SYSTEM_INTERRUPT_CLEAR` (0x0B).

Out of range / blocked targets show as 8190 mm in the raw register; we surface that as -1 to match the `DistanceSensor` contract.

This driver implements the minimum register sequence to get usable single-shot readings on a power-on-default VL53L0X. ST’s reference API ships an extensive calibration / VHV / measurement-budget setup that improves accuracy on tuned hardware — that’s a follow-up patch when we have boards to validate against. The default-init ranging here is what most off-the-shelf MicroPython VL53L0X drivers use, and is good for the WRO use cases (line-of-sight, 30–1500 mm).

```
class openbricks.drivers.vl53l0x.VL53L0X (i2c, address=_DEFAULT_ADDR,
                                         timeout_ms=200)
```

Bases: *DistanceSensor*

ST VL53L0X laser time-of-flight distance sensor.

Parameters

- **i2c** – a `machine.I2C` (or compatible) instance.
- **address** – 7-bit I2C address (default 0x29). XSHUT-strapping multiple sensors onto one bus requires assigning each a unique address before constructing — outside this driver’s scope.
- **timeout_ms** – how long to poll for a measurement to finish. Default 200 ms; the chip is typically done in 33–50 ms.

```
distance_mm()
```

Distance ahead in millimetres; -1 if no echo / out of range.

8.4.7 VL53L1X (laser distance, long range)

ST VL53L1X laser time-of-flight distance sensor.

Successor to the VL53L0X with 4 m range (vs 2 m on the L0X) and a different register map. I2C default address 0x29 (same as L0X — XSHUT strapping needed if both share a bus).

The chip-ID lives at 16-bit register 0x010F and reads back 0xEACC for a VL53L1X (or 0xEBAA for VL53L4CD, an L1X-pin-compatible variant).

Like the L0X driver this module ships the minimum register sequence to get usable single-shot ranging on a power-on-default chip. ST's reference API does an extensive calibration / VHV / SPAD-array selection pass for tuned operation; that lives behind a separate `calibrate()` method we'll add when we have hardware to validate against.

Measurement cycle:

1. Write 0x40 to `SYSTEM_INTERRUPT_CLEAR` (16-bit reg 0x0086).
2. Write 0x40 to `SYSTEM_MODE_START` (16-bit reg 0x0087) — kicks a single-shot ranging.
3. Poll `GPIO_TIO_HV_STATUS` (16-bit reg 0x0031) bit 0 until clear.
4. Read `RESULT_FINAL_CROSSTALK_CORRECTED_RANGE_MM_SD0` (16-bit reg 0x0096) as a 16-bit big-endian value — distance in mm.
5. Write 0x01 to `SYSTEM_INTERRUPT_CLEAR` (16-bit reg 0x0086).

VL53L1X registers are 16-bit (vs L0X's 8-bit) — addresses go through two byte-swaps on the I2C wire. The driver wraps that in `_read_u8_16` / `_write_u8_16` helpers.

```
class openbricks.drivers.vl53l1x.VL53L1X(i2c, address=_DEFAULT_ADDR,
                                         timeout_ms=200)
```

Bases: *DistanceSensor*

ST VL53L1X laser time-of-flight distance sensor (4 m range).

Parameters

- **`i2c`** – a `machine.I2C` (or compatible) instance.
- **`address`** – 7-bit I2C address (default 0x29).
- **`timeout_ms`** – how long to poll for a measurement to finish. Default 200 ms; the chip is typically done in ~50 ms.

`distance_mm()`

Distance ahead in millimetres; -1 if no echo / out of range.

8.5 Display & bus drivers

8.5.1 SSD1306 (OLED display)

SSD1306 OLED driver — thin wrapper around `micropython-lib`'s `ssd1306.SSD1306_I2C`.

The `micropython-lib` driver is already battle-tested and subclasses `framebuf.FrameBuffer`. We re-expose its core methods (`text`, `pixel`, `fill`, `show`) explicitly and delegate the rest (`rect`, `line`, `hline`, `scroll`, `contrast`, ...) via `__getattr__`. An extra `clear()` convenience erases the buffer and pushes a blank frame.

The `ssd1306` module is frozen into each board's firmware image (see `native/boards/*/manifest.py`), so `import` works out of the box on flashed hardware.

A Display is an external I2C component, not part of the hub — instantiate one directly wherever you want to draw text / pixels.

class `openbricks.drivers.ssd1306.SSD1306` (*i2c, addr=0x3C, width=128, height=64*)

Bases: `object`

128×64 (or 128×32) SSD1306 OLED on I2C.

text (*s, x, y, c=1*)

Draw string *s* with the 8x8 font at pixel (*x, y*); *c*=1 lit, *c*=0 dark. Buffered — call `show()`.

pixel (*x, y, c=None*)

Set pixel (*x, y*) to *c* (1 lit / 0 dark), or return its current value when *c* is `None`.

fill (*c*)

Fill the whole buffer with *c* (1 lit / 0 dark).

show ()

Push the buffer to the panel (nothing appears until this).

clear ()

Blank the display immediately (fill 0 + show).

8.5.2 WS2812 / WS2812B (RGB LED strip)

WS2812 / WS2812B RGB LED strip driver (NeoPixel protocol).

Targets the common addressable-LED modules — the 8-LED “stick” (WS2812B x8), rings, and cut-to-length strips — on a single data GPIO. The bit-banged 800 kHz protocol itself comes from MicroPython's built-in `neopixel` module; this wrapper adds what user code actually wants on top of it:

- **Brightness scaling.** Raw WS2812s at full duty are dazzling and hot; `brightness=0.2` is a comfortable indoor default (same convention as the hub's onboard status LED). Colors are stored unscaled and multiplied only at `show()`, so `strip[i]` reads back exactly what you assigned and changing `brightness` later re-scales everything on the next `show()`.
- **Buffered updates.** Item assignment only touches the buffer; `show()` pushes the whole strip in one wire transaction — an animation frame is *N* assignments + one `show()`, not *N* flickery writes. `fill()` / `clear()` are one-call conveniences that push immediately.

Usage:

```
from openbricks.drivers.ws2812 import WS2812

strip = WS2812(pin=21, n=8)           # WS2812B x8 stick
strip.fill((0, 60, 0))                # everything green (pushed)
strip[0] = (255, 0, 0)                 # buffer only...
strip[7] = (0, 0, 255)
strip.show()                           # ...pushed together
```

Wiring the x8 stick: DIN → the data GPIO, 5V → 5 V supply, GND → common ground. The ESP32's 3.3 V data line is out of spec for a 5 V-supplied WS2812B ($V_{IH} = 0.7 \times VDD = 3.5$ V) but works with virtually every module in practice; if you see glitches, power the stick from 3.3 V (fine for the small x8 boards) or add a level shifter.

```
class openbricks.drivers.ws2812.WS2812 (pin, n=8, brightness=0.2)
```

Bases: `object`

A strip of *n* WS2812/WS2812B RGB LEDs on one data pin.

show()

Push the buffered colors to the strip (one transaction), applying the current `brightness` scale.

fill (*color*)

Set every LED to `color` (an (*r*, *g*, *b*) tuple) and push immediately.

clear()

All LEDs off, pushed immediately.

property brightness

Global brightness scale 0.0 – 1.0. Assigning re-scales the whole strip on the next `show()` (or right now via `show()` — colors are stored unscaled).

8.5.3 TCA9548A (I2C multiplexer)

TI TCA9548A 8-channel I2C multiplexer / switch.

Some I2C sensors have a fixed address with no way to change it — the TCS34725 colour sensor is stuck at `0x29`, for instance — so you cannot put two of them on one bus without an address collision. The TCA9548A sits between the host and the sensors and fans one bus out to eight electrically-isolated channels (SD0/SC0 .. SD7/SC7); each channel can host its own copy of the same-address device.

The mux itself answers at `0x70` by default (`0x70..0x77` via the A0/A1/A2 address pins). Channel selection is a single-byte write to that address: bit *N* enables channel *N*. Multiple bits enable several channels at once; `0x00` disables all of them.

The drop-in path is `mux[n]`: it returns an object that quacks like a `machine.I2C` bus but transparently selects channel *n* before every operation, so existing drivers construct against it unchanged:

```
from machine import I2C, Pin
from openbricks.drivers.tca9548a import TCA9548A
from openbricks.drivers.tcs34725 import TCS34725

i2c = I2C(0, sda=Pin(21), scl=Pin(22), freq=400_000)
mux = TCA9548A(i2c)
left  = TCS34725(mux[0])    # 0x29 on channel 0
right = TCS34725(mux[1])    # 0x29 on channel 1
```

Reference: TCA9548A datasheet (Texas Instruments), section 8.3.

```
class openbricks.drivers.tca9548a.TCA9548A (i2c, address=_DEFAULT_ADDR)
```

Bases: `object`

8-channel I2C multiplexer for same-address sensors.

Index it like a list: `mux[n]` returns an I2C-compatible handle for channel *n* (0-7) that selects the channel transparently before every transaction — pass it to any driver in place of the real `machine.I2C`.

Parameters

- **i2c** – the upstream `machine.I2C` bus the mux sits on.

- **address** – mux’s own I2C address, 0x70-0x77 via A0/A1/A2 straps (default 0x70).

select (*channel*)

Enable exactly *channel* (0..7), disabling the rest.

disable ()

Disable every channel (control byte 0x00).

8.6 Interfaces

8.6.1 Core interfaces

Abstract interfaces for openbricks components.

MicroPython doesn’t ship full `typing.Protocol` support, so these are plain base classes. Drivers should subclass the appropriate interface and fill in every method. The higher-level modules (`robotics`, `config`) only depend on these interfaces, never on concrete drivers — that’s what makes the system plug-and-play.

If you add a new category of component (e.g. a distance sensor), add its interface here.

class `openbricks.interfaces.Motor`

Bases: `object`

A bidirectional motor.

Implementations range from an open-loop H-bridge driver (L298N) to a closed-loop geared motor with quadrature encoder (JGB37-520).

The method names and semantics follow the Pybricks Prime Motor API (1.21.0): `run(speed)` is degrees per second, closed loop; the raw-duty command is `dc(duty)`. Additional openbricks methods (`coast`, `run_speed`) remain as aliases.

Units

- `duty` is -100..100 (percent duty cycle, sign = direction).
- `speed` is degrees per second at the output shaft (closed-loop only).
- `angle` is degrees at the output shaft (closed-loop only).

run (*speed*)

Run at *speed* degrees per second, closed loop — `Pybricks Motor.run()`. Non-blocking. Concrete: delegates to `run_speed()` (the openbricks alias), so open-loop drivers surface its `NotImplementedError`.

BREAKING (1.21.0): before Pybricks parity this method took percent power. That command is now `dc(duty)` — a script still calling `run(30)` for power gets 30 deg/s instead (slow, not dangerous) or `NotImplementedError` on open-loop drivers.

dc (*duty*)

Run at a fixed raw duty cycle (-100..100), open loop — `Pybricks Motor.dc()`. Non-blocking. This is the pre-1.21.0 `run()`.

stop()

Stop and let the motor spin freely; it gradually stops from friction. `Pybricks Motor.stop()` semantics — the default of the three stop flavours (stop/brake/hold), and a concrete method: it delegates to `coast()`, so every driver gets it for free.

brake()

Stop with active braking (both terminals shorted).

coast()

Stop by cutting drive power (motor free-wheels).

hold()

Stop and actively hold the current shaft angle via closed-loop control. Only motors with position-mode hardware (e.g. ST-3215) or a software position loop implement this; open-loop drivers raise `NotImplementedError` — pick `brake` or `coast` instead.

angle()

Return the current shaft angle in degrees.

reset_angle(*angle=0*)

Set the current angle to `angle` degrees.

run_speed(*deg_per_s*)

Hold a target speed (closed loop).

run_angle(*deg_per_s, target_angle, wait=True*)

Rotate by `target_angle` degrees at `deg_per_s`. Blocks if `wait`; otherwise returns immediately and the caller polls `done()` to advance the move and detect completion.

done()

Return `True` if no non-blocking move is in flight or the active `run_angle(wait=False)` move has reached its target. Drivers that don't support non-blocking moves always return `True` (a `wait=True` call is finished before returning to the caller, by definition).

speed()

Measured shaft speed in degrees per second — `Pybricks Motor.speed()`. Closed-loop drivers implement it (encoder observer / servo present-speed register).

load()

Measured torque at the shaft in mNm — `Pybricks Motor.load()`. Drivers with load feedback (serial servos) implement it; the value is derived from the servo's load register and its datasheet stall torque, so treat it as an estimate.

stalled()

`True` when the motor is pushing as hard as it can but cannot reach its commanded speed — `Pybricks Motor.stalled()`. Drivers with load feedback implement it.

run_time(*speed, time_ms, then=Stop.HOLD, wait=True*)

Run at speed `deg/s` for `time_ms` ms, then stop with the `then` flavour — `Pybricks Motor.run_time()`. `wait=False` is not supported (no background timer is allocated for it); pass `wait=True` or sequence it yourself.

run_target(*speed, target_angle, then=Stop.HOLD, wait=True*)

Run to the ABSOLUTE `target_angle` (degrees, in the `reset_angle` frame) at up to

`speed deg/s` — `Pybricks Motor.run_target()`. Built on the relative `run_angle`: the delta is measured from `angle()` at call time.

`run_until_stalled` (*speed, then=Stop.COAST, duty_limit=None*)

Run at `speed deg/s` until `stalled()`, apply the `then` flavour, and return the angle where it stalled — `Pybricks Motor.run_until_stalled()` (its default `then == Stop.COAST`).

`duty_limit` (percent, $0 < \text{limit} \leq 100$) caps the motor's torque for the duration of the run — the Pybricks gripper- homing pattern: drive gently into the end stop without crushing it. The cap is applied before the motion starts and restored afterwards, stall or not. Drivers opt in via `_duty_limit_push / _duty_limit_pop` (the ST3215/ ST3032 serial servos implement it as a temporary torque- limit register write; their stall detection scales to the cap).

class `openbricks.interfaces.Servo`

Bases: `object`

A position-controlled servo (angle-addressable).

`move_to` (*angle_deg, speed=None, wait=True*)

Move to absolute angle in degrees.

`angle` ()

Read back the current angle.

class `openbricks.interfaces.IMU`

Bases: `object`

A 3-axis inertial measurement unit.

The expected unit convention is:

- heading/yaw/pitch/roll in degrees
- angular_velocity in degrees / second
- acceleration in m / s²

`heading` ()

Return heading (yaw) in degrees, wrapped to [-180, 180).

`angular_velocity` ()

Return (wx, wy, wz) in deg/s.

`acceleration` ()

Return (ax, ay, az) in m/s².

class `openbricks.interfaces.ColorSensor`

Bases: `object`

An RGB-ish color sensor.

`rgb` ()

Return (r, g, b) each in 0..255.

`ambient` ()

Return ambient / clear-channel intensity in 0..100.

8.6.2 Distance sensors

Distance-sensor interface, kept separate from `interfaces.py`.

Why split: `openbricks/__init__.py` eagerly imports the four core interfaces (Motor / Servo / IMU / ColorSensor) so that every `import openbricks` pays them. The observer's variance test runs close to the MicroPython heap ceiling, so adding even a small class to `interfaces.py` tips it over. Distance-sensor users explicitly import this module.

```
class openbricks.distance.DistanceSensor
    Bases: object

    A forward-facing range sensor (HC-SR04 / VL53L0X).

    distance_mm()
        Distance ahead in millimetres; -1 if no echo / out of range.
```

8.7 Hubs

Board-level peripherals — the onboard LED and buttons — behind one `Hub` object per supported devkit.

```
from openbricks.hub import ESP32S3DevkitHub

hub = ESP32S3DevkitHub()
hub.led.rgb(0, 80, 0)           # onboard NeoPixel: dim green
if hub.bluetooth_button.pressed():
    print("button held at boot")
```

Hub — board-level peripherals (status LED, user button) for each supported MCU devkit.

Concrete hubs:

- `ESP32DevkitHub` — ESP32 DevKitC-V4: blue LED on GPIO 2.
- `ESP32S3DevkitHub` — ESP32-S3 DevKitC-1: onboard WS2812 RGB LED on GPIO 48.

Both hubs expect **two** momentary pushbuttons, each wired between a GPIO and GND:

- `bluetooth_button_pin` (classic ESP32 default **GPIO 5**; ESP32-S3 default **GPIO 38** — the S3 has only ten analog-capable pins, ADC1 = GPIO 1-10, and a button doesn't need one) — short-press toggles BLE on/off. Watched by `openbricks.bluetooth_button.BluetoothToggleButton` which the hub auto-wires when `bluetooth=True` (the default).
- The **program button** (default **GPIO 39**) — short-press starts or stops `/program.py`. Watched by `openbricks.launcher` directly, which the frozen default `main.py` starts via `launcher.run()`.

Two pins → no duration-based dispatch. Each button does one thing on every short press.

We deliberately avoid GPIO 0 (BOOT) for either role: holding it during reset puts the ESP32 into the ROM bootloader, so sharing that pin with an application button would turn an accidental power-glitch into a flash-mode drop.

Both hubs auto-wire the BLE toggle button by default (`bluetooth=True`): **constructing a hub** restores the persisted BLE state, installs a short-press handler on the `bluetooth_button_pin`, and — on the S3 — paints the WS2812 blue (BLE on) or yellow (BLE off). While a user program runs, the same LED **flashes** its state colour at 1 Hz as a “robot is running” indicator (plain on/off blink on the classic ESP32's single-colour LED), returning to the solid idle colour when the program stops. Put `hub =`

`ESP32S3DevkitHub()` in your `main.py` to turn that on; omit the call (or pass `bluetooth=False`) to keep the button free for your own use.

External I2C components like SSD1306 OLEDs are **not** part of the hub — they're wired to any I2C bus the user chooses, instantiated directly from `openbricks.drivers.ssd1306` or similar, and used alongside the hub rather than through it.

class `openbricks.hub.StatusLED`

Bases: `object`

A user-visible status LED on the hub.

Plain single-colour LEDs implement `on / off`. Addressable RGB LEDs (WS2812 and friends) additionally implement `rgb`.

on()

Turn the LED on (RGB LEDs restore their last colour).

off()

Turn the LED off.

rgb(r, g, b)

Set colour (each channel 0..255). Raises on non-addressable LEDs.

class `openbricks.hub.Button`

Bases: `object`

A momentary pushbutton on the hub.

pressed()

Return `True` while the button is held down.

class `openbricks.hub.Hub`

Bases: `object`

Board-level peripherals baked into a specific MCU devkit.

led = None

bluetooth_button = None

class `openbricks.hub.SimpleLED(pin, active_high=True)`

Bases: `StatusLED`

Single-colour LED driven by one digital pin.

on()

Turn the LED on (RGB LEDs restore their last colour).

off()

Turn the LED off.

class `openbricks.hub.NeoPixelLED(pin, brightness=0.2)`

Bases: `StatusLED`

Single-pixel WS2812 / NeoPixel driver — the onboard LED on ESP32-S3 DevKitC-1 and most modern ESP32-S3 compact boards.

Unlike `SimpleLED` this one implements `rgb(r, g, b)` too, which is what the Bluetooth-toggle button uses for blue / yellow feedback.

`brightness` scales every channel on write (0.0 – 1.0). 0.2 is a comfortable indoor default — the raw WS2812 at full brightness is dazzling.

on()

Turn the LED on (RGB LEDs restore their last colour).

off()

Turn the LED off.

rgb(r, g, b)

Set colour (each channel 0..255). Raises on non-addressable LEDs.

class `openbricks.hub.PushButton` (*pin*, *active_low=True*)

Bases: `Button`

Digital pushbutton with configurable active level and internal pull.

pressed()

Return `True` while the button is held down.

class `openbricks.hub.ESP32DevkitHub` (*led_pin=2*, *bluetooth_button_pin=5*, *bluetooth=True*)

Bases: `Hub`

ESP32 DevKitC-V4 onboard hub: blue LED on GPIO 2, BLE-toggle button on GPIO 5.

`bluetooth` default `True` wires the button to a short-press BLE toggle and restores the persisted state at boot (see `openbricks.bluetooth_button` / `openbricks.bluetooth`). The on-board LED is single-colour so no colour feedback on toggle, but it still blinks at 1 Hz while a user program runs (dark at idle). Pass `bluetooth=False` to skip the wiring entirely, or `bluetooth_button_pin=<N>` to use a different GPIO. Wire the button between the chosen GPIO and GND; an internal pull-up is enabled.

class `openbricks.hub.ESP32S3DevkitHub` (*led_pin=48*, *bluetooth_button_pin=38*,
brightness=0.2, *bluetooth=True*)

Bases: `Hub`

ESP32-S3 DevKitC-1 onboard hub: WS2812 RGB LED on GPIO 48, BLE-toggle button on GPIO 38.

`led_pin` defaults to 48 (the DevKitC-1 onboard WS2812). Pass `led_pin=None` to disable the LED entirely, or any other GPIO for a WS2812 wired elsewhere. `brightness` (0.0 – 1.0) scales channel values on write; default 0.2 is comfortable indoors.

`bluetooth_button_pin` defaults to 38 (was 5 until 1.66.3): the S3 has exactly ten analog-capable pins (ADC1, GPIO 1-10) and a button needs none of them — parking the default there cost an analog channel the moment a sensor array arrived. GPIO 38 is free, non-strapping and digital-only. Wire a momentary switch between GPIO 38 and GND; the pin is configured `Pin.IN` with `PULL_UP` so no external resistor is needed. Override via `bluetooth_button_pin=<N>` (a hub wired to the old default passes 5).

8.8 Tools

Small user-facing utilities mirroring `pybricks.tools`. Small utilities mirroring the `pybricks.tools` module.

`openbricks.tools.wait (milliseconds)`

Pause for milliseconds ms.

class `openbricks.tools.StopWatch`

Elapsed-time stopwatch in milliseconds.

`time()`

`reset()`

`pause()`

`resume()`

8.9 Runtime services

These modules run behind the scenes on the hub — the frozen `main.py` wires them up at boot. They're documented here because their behavior (button semantics, log rotation, BLE persistence) is user-visible.

8.9.1 Reserved-GPIO guard

Reserved-GPIO guard: fail pin misuse at construction time, loudly.

Drivers that take raw GPIO numbers call `check()` before touching `machine.Pin`. Two kinds of mistakes are caught:

- **Chip-reserved pins** — pins that can never work for user wiring on the running chip: non-existent numbers, the SPI-flash pins, the ESP32-S3's native-USB pair, and (for output roles) the classic ESP32's input-only range. Claiming these either crashes the chip, bricks the USB connection, or fails somewhere far from the real mistake; the guard names the pin and the reason instead.
- **Pins the firmware runtime already owns** — the program button, the Bluetooth-toggle button, and the status LED register themselves via `claim()` when the hub / launcher wire them at boot. A driver constructed on one of those pins gets an error naming the owner ("in use as the program button") plus the constructor argument that moves the owner elsewhere. A driver's own claims (a QTR array's pins) last for the program that made them: the launcher releases them at the next program boundary.

Chip detection reads `os.uname().machine` and recognizes the two supported targets (ESP32, ESP32-S3). Off-chip — CPython tests, the MuJoCo sim, unix MicroPython — no chip is detected and the chip-reserved rules are skipped (the runtime-claims check still applies); pins there are fakes, not wiring. Tests pin a chip explicitly with `set_chip()`.

Strapping pins (0/3/45/46 on the S3) are deliberately *not* blocked: they work as regular GPIOs after boot and are routinely used. The wiring guides in `docs/hardware.md` cover the soft cases.

exception `openbricks.pins.ReservedPinError`

A GPIO was requested that can't (or shouldn't) be user-wired.

`openbricks.pins.set_chip(chip)`

Force the chip model the guard validates against.

`chip` is "esp32", "esp32s3", or `None` to return to autodetection. Used by tests; harmless elsewhere.

`openbricks.pins.claim(pin, role, hint="", program=False)`

Record that the firmware runtime owns `pin` (e.g. a button).

`program=True` marks a claim a driver makes for the running program only: `release_program_claims()` hands it back when the program ends, so the next program may wire that pin anew.

`openbricks.pins.release(pin)`

Forget a claim — for callers that hand a pin back (e.g. after `hub.bluetooth_toggle.stop()`). Unknown pins are a no-op.

`openbricks.pins.release_program_claims()`

Forget every claim made with `program=True` — the launcher calls this at the program boundary. Boot-time claims stay.

`openbricks.pins.check(pin, role, output=True)`

Validate that `pin` is usable for `role` on this chip.

Parameters

- **pin** – GPIO number the caller is about to wire.
- **role** – human-readable purpose (“L298N IN1”, “HC-SR04 echo”) — appears in the error message.
- **output** – the pin must drive (`True`) or only read (`False`). Only matters on the classic ESP32, where GPIO 34-39 are input-only.

Raises

`ReservedPinError` – with the pin, the role, and the reason.

8.9.2 Program launcher

Button-gated user-program launcher.

Pybricks-style workflow:

- `openbricks upload` stages a script at `/program.py` but does not run it — the user presses the program button to launch.
- `openbricks run` stages the same script and triggers the launcher immediately. Output streams back to the client; pressing the program button stops the program; when the program stops, the terminal exits.

Each press is a full press-release cycle. The program button has its own GPIO (default 39), separate from the BLE-toggle button watched by `openbricks.bluetooth_button` (default 38). Two pins → no duration-based dispatch — every press on the program pin means start-or-stop, and every press on the BLE pin means toggle-BLE.

Wiring:

- Press while idle → start `/program.py` (on release, with a post-stop lockout against bounce).

- Press while running → the stop fires on press-DOWN: the e-stop latch engages (motors halt + motion commands raise — see `openbricks.estop`), and a `KeyboardInterrupt` injection is requested and *retried* until the program is actually dead.

The watcher runs off a `machine.Timer` kept alive for the whole hub uptime (we never `deinit` it), so `button-press-to-run` survives `openbricks run` interrupting the main idle loop.

Typical `main.py` (the firmware ships a frozen default; users can override by writing to `/main.py` in VFS):

```
from openbricks import bluetooth, launcher
bluetooth.apply_persisted_state()
launcher.run()
# installs watcher + blocks on the idle loop
```

```
class openbricks.launcher.Launcher (button, program_path=DEFAULT_PROGRAM_PATH,
                                     poll_ms=DEFAULT_POLL_MS)
```

Shared state for the program-button watcher.

Tests instantiate this directly and drive `_tick` with a fake `Pin`; production code uses `_ensure_launcher()` below, which installs a singleton + `machine.Timer`.

```
START_LOCKOUT_MS = 500
```

```
STOP_RETRY_MS = 300
```

```
STARVE_NOTE_MS = 5000
```

```
DEBOUNCE_TICKS = 2
```

```
RUN_START_GRACE_MS = 400
```

```
START_PRESS_OPEN_MS = 600
```

```
RELEASE_CHATTER_MS = 500
```

```
note_external_stop()
```

Attribute a stop delivered OUTSIDE this watcher's own machinery — the hard-button path, or a REPL Ctrl-C.

The 1.48.2 start gates check state (`_lockout_until_ms`, press lifecycle) that only the watcher's stop path armed. When the hard path wins the race (stop in ~2 ms, before the watcher's next 50 ms tick), that state never arms: the stopping press's echoes — PCNT edges, the hard latch's post-disarm confirmation — read as fresh idle presses and dispatch a phantom start (bench 2026-08-03, second occurrence: gates in place, lockout never armed, next BLE session dead again). Called from the program teardown, so EVERY interrupt-unwound run arms the same suppression.

```
openbricks.launcher.dump_events()
```

Print the launcher button-event ring, oldest first.

```
openbricks.launcher.program_running()
```

True while a user program is executing — whatever started it (button press, `openbricks run`, or a scheduled start; all paths maintain the same flag).

This is the hub-wide “robot is running” signal. The BLE toggle watcher polls it from its 50 ms tick to flash the status LED while a program runs (`openbricks.bluetooth_button`), which is why it must stay cheap: one attribute read, no allocation.

```
openbricks.launcher.run(program_path=DEFAULT_PROGRAM_PATH,
                        button_pin=DEFAULT_BUTTON_PIN, poll_ms=DEFAULT_POLL_MS,
                        timer_id=0)
```

Install the button watcher and block on the cooperative drain loop. Called from the frozen `main.py`.

`timer_id=0` is the first ESP32-S3 hardware timer. The previous default `-1` (virtual timer) was supported by older MicroPython builds but raises `ValueError: invalid Timer number` on the v1.27+ MP we vendor — esp32-s3 only exposes hardware timers 0..3.

Intentionally blocks forever. If `openbricks run` later sends a Ctrl-C over the REPL to interrupt this loop, the Timer stays alive (we never `deinit` it) so subsequent `run_program` / `button-press` start continue to work.

```
openbricks.launcher.run_program(program_path=DEFAULT_PROGRAM_PATH)
```

Client-triggered entry for `openbricks run`.

Sets the `_running` flag, then `exec`'s the program in the main thread (`_exec_program_raw` arms the native stop button for the duration). Propagates `KeyboardInterrupt` so the raw-REPL disconnect signals “stopped” back to the client (which then exits).

The program-boundary native-state wipe (motor_process callbacks + `st_bus` runtime) happens inside `_exec_program_raw`, shared with the button-press paths.

8.9.3 Run logs

Per-run log capture: `tee print(...)` output to a file on flash so untethered runs can be inspected later via `openbricks log`.

The launcher wraps every program execution with `log.session()` so the user's `print` output streams to *both* the live USB / BLE console (when one's listening) and a rotating file on flash. With nobody listening on the live channel, the file is the only record. `openbricks log` reads the most recent files back over BLE.

Storage layout:

```
/openbricks_logs/run_0.log
/openbricks_logs/slot_1.log
/openbricks_logs/slot_2.log
```

Each run gets the next index; the index lives in the file's header line ("`<epoch> -- run_N --`"), NOT the filename. The `MAX_RUNS` slot files are reused in place (`run N` overwrites `slot_(N % MAX_RUNS)`) — the earlier delete+create rotation churned littlefs directory metadata until every commit's allocator traversal cost ~400 ms (bench 2026-08-09); truncate-reuse keeps commits at fresh-filesystem cost. Flash usage stays bounded and indices grow monotonically, exactly as before.

Every line is prefixed with a raw int64 **UTC Unix epoch in milliseconds** (e.g. `1783950123456 left ambient: 33`). No formatting, no timezone on the hub — the host CLI converts to the user's local time at display. The ESP32 RTC starts at 2000-01-01 on power-up; the CLI syncs it from the host clock on every connect, so runs started after any `openbricks run / log / upload` carry real wall-clock stamps (an unsynced run shows year-2000 dates, which is self-diagnosing).

The session is also bytes-capped: once a run's log file passes `MAX_BYTES` bytes, further writes are dropped from the file (the live console still gets them). This keeps a runaway `while True: print(...)` from filling the entire flash partition.

Implementation note: MicroPython doesn't expose `sys.stdout` as a re-bindable attribute on every port, so we tee at the `builtins.print` level instead. This catches every `print(...)` call — including ones with `file=sys.stderr` — but does not catch direct `sys.stdout.write()` calls. User code on the firmware path overwhelmingly goes through `print()`, so this trade-off is fine. The launcher additionally calls `log.write_text(...)` from its exception handler so tracebacks are captured.

`openbricks.log.note(text)`

Write one stamped line to the active run's log file.

No-op when no program is running (there is no file to write to). File only — the live console is deliberately not touched; callers that want a console message print one themselves. Used by the launcher so every button press that starts or stops a run leaves a timestamped entry in that run's log.

`openbricks.log.worst_write_ms()`

Slowest single filesystem call of the active session, in ms. 0 when nothing has been written or no session is active. The launcher includes this in its tick-starvation note: if the two numbers are of the same order, littlefs (block erase under a write) owns the starvation; if this stays small while the gaps are large, the blocker is elsewhere.

`openbricks.log.pump()`

Drain the active session's buffered output to its file.

Called from the launcher's Timer tick, which is what makes logging asynchronous: `print` appends to RAM and returns, this moves the bytes to flash off the program's hot path. No-op when no program is running. Never raises — the tick that calls this also owns the stop button.

`openbricks.log.flush()`

Commit the active session's buffered output NOW.

For the moments durability beats speed: the stop button firing, or any other point where the next thing that happens might be a reset. No-op when no program is running.

`openbricks.log.session()`

Construct a fresh `_LogSession`.

Use as a context manager:

```
with log.session() as sess:
    run_user_program()
    # sess.write_text(extra) for non-print output if needed.
```

`openbricks.log.list_runs()`

Return a list of `(index, full_path)` tuples, oldest first. Used by the on-hub helper that `openbricks log` invokes via raw-paste to enumerate available runs. Slot files are listed by their header index; legacy `run_N.log` files (pre-slot firmware) still appear until the next run's migration removes them.

`openbricks.log.read_run(index)`

Read a single run's log by index. Raises `OSError` if no such run exists (the slot was reused by a newer run, or the index never happened).

8.9.4 Bluetooth (BLE REPL & console)

Persistent Bluetooth on/off state for the hub.

The firmware ships with BLE compiled in (so `mpremote-over-BLE` code transfer works). At runtime we let the user toggle the advertising stack on and off; the choice is persisted in NVS so reboots don't lose the state. Default (no key in NVS yet) is **on**.

Typical usage from `main.py` right after the firmware boots:

```
from openbricks import bluetooth
bluetooth.apply_persisted_state()
```

Then toggle programmatically (or from the hub's button — see `openbricks.hub.ESP32DevkitHub` once that integration lands):

```
bluetooth.toggle() # flip current state, persist, apply
bluetooth.set_enabled(False) # explicit off
```

Activating BLE requires a hub name (`openbricks.HUB_NAME`) — flash the image with `scripts/flash_firmware.py --name NAME` first. We refuse to advertise with no name rather than defaulting to a shared value, since two hubs with the same advertising name can't be individually addressed.

`esp32.NVS` and `bluetooth.BLE` are imported lazily so desktop tests running under unix MicroPython don't need to have the firmware-only modules present — they install fakes in `tests/_fakes_ble.py`.

exception `openbricks.bluetooth.HubNameNotSetError`

Raised when the firmware was flashed without `--name NAME`.

`openbricks.bluetooth.is_enabled()`

Return the persisted flag (`True` if no value has ever been written).

`openbricks.bluetooth.set_enabled(enabled)`

Persist enabled and apply it to the BLE stack immediately.

Raises `HubNameNotSetError` when turning BLE on with no hub name flashed. Turning BLE off is always allowed — a nameless hub can still be silenced.

When enabling, also starts the NUS REPL bridge (`openbricks.ble_repl`) so `openbricks run / stop` can push scripts over BLE. When disabling, the bridge is torn down first so `dupterm` isn't left pointing at an inactive stack.

`openbricks.bluetooth.add_state_listener(callback)`

Register `callback(enabled)` to run after each state change. Adding the same callback twice is a no-op.

`openbricks.bluetooth.remove_state_listener(callback)`

Unregister a callback added with `add_state_listener()`. Unknown callbacks are a no-op.

`openbricks.bluetooth.toggle()`

Flip the current state. Returns the new (post-toggle) value.

`openbricks.bluetooth.apply_persisted_state()`

Read the persisted flag and apply it to the BLE stack.

Call this at boot (e.g. top of `main.py`) so the BLE radio reflects whatever the user last chose before the reboot — or comes up enabled on the very first boot.

Tolerant of a missing hub name: if the persisted state says BLE-on but no hub name is flashed (typical on a freshly-flashed chip before the first `openbricks flash --name ... run`), prints a one-line warning and skips activation. Better than raising `HubNameNotSetError` from `main.py`,

which (per the 1.0.4 silent-boot bug) could lose its traceback to USB-Serial-JTAG timing and brick the boot.

Nordic UART Service bridge for the MicroPython REPL over BLE.

Vendored from upstream MicroPython's `examples/bluetooth/ble_uart_peripheral.py` (the `BLEUART` class) and `ble_uart_repl.py` (the `BLEUARTStream` dupterm adapter). Both files are MIT-licensed under MicroPython's project LICENSE. The one structural change is the advertising payload — upstream advertises only the device name + appearance; we add the NUS 128-bit service UUID so clients filtering by service can find us. Everything else (connection-set tracking, append-mode rx buffer, scheduled-flush write batching, `io.IOBase` inheritance) is the upstream pattern, unmodified.

Why vendored: writing this from scratch using only the docs (the 1.0.0-1.1.1 history) produced a long parade of “invisible until hardware” bugs — each fix required a hardware reflash + paste diagnostic round. Starting from upstream's working example would have caught all of them at once.

Public surface (what `openbricks.bluetooth.apply_persisted_state` calls):

- `start()` — bring up the NUS bridge. Idempotent.
- `stop()` — tear down. Idempotent.
- `is_running()` — query.

`openbricks.ble_repl.host_active(window_ms=HOST_ACTIVE_MS)`

True while a BLE central has written to the hub within `window_ms` — an upload, a run's staging paste, a keystroke.

`openbricks.ble_repl.log_entries()`

The recorded events, oldest first (unwraps the ring).

`openbricks.ble_repl.dump_log()`

Print the in-memory BLE event log over USB-Serial-JTAG.

Why a helper rather than `print(_LOG)`: `print` routes through dupterm which routes through this module's stream, so a long log print adds entries to itself while running. Iterating + printing item by item is bounded; the bytes added during the print sit in `_tx_buf` until later and don't grow the log.

`openbricks.ble_repl.clear_log()`

`openbricks.ble_repl.is_running()`

`openbricks.ble_repl.pump_tx()`

Re-arm the TX flush if bytes are sitting in the buffer.

Liveness backstop, called from the launcher's poll tick. A flush chain dies in two ways: `micropython.schedule`'s queue was full at re-schedule time, or a notify failed and the retry is deliberately paced (see `_flush`). In both cases the buffered bytes wait for the *next* `write()` — which never comes when the writer already finished. That was the `openbricks log` end-of-dump stall: the file content and the raw-REPL terminator sat in `_tx_buf` forever while the host timed out. The pump gives the chain a fresh start at tick cadence; it is idempotent and no-ops when the buffer is empty or the bridge is down.

`openbricks.ble_repl.start()`

Bring up the NUS bridge. Idempotent — second call is a no-op if already running.

Caller must have `ble.active(True)` before calling. Hub name comes from `openbricks.HUB_NAME` (NVS-backed); we refuse to advertise without one.

```
openbricks.ble_repl.stop()
```

Tear down the NUS bridge. Idempotent.

8.9.5 BLE toggle button

Short press on the BLE-toggle button toggles BLE on/off.

Wires a `machine.Timer`-driven poll loop (default 50 ms) against a `Button`-conformant object and calls `openbricks.bluetooth.toggle()` once per press-release cycle. State is persisted via NVS by the `bluetooth` module, so the new value survives reboots.

The same poll loop doubles as the **run indicator**: while a user program is executing (`launcher.program_running()`), the status LED flashes at 5 Hz instead of holding a solid colour — blue when BLE is on, yellow when it's off, plain on/off blinking on single-colour LEDs. When the program stops, the LED returns to its idle state (solid state colour on RGB hubs, dark on single-colour hubs). It renders the **transfer indicator**: while a host tool is writing to the hub over BLE (`openbricks.upload/run` staging a program — `ble_repl.host_active()`), the LED flashes purple at 10 Hz, outranking the run blink. And it renders the **press acknowledgment**: every program-button press flashes the LED for a moment — red for the press that starts a run, green for the press that stops one (`notify_press()`, called by the launcher's press detectors) — outranking both.

This is a different physical button from the one `openbricks.launcher` watches for program start/stop — the BLE toggle lives on its own GPIO (default 38 on the S3, 5 on the classic ESP32 — see [openbricks.hub.Hub](#)) while the program button is on GPIO 39. Two pins → no duration-based dispatch, every press on this pin means “flip BLE”.

Usage from `main.py`:

```
from openbricks import bluetooth
from openbricks.bluetooth_button import BluetoothToggleButton
from openbricks.hub import ESP32DevkitHub

bluetooth.apply_persisted_state()
hub = ESP32DevkitHub()
BluetoothToggleButton(hub.bluetooth_button).start()
```

The helper is deliberately standalone (not baked into `Hub`) so tests can exercise it in isolation, and so boards without a button — or users who want to drive the toggle from something other than a physical press — can skip it.

```
openbricks.bluetooth_button.notify_press(stop=False)
```

Record a program-button press. `stop=True` marks it as the press that stops a run (green flash); the default is a start press (red). The active `BluetoothToggleButton` renders it on its next poll tick. Safe from any context — it only sets two module variables.

```
class openbricks.bluetooth_button.BluetoothToggleButton(button, led=None,
                                                         poll_ms=DEFAULT_POLL_MS,
                                                         timer_id=1,
                                                         color_on=DEFAULT_COLOR_ON,
                                                         color_off=DEFAULT_COLOR_OFF,
                                                         program_running=None,
                                                         host_active=None)
```

Polls a button and toggles BLE on each press-release cycle.

Optional RGB LED feedback (blue = BLE on, yellow = off). Call `start()` to begin polling on a `machine.Timer`; `stop()` releases the timer. The toggled state persists across reboots.

start()

Begin polling. Safe to call repeatedly — the second call is a no-op.

On first call, paints the LED (if one was provided) to reflect the current persisted BLE state, and registers with `bluetooth.add_state_listener` so the LED also follows state changes made *without* the button — `bluetooth.set_enabled` from user code or a tool over the REPL.

stop()

Stop polling, release the timer, and unregister the LED state listener.

ARCHITECTURE

A short tour of how `openbricks` is organized and why. If you’ve read Pybricks’ `pbio` codebase, a lot of this will look familiar — the layering is borrowed directly, and for the same reason: `openbricks` ships as a **custom MicroPython firmware**, not a library you install on top of stock MicroPython. Pybricks does exactly this for LEGO hubs; we do it for commodity MCUs.

Owning the firmware shapes several decisions:

- Background control loops (`MotorProcess`) can run always-on off a hardware timer — nobody else is contending for that peripheral.
- Platform selection means picking which firmware image to flash, not runtime-dispatching between adapters.
- Hot control code can be compiled in as a native C extension later without a separate install step.
- We can extend or add machine-level primitives (custom timers, a hub abstraction) because we build the `machine` module.

9.1 Four layers

User code	(<code>main.py</code> , <code>robotics.DriveBase</code> , ...)
Abstract interfaces	(<code>Motor</code> , <code>Servo</code> , <code>IMU</code> , <code>ColorSensor</code>)
Concrete drivers	(<code>st3032</code> , <code>icm45686</code> , <code>tcs34725</code> , ...)
MicroPython HAL	(<code>machine.Pin</code> , <code>I2C</code> , <code>UART</code> , <code>PWM</code>)
openbricks firmware image — custom MicroPython build for this specific MCU, with all the above baked in	

The two middle layers are what make this different from “a pile of MicroPython scripts.” Interfaces (`openbricks/interfaces.py`) define the contract each family of component obeys; drivers implement that contract; everything above the interface line depends only on interfaces, not on specific chips. That’s why swapping a JGB37-520 DC motor for an ST-3032 serial servo only changes the driver you instantiate — the `DriveBase` class asks for “a `Motor`” and doesn’t know or care what’s underneath.

This is the same split Pybricks has: `pbio/include/pbio/*.h` is the interface, `pbio/src/*.c` is the library, `pbio/drv/*` is the driver layer. We take the same approach in C — `native/user_c_modules/`

`openbricks/` holds the hot control code that runs at the scheduler tick rate. Targeted pbio-parity on control quality is the reason that code is C and not Python.

9.2 Pybricks-parity control, in C

All four of the big pbio control-quality items are ported and shipped in `native/user_c_modules/openbricks/`. Each corresponds to a pbio source file and keeps its structure close — pbio is MIT-licensed so the ports are direct where they can be.

1. **State observer** (`pbio observer.c`) — our `observer.c` is a two-state α - β filter. Less capable than pbio's full model-based observer (no motor model, no PWM coupling, no current/flux estimation) but a $\sim 60\times$ variance reduction over raw finite-differencing for little code. Upgrading to a model-based observer is later roadmap work.
2. **Trajectory planning** (`pbio trajectory.c`) — our `trajectory.c` computes trapezoidal (and triangular fall-through) speed profiles with explicit accel / cruise / decel phases. `servo.run_target()` and `DriveBase.straight()` / `.turn()` sample it each tick.
3. **Cooperative multitasking** (`pbio motor_process.c + os.c`) — our `motor_process.c`. Always-on 1 kHz tick off a `machine.Timer`. Native subscribers (`Servo`, `DriveBase`) register via a fast C-function-pointer path ($\sim 1\ \mu\text{s}/\text{tick}$); Python callables are still accepted on a slower dispatch path for user extensibility. Honesty note: on `esp32`, `machine.Timer` callbacks are dispatched through `micropython.schedule`'s bounded queue — a main thread blocked in one long C call delays or silently drops ticks, so 1 kHz is nominal, not guaranteed (unlike pbio, whose loop runs under the VM). Since 1.37.0 the controllers' **clock** is immune to that: on real hardware the tick advances `now_ms` by measured `mp_hal_ticks_ms` deltas (wall time, enabled by the frozen `boot.py`), so dropped ticks cost control updates but no longer dilate trajectory time. Since 1.38.0 the firmware also carries a **hard tick** (`native/patches/esp32-openbricks-hard-tick.patch`): a periodic C hook on the `esp_timer` service task, below the Python scheduler entirely — verified on hardware via `motor_process.hard_tick_selftest()` / `hard_tick_count()`, and watched from then on by `hard_tick_stats()` (3.3.0: fires, late fires, worst gap in μs — the regression alarm for the next peripheral that joins the tick; a fire more than $250\ \mu\text{s}$ past its 1 ms slot counts as late). Existing controllers still dispatch through the scheduler (their encoder reads call into Python objects, which the hard context must never do); the serial-bus motor path lives there: since 1.45.0 `DriveBase` (`openbricks.robotics`) **adopts** serial-bus Motor objects transparently onto the hard-tick engine — it releases their `machine.UART` (explicit ownership handover, no peripheral double-claim) and runs the 2-DOF controller inside the hard tick on serial-bus servo slots ($\sim 220\ \text{Hz}$ odometry per wheel, floor-verified 0.3% odometry closure on a square), with `use_gyro(True)` fed by the ICM-45686 read inside the hard tick itself (1 kHz heading correction, no Python in the loop; bench-verified $+0.6^\circ$ over a four-turn square) — or, for an I2C IMU like the BNO055, by a Python outer loop at ~ 50 -100 Hz. Since 1.89.0 the wheels themselves run “dumb mode” by default: the servo is switched to its open-loop duty mode and the engine's own integer FF+PI closes the speed loop over raw duty sync-packets — every layer of the drive loop is openbricks code (`DriveBase(..., drive=DriveMode.WHEEL)` restores the servo's internal speed controller). There is exactly ONE drivebase class and NO Python control loop: on a runtime with neither the native bus nor the sim's emulated bus, constructing a serial-bus `DriveBase` raises instead of silently degrading. The sim emulates the `st_bus` surface (`_SimStBus` over `MuJoCo` wheels), so the same controller code path runs everywhere.
4. **Drivebase coupling** (`pbio drivebase.c`) — our `drivebase.c` runs two coupled controllers in (sum, diff) coordinates with position feedback on both. Exit criterion: asymmetric-friction test (one wheel at $0.9\times$ commanded speed) keeps heading error under 5% of forward distance — the pure-Kp M1 fallback fails it.

9.3 Host tooling

Everything above describes what runs on the hub. There's a parallel host-side surface — a single PyPI package called `openbricks` that ships:

- A console CLI: `openbricks flash | list | run | upload | stop | log` for hub interaction over BLE / USB. See `tools/openbricks/openbricks_dev/`.
- A MuJoCo-backed simulator: `openbricks sim {preview, run}` opens a physics sim with the same C control cores as the firmware (`*_core.c` files compile into both targets, so the sim's hot-path math is byte-identical). Lives under `tools/openbricks/openbricks_sim/`. Optional via `pip install openbricks[sim]`.
- A driver shim that lets from `openbricks.drivers.st3032 import ST3032Motor` (and `ST3215Motor / JGB37Motor / BNO055 / TCS34725 / HC-SR04 / VL53L0X / VL53L1X`) run unchanged in MuJoCo — `openbricks sim run main.py` installs no-op machine fakes and replaces the I2C driver classes with sim-aware versions.
- Per-run log capture on the hub: every program execution tee'd to `/openbricks_logs/slot_N.log` (10 slot files reused in place, 64 KB each; run `N` overwrites slot `N % 10` and carries its run index in the file's header line — truncate-reuse instead of delete+create keeps littlefs commits at fresh-filesystem cost. Ten slots are enough that a few diagnostic `openbricks run -c` sessions don't rotate away the failing run they're investigating). Each line is prefixed with a raw UTC Unix-epoch-milliseconds stamp; `openbricks log -n NAME` reads the file back over BLE and renders the stamp as `[YYYY-MM-DD HH:MM:SS.mmm]` in your local timezone — useful for untethered runs where no live console was attached. The CLI syncs the hub's RTC from the host clock on every connect (run / upload / log), so runs started after any connect carry real wall-clock time; a hub that powered up and never saw a connect stamps from 2000-01-01, which is self-diagnosing. Button presses leave stamped entries too: a run's log opens with a header line — `started: button press | firmware 1.12.0 | program /program.py | uptime N ms | free N B` — and a stop press writes `button pressed -> stop` the moment it lands, followed by `estop engaged` and a final `stopped: KeyboardInterrupt (N ms after press, M retries) debrief`, so a misbehaving stop chain is diagnosable from the log alone. Since 1.44.0 the stop chain is also **bounded**: the program button is sampled on the hard tick (core 0) — a debounced press while a program runs fires the `KeyboardInterrupt` injection and the native-bus torque-off from C within ~2 ms, regardless of scheduler state (the Python watcher, which stays as defence in depth and as the classic-bus e-stop, measured gaps to 981 ms under load). An uncaught exception writes its **full traceback**, not just the exception's repr — on an untethered run the file is the only record, and a bare `OSError(19,)` doesn't say which call failed. Since 3.4.0 a program that ends by itself (clean return or uncaught exception) first **brakes an adopted drive base to rest** under the controller — heading held by the gyro when one is in use — and logs the outcome (`brake: wheels to rest in N ms`) before the torque-off that ends every run; the button's e-stop path keeps its instant torque-off. Since 3.9.0 that torque-off is **verified per servo**: the broadcast that kills every servo in one packet gets no reply by protocol, and it is fired into whatever transaction the 1 kHz pump has in flight — a servo mid-reply on the half-duplex line cannot hear it, and one that missed it keeps its last speed under torque (bench 2026-09-07: a task motor crept on for minutes after `finished: clean exit`). The broadcast stays as the first strike; the pump then writes torque-off to each configured servo individually (a unicast write is acknowledged) and reads the register back until the wire says 0, retrying and finally latching like a config write. The run's last line is the outcome — `torque-off confirmed: servo ids 2, 1, 3, 4 in 9 ms`, or `torque-off NOT confirmed: servo id 4 answered torque register 0x28 = 1 after 8 attempts - check power and the servo bus wiring`, also printed to the console — so a servo that did not die is named, never silent. The same rule now governs `run_angle(..., wait=False)` on an adopted motor:

the `then=` end-state rides with the move and the C tick applies it the moment the move arrives, so a fire-and-forget move that the program never polls ends in the state it asked for instead of holding under power until the kill.

- Log writes are **asynchronous**. `print` only appends to a RAM buffer; the bytes reach flash from the launcher's Timer tick (`log.pump()`). A `flush()` on `littlefs` forces a metadata commit measured at ~60-90 ms on the ESP32 bench, and doing that per line ran synchronously on the main thread between the user program's own bytecodes — logging cost more than the work it logged and distorted the timing of whatever the robot was controlling. Real commits are paid where durability matters: program end, the stop button, and every `write_text` (the `started: / stopped: / Exception:` lines and button notes). So a hard reset can lose recent `print` output — never the run's framing. Measure it on your own hub with `openbricks run -n NAME examples/log_write_benchmark.py`.
- The **wired UART console is asynchronous too** (a build-time patch, `native/patches/esp32-uart-repl-tx-nonblocking.patch`). Upstream's UART stdout busy-waits until every byte has left the wire — ~5.1 ms for a typical line at 115200, paid by `print()` on the calling thread even with nothing attached to the UART pins. With the patch, `print` copies into a 2 KB ring and the UART interrupt drains it in the background; a print storm deeper than the ring drops the remainder **on the wired console only** — BLE and the run log (both already asynchronous) keep every line. Net effect: a `print` costs string formatting plus three RAM buffer appends, ~1 ms, regardless of what is or isn't listening on any transport.

The Python module names on the host are deliberately split (`openbricks_dev` for the CLI, `openbricks_sim` for the sim) so they don't shadow the firmware-side `openbricks` package, which is sometimes imported on the host by the sim's driver shim.

9.4 Status

All foundational milestones are landed. Roadmap items completed:

- **M1** — always-on 1 kHz scheduler in C (`motor_process.c`).
- **M2** — observer + trajectory + servo state machine, all in C.
- **M3** — 2-DOF coupled drivebase in C, with optional gyro-feedback (`use_gyro(True)`) for slip-immune heading via an attached IMU.
- **M4** — hub abstraction (status LED, user button) + SSD1306 OLED. ESP32 + ESP32-S3 firmware images both build from the same codebase.
- **M5** — per-platform firmware images auto-published on every push to `main` (rolling `latest`) and on `v*` tags (versioned).

Sim phases (host-side): A (chassis + worlds) → B (shared C cores) → C (runtime + driver shim) → D (sensors + scenario reset / scoring) all landed. Phase E1 — pixel-accurate colour-sensor texture sampling — landed via CPU-side sampling: the sensor reads `model.tex_data` directly, computes UV from the geom-local hit point, and indexes the texel. No offscreen GL context, no platform divergence, works on macOS / Linux / Windows. Originally scoped as “Linux EGL headless rendering” but the EGL machinery is only needed for scenes with shadows / lighting / overlays over the textured plane — the WRO use case is a flat printed mat where the texture IS the answer.

Phase F (WRO 2026 RoboMission, 0.10.8 → 0.10.12) is feature-complete:

- **F1** — high-fidelity mat textures rasterised at 150 dpi (~14000×6750 px) from the official “Game Mat

Printing File” PDFs. Drives Phase E1’s sensor sampling against the real printed artwork. `scripts/regen-wro-mat-textures.sh` re-fetches and re-rasterises when WRO updates the source PDFs.

- **F2** — every visible LEGO prop in all three age categories (Elementary, Junior, Senior) modelled as LDraw assemblies. Per- prop `.ldr` files are the source of truth; a map’s LDraw props (`{"ldr": ".../*.ldr"}` in its `map.json`) become MJCF bodies at load time (`mapfile.to_mjcf` via `openbricks_sim.lego_mjcf`; maps are JSON since 4.32.0, their MJCF made in memory). 13 LDraw part types in the registry today; new parts plug in by adding one `_PartSpec` entry. Senior also wires the WRO-published 3D-printed “mosaic frame” STL as a static MuJoCo `<mesh>`. Elementary’s props are Workbench builds since 4.34.0 (`{"file": "props/*.assembly.json"}`), brick for brick from the building instructions.
- **F3** — per-round randomization (WRO General Rules glossary “Robot Round” definition). Same seed → same layout. Specs are per-world tuples of `_RandomizationSpec` driven by one shared seeded RNG, so a Senior round shuffles all four cement colour groups deterministically from a single `seed=N`. A shuffled element keeps the map’s heading and lands with its footprint centred on its slot (a build’s origin is its first brick’s, often off its middle).
- **F4 + F5** — closed the F2 deferreds (mosaic frame mesh, dual- colour Senior barriers) and lifted Junior + Senior randomization slot coordinates from estimates to mat-extracted positions (same pixel-inspection flow Elementary used in 0.10.10).

Remaining in Phase E: broader worlds library, more example walkthroughs. EGL offscreen rendering would unlock simulation of scenes more complex than a printed mat (e.g. coloured 3D obstacles that cast shadows onto the colour sensor’s view); not yet prioritised.

Upgrading the α - β observer to a pbio-style model-based observer (voltage/current coupling + motor model) is on the longer-term list — a precision lift we pick up once we have real hardware to measure against.

BUILDING THE FIRMWARE

openbricks is a custom MicroPython firmware. Users flash the resulting image to their MCU; the openbricks Python package is frozen into the image and the motor-control hot path (scheduler, trajectory, observer, servo, drivebase) lives in a compiled C extension.

See `native/README.md` for the directory layout.

10.1 One-time setup

10.1.1 1. Initialise all submodules

```
git submodule update --init --recursive
```

The MicroPython source tracks a master commit under `native/micropython/`. Bumps should be deliberate and tested on real hardware — `motor_process.c` and the other native modules assume a specific MP ABI. The current pin is documented in `.gitmodules / native/micropython HEAD`.

10.1.2 2. Install the target toolchain

ESP32 / ESP32-S3 (current targets)

Install **ESP-IDF v5.5.4** (or newer in the 5.5 line). MicroPython master supports ESP-IDF 5.3 through 5.5; we recommend 5.5.4 because that's what CI builds against and what the maintainer tests on hardware.

```
mkdir -p ~/esp && cd ~/esp
git clone -b v5.5.4 --recursive https://github.com/espressif/esp-idf.git
cd esp-idf-v5.5 && ./install.sh esp32,esp32s3
source export.sh
```

sets \$IDF_PATH; needs to happen in every shell

Verify:

```
echo $IDF_PATH          # /Users/<you>/esp/esp-idf-v5.5
idf.py --version        # ESP-IDF v5.5.4
```

Note. If you're running an older MicroPython pin (e.g. v1.28.0), you'll need ESP-IDF v5.2.x — v1.28 pre-dates the MP commits that add 5.4+ support. Check `git -C native/micropython log --oneline -1 first`.

10.2 Building

From the repo root:

```
./scripts/build_firmware.sh esp32      # original ESP32 (Xtensa LX6)
./scripts/build_firmware.sh esp32s3    # ESP32-S3 (Xtensa LX7, native USB)
```

The script checks `native/micropython` is populated and `$IDF_PATH` is set, then builds `mpy-cross` once and produces the image for the selected target with `BOARD=openbricks_<target>` and `USER_C_MODULES=$(pwd)/native/user_c_modules`.

Output tree: `native/micropython/ports/esp32/build-openbricks_<target>/`

File	Purpose
<code>firmware.bin</code>	complete flash image (everything below combined)
<code>bootloader/bootloader.bin</code>	second-stage bootloader
<code>partition_table/partition-table.bin</code>	partition map
<code>micropython.bin</code>	application partition only

10.3 Flashing

Use `openbricks flash`. It drives `esptool.py` to write the image and then writes the hub's BLE advertising name into NVS. The name is **per-hub**, set at flash time (not build time) — one firmware image is reused across every hub, and each hub gets its own identity here. `--name` is mandatory: two hubs that answer to the same name can't be individually addressed over BLE.

```
pipx install openbricks          # one-time; or `pip install openbricks`

openbricks flash \
  --name RobotA \
  --firmware native/micropython/ports/esp32/build-openbricks_esp32s3/
↪ firmware.bin
```

`--port` is auto-detected when exactly one ESP device is connected; pass it explicitly with several devices attached. (`--firmware` is optional too — omitted, the newest **released** image for the detected chip is downloaded — but when flashing a local build like here, point it at your build output.) A build's `firmware.bin` carries no version in its name, so when the hub already runs `openbricks` the flash asks for confirmation before overwriting it; pass `--yes` in scripts.

The command erases flash, writes `firmware.bin` at the offset the image was built for — `0x0` on the S3, `0x1000` on the classic ESP32, detected from the `partition-table` position inside the image itself — waits for the device to boot, then pokes the name into `esp32.NVS("openbricks").hub_name` via `mpremote` and reads it back to verify. Cross-platform — works on macOS, Linux, Windows (use `COM5` etc. for `--port`).

If you'd rather use `esptool.py` directly (e.g. mass-flashing with a fixture, no name needed yet), the raw commands:

```
# Classic ESP32 - bootloader at 0x1000
esptool.py --chip esp32 --port /dev/tty.usbserial-XXXX erase_flash
esptool.py --chip esp32 --port /dev/tty.usbserial-XXXX --baud 460800 \
```

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```

write_flash -z 0x1000 openbricks-esp32-firmware-v0.9.0.bin

# ESP32-S3 - bootloader at 0x0
esptool.py --chip esp32s3 --port /dev/tty.usbmodemXXXX erase_flash
esptool.py --chip esp32s3 --port /dev/tty.usbmodemXXXX --baud 460800 \
    write_flash -z 0x0 openbricks-esp32s3-firmware-v0.9.0.bin

```

A hub flashed this way boots fine but has no name; `openbricks.bluetooth.set_enabled(True)` will raise `HubNameNotSetError` until you write one via `mpremote`:

```

mpremote connect PORT exec "
import esp32
nvs = esp32.NVS('openbricks')
nvs.set_blob('hub_name', b'RobotA')
nvs.commit()
"

```

Common footgun. Flashing the S3 `firmware.bin` at `0x1000` (classic-ESP32 offset) leaves `0x0..0xFFF` untouched; the S3 ROM boots from `0x0` and fails with `Invalid image block, can't boot`. Always match the offset to the chip.

Or, if you built with `idf.py`:

```

cd native/micropython/ports/esp32/build-openbricks_esp32
idf.py -p /dev/tty.usbserial-XXXX flash monitor

```

10.4 Verifying the image at the REPL

Connect over the USB UART (e.g. `mpremote connect /dev/tty.usbserial-XXXX`) and at the REPL:

```

from openbricks._native import motor_process, Servo, TrapezoidalProfile, \
    Observer, DriveBase
motor_process.is_running()    # False initially; True once a motor attaches

```

The `openbricks` Python package is frozen into the image, so `import openbricks.*` works without copying any files. The first time a closed-loop motor calls `run_speed`, it registers its control step with the scheduler and the 1 kHz timer ISR comes online.

10.5 Running tests

Tests exercise the real C module under the unix MicroPython binary — the same build as firmware, minus the ESP32 port:

```

make -C native/micropython/mpy-cross -j
make -C native/micropython/ports/unix \
    VARIANT=standard \
    USER_C_MODULES=$(pwd)/native/user_c_modules -j
./native/micropython/ports/unix/build-standard/micropython tests/run.py

```

The runner spawns one MP subprocess per test module for state isolation (the module list lives in `tests/run.py::_TEST_MODULES`). Expected final line: `all modules passed`.

10.6 CI

GitHub Actions runs several job groups on every push / PR (see `.github/workflows/ci.yaml`):

- **test** — builds the unix MP binary with the `_openbricks_native user_c_module` and runs `tests/run.py`. No ESP-IDF needed. This is the merge gate for firmware changes.
- **cpython-tests / openbricks-py** — the CPython-compatible subset of the firmware tests under stock CPython (catches MP/CPython interpreter divergence), the latter with coverage upload.
- **openbricks-host** — the host CLI + sim test suite (`tools/openbricks/tests`) with coverage, plus a `--help` smoke test of every subcommand and a headless preview of every shipped sim world.
- **coverage** — the firmware suite again on the gcov-instrumented unix MP build; uploads C-core coverage.
- **firmware** — a matrix job (targets: `esp32`, `esp32s3`) that builds each image inside the `espressif/idf:v5.5.4` container and uploads `firmware.bin` + `bootloader` + `partition-table` per target as a workflow artifact.
- **qemu-smoke** — boots the just-built ESP32-S3 image in Espressif's QEMU and asserts the bootloader reaches app entry with no panic markers.
- **sim-rs** — the native sim app (`tools/sim`): `fmt` + `clippy` + tests under coverage on Linux for PRs and tags, and a release build + archive on all four platforms.
- **release / build-openbricks-sdist / build-openbricks-wheels / publish-openbricks** — firmware GitHub Releases (rolling latest on main pushes, versioned on `v*` tags) and the PyPI sdist + cibuildwheel wheels published on `cli/v*` tags (releases up to 0.10.24 used `openbricks/v*`). A main push re-runs no test job (each was proven on the PR that produced the commit) and only builds `firmware` + `sim-rs` for the rolling release. On a tag the whole suite runs and gates the publish: a failed test, `c-unit`, `cpython-tests`, `openbricks-py`, `openbricks-host`, `firmware` or `sim-rs` stops a `v*` release, and a failed test, `c-unit`, `cpython-tests`, `openbricks-py`, `openbricks-host` or `sdist/wheel` build stops a `cli/v*` PyPI upload. `qemu-smoke` is `continue-on-error` and reports without gating. `tools/openbricks/tests/test_release_tags.py` plays this graph out per event.

Successful PRs produce a flashable image downloadable from the Actions run.

10.7 Troubleshooting

“IDF_PATH is not set” — source `export.sh` in the shell that runs the build script. Each new shell needs it.

Build fails at `modbluetooth_nimble.c` — BLE (NimBLE) is *enabled* on both boards via `boards/sdkconfig.ble` in each board's `mpconfigboard.cmake` `SDKCONFIG_DEFAULTS` list; it's what `openbricks run / upload / stop` ride on, so don't disable it. A failure here usually means an ESP-IDF / MicroPython version mismatch — see the `WIFI_AUTH_MAX` entry below. (WiFi, by contrast, *is* deliberately disabled in `sdkconfig.board + mpconfigboard.h`.)

Build fails at `network_wlan.c` with a `_Static_assert` about `WIFI_AUTH_MAX` — ESP-IDF / MP version mismatch. Either bump ESP-IDF into the supported range (5.3–5.5 for current MP master) or bump the MP submodule.

Submodule is empty — `git submodule update --init --recursive`.

Link error “undefined reference to `mp_register_module__openbricks_native`” — the `user_c_module` isn’t being built into the image. Check `./scripts/build_firmware.sh` is passing `USER_C_MODULES=$(pwd) /native/user_c_modules` to `idf.py`.

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